TayChenJiangNguyenLi-stat443project

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Loading packages

```
library(readr)
library(dplyr)
library(lubridate)
```

Reading the data

```
# Covering 1981-01 to 2025-12
nhpi_full <- read.csv("18100205.csv", na.strings = c("..", "NA", "")) %>%
  mutate(REF_DATE = as.Date(paste0(REF_DATE, "-01")),
         REF_DATE_ym = format(REF_DATE, "%Y-%m"))
# head(nhpi_full)

# Covering 1981-01 to 2023-12
ir <- read_csv("IRSTCB01CAM156N.csv",show_col_types = FALSE) %>%
  rename(REF_DATE = observation_date,
         IR = IRSTCB01CAM156N) %>%
  mutate(REF_DATE = as.Date(REF_DATE),
         REF_DATE_ym = format(REF_DATE, "%Y-%m"),
         IR = as.numeric(IR)) %>%
  filter(!is.na(IR))
# head(ir)

# Combining the dataset for ARMAX (cover the overlap only)
nhpi_ir <- nhpi_full %>%
  left_join(ir %>% select(REF_DATE_ym, IR),
            by = "REF_DATE_ym") %>%
  filter(REF_DATE <= as.Date("2023-12-01"))
# head(nhpi_ir)

# Getting lagged interest rate
nhpi_ir <- nhpi_ir %>%
  group_by(GEO, New.housing.price.indexes) %>%
  arrange(REF_DATE) %>%
  mutate(IR_lag1 = lag(IR, 1),
         IR_lag2 = lag(IR, 2)) %>%
  ungroup()
```

```r
# head(nhpi_ir)

# Checking date coverage
range(nhpi_full$REF_DATE)  # should include 1981-01-01 to 2025-12-01 (or close)
range(ir$REF_DATE)         # should include 1981-01-01 to 2023-12-01
range(nhpi_ir$REF_DATE)    # should include 1981-01-01 to 2023-12-01

# Checking if there's any missing IR after merge? (should be basically none within overlap)
mean(is.na(nhpi_ir$IR))

# Checking lags (first month per series will be NA)
nhpi_ir %>%
  filter(GEO == "Canada", New.housing.price.indexes == "Total (house and land)") %>%
  select(REF_DATE, VALUE, IR, IR_lag1, IR_lag2) # %>%
  # head(5)
```

Defining functions

Baseline 1: Naive/Persistence

```r
persist_fc <- function(train, holdout, iprint = FALSE) {
  n <- length(train)
  n_holdout <- length(holdout)
  mse <- 0
  fcvec <- numeric(n_holdout)

  # first forecast
  fc <- train[n]
  yt <- holdout[1]
  fcvec[1] <- fc
  fcerror <- yt - fc
  mse <- mse + fcerror^2

  # remaining forecasts
  for (i in 2:n_holdout) {
    yt <- holdout[i]
    fc <- holdout[i-1]
    fcvec[i] <- fc
    fcerror <- yt - fc
    mse <- mse + fcerror^2
  }

  if (iprint)
    print(cbind(holdout, fcvec))

  rmse <- sqrt(mse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

Baseline 2: Training-Set Average

```
iid_fc <- function(train, holdout, iprint = FALSE) {
  n <- length(train)
  n_holdout <- length(holdout)
  average <- mean(train)
  mse <- 0
  fcvec <- numeric(n_holdout)

  for (i in 1:n_holdout) {
    yt <- holdout[i]
    fc <- average
    fcvec[i] <- fc
    fcerror <- yt - fc
    mse <- mse + fcerror^2
  }

  if (iprint)
    print(cbind(holdout, fcvec))

  rmse <- sqrt(mse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

Method 1: Best Exponential Smoothing (allowing for trend and seasonality, if relevant)

```
# Simple Exponential Smoothing
esm_fc <- function(train, holdout, alpha, level, iprint = FALSE) {
  n_holdout <- length(holdout)
  sse <- 0
  fcvec <- numeric(n_holdout)

  fc <- level
  yt <- holdout[1]
  level_new <- level
  fcvec[1] <- fc
  fcerror <- yt - fc
  sse <- sse + fcerror^2

  for (i in 2:n_holdout) {
    level_new <- alpha * holdout[i-1] + (1 - alpha) * level_new

    fc <- level_new
    fcvec[i] <- fc
    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + fcerror^2

    if (iprint) {
      cat("i =", i, "| fc =", fc, "| y =", yt, "| level =", level_new, "\n")
    }
```

```r
  }

  rmse <- sqrt(sse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

```r
# Linear Exponential Smoothing
lholt_fc <- function(train, holdout, alpha, beta, level, slope, iprint = FALSE) {
  n_holdout <- length(holdout)
  sse <- 0
  fcvec <- numeric(n_holdout)

  # first forecast
  fc <- level + slope
  yt <- holdout[1]
  newfc <- fc
  fcvec[1] <- fc
  fcerror <- yt - fc
  sse <- sse + fcerror^2

  level_prev <- level
  slope_prev <- slope

  if (iprint) {
    cat("i = 1 | fc =", fc, "| y =", yt, "| level =", level_prev, "| slope =", slope_prev, "\n")
  }

  # remaining forecasts
  for (i in 2:n_holdout) {
    level_new <- alpha * holdout[i-1] + (1 - alpha) * fc
    slope_new <- beta * (level_new - level_prev) + (1 - beta) * slope_prev

    fc <- level_new + slope_new
    fcvec[i] <- fc
    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + fcerror^2

    if (iprint) {
      cat("i =", i, "| fc =", fc, "| y =", yt, "| level =", level_new, "| slope =", slope_new, "\n")
    }

    level_prev <- level_new
    slope_prev <- slope_new
  }

  rmse <- sqrt(sse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

```r
# Additive Seasonal Exponential Smoothing
aseason_fc <- function(train, holdout, alpha, beta, gamma, level, slope, season, iprint = FALSE) {
  n_holdout <- length(holdout)
```

```r
m <- length(season)
sse <- 0
fcvec <- numeric(n_holdout)

# prev states from training end
level_prev <- level
slope_prev <- slope

# helper: map i -> seasonal position in 1..d
spos <- function(i) ((i-1) %% m) + 1

# First forecast
fc <- level_prev + slope_prev + season[spos(1)]
fcvec[1] <- fc
yt <- holdout[1]
fcerror <- yt - fc
sse <- sse + fcerror^2

if (iprint) {
  cat("i = 1 | fc =", fc, "| y =", yt, "| level =", level_prev, "| slope =", slope_prev, "| season ="
}

# Remaining forecast
for (i in 2:n_holdout) {
  # Seasonal of previous observation (i-1)
  k_prev <- spos(i-1)
  s_prev <- season[k_prev]
  # Update using previous observation holdout[i-1]
  level_new <- alpha * (holdout[i-1] - s_prev) + (1 - alpha) * (level_prev + slope_prev)
  slope_new <- beta * (level_new - level_prev) + (1 - beta) * slope_prev
  s_new <- gamma * (holdout[i-1] - level_new) + (1 - gamma) * s_prev

  # store updated seasonal state one full cycle later
  season[k_prev] <- s_new

  # Seasonal of current forecast (i)
  k_cur <- spos(i)
  s_cur <- season[k_cur]

  # Forecast current observation using updated states + current seasonal
  fc <- level_new + slope_new + s_cur
  fcvec[i] <- fc
  yt <- holdout[i]
  fcerror <- yt - fc
  sse <- sse + fcerror^2

  if (iprint) {
    cat("i =", i, "| fc =", fc, "| y =", yt, "| level =", level_new, "| slope =", slope_new, "| seaso
  }

  # Update level and slope
  level_prev <- level_new
  slope_prev <- slope_new
```

```
  }

  rmse <- sqrt(sse / n_holdout)
  return(list(fc = fcvec, rmse = rmse))
}
```

Method 2: ARIMA

```
arma_fc <- function(tsdata, ntrain, order, seasonal, method, traincoef, include.mean, iprint=F) {
  obj = arima(tsdata, order=order, seasonal = seasonal, init=traincoef, fixed=traincoef,
              method=method, include.mean=include.mean, optim.control=list(maxit=0)) # no log-likelihoo
  fc = tsdata - obj$residuals

  ntotal = length(tsdata)
  holdout_fc = fc[(ntrain+1):ntotal]
  holdout = tsdata[(ntrain+1):ntotal]

  if(iprint) print(cbind(holdout,holdout_fc))

  rmse = sqrt(mean((holdout-holdout_fc)^2))
  return(list(rmse=rmse, fc=holdout_fc, residuals = obj$residuals))
}
```

Method 3: ARIMAX with lagged interest

```
armax_fc <- function(tsdata, ntrain, order, seasonal, xreg, method, traincoef, include.mean, iprint=F) {
  obj = arima(tsdata, order=order, seasonal = seasonal, init=traincoef,
              fixed=traincoef, method=method, include.mean=include.mean,
              xreg=xreg, optim.control=list(maxit=0))

  fc = tsdata - obj$residuals

  ntotal = length(tsdata)
  holdout_fc = fc[(ntrain+1):ntotal]
  holdout = tsdata[(ntrain+1):ntotal]

  if(iprint) print(cbind(holdout,holdout_fc))

  rmse = sqrt(mean((holdout-holdout_fc)^2))
  return(list(rmse=rmse, fc=holdout_fc, residuals = obj$residuals))
}
```

Analyzing Montreal

Setting up

```r
# Reading the data
montreal_nhpi_data <- read.csv("Full_Cities/montreal_full.csv")
montreal_nhpi_ir_data <- read.csv("Extracted_Cities/montreal.csv")

# Extracting the NHPI
montreal_nhpi <- montreal_nhpi_data$VALUE

# Extracting the one-month lag interest rate
montreal_irlag1 <- montreal_nhpi_ir_data$IR_lag1

# Setting the time window from 1981-2023
montreal_ts <- ts(montreal_nhpi, start = c(1981,1), end = c(2023, 12), frequency = 12)
montreal_ir_ts <- ts(montreal_irlag1, start = c(1981,1), end = c(2023, 12), frequency = 12)

# Applying a transformation for NHPI
montreal_transformed_nhpi <- diff(log(montreal_ts))

# Setting a training set from 1981-2021
montreal_train_ts <- window(montreal_transformed_nhpi, end = c(2021,12))
montreal_train_ir_xreg <- window(montreal_ir_ts, start = c(1981,2), end = c(2021,12))

# Setting a holdout set from 2022-2023
montreal_holdout_ts <- window(montreal_transformed_nhpi, start = c(2022,1))
montreal_holdout_ir_xreg <- window(montreal_ir_ts, start = c(2022,1))
```

Doing the baseline forecasts

```r
# Baseline 1: Naive/Persistence
montreal_persist_fc <- persist_fc(montreal_train_ts, montreal_holdout_ts, iprint = F)

# Baseline 2: Training-Set Average
montreal_iid_fc <- iid_fc(montreal_train_ts, montreal_holdout_ts, iprint = F)
```

Doing the exponential smoothing forecasts

```r
# Method 1
ntrain <- length(montreal_train_ts)
ntotal <- length(montreal_ts)-1

montreal_transformed_nhpi_train <- montreal_transformed_nhpi[1:ntrain]
montreal_transformed_nhpi_holdout <- montreal_transformed_nhpi[(ntrain+1):ntotal]

# Simple Exponential Smoothing
fit_expsmo <- HoltWinters(montreal_train_ts, beta = F, gamma = F)
montreal_esm_fc <- esm_fc(montreal_transformed_nhpi_train, montreal_transformed_nhpi_holdout,
                          alpha = fit_expsmo$alpha, level = fit_expsmo$coefficients["a"],
                          iprint = F)

# Linear Exponential Smoothing
fit_hw <- HoltWinters(montreal_train_ts, gamma = F)
```

```r
montreal_holt_fc <- lholt_fc(montreal_transformed_nhpi_train, montreal_transformed_nhpi_holdout,
                             alpha = fit_hw$alpha, beta = fit_hw$beta,
                             level = fit_hw$coefficients["a"],
                             slope = fit_hw$coefficients["b"], iprint = F)

# Additive Seasonal Exponential Smoothing
wafit <- HoltWinters(montreal_train_ts, seasonal = "additive")
montreal_aseason_fc <- aseason_fc(montreal_transformed_nhpi_train,
                                  montreal_transformed_nhpi_holdout,
                                  alpha = wafit$alpha, beta = wafit$beta,
                                  gamma = wafit$gamma, level = wafit$coef[1],
                                  slope = wafit$coef[2], season = wafit$coefficient[3:14],
                                  iprint = F)

# Comparing RMSE
rmse_esm <- cbind(montreal_esm_fc$rmse, montreal_holt_fc$rmse, montreal_aseason_fc$rmse)
colnames(rmse_esm) <- c("esm", "lholt", "aseason") # RMSE: 0.0079, 0.0077, 0.0088
rmse_esm
```

```
##           esm       lholt     aseason
## a 0.007918044 0.007662966 0.008806579
```

```r
# Best exponential smoothing method: Linear Exponential Smoothing
```

Doing the ARIMA forecasts

```r
# Method 2
montreal_transformed_nphi_auto <- forecast::auto.arima(montreal_train_ts)
```

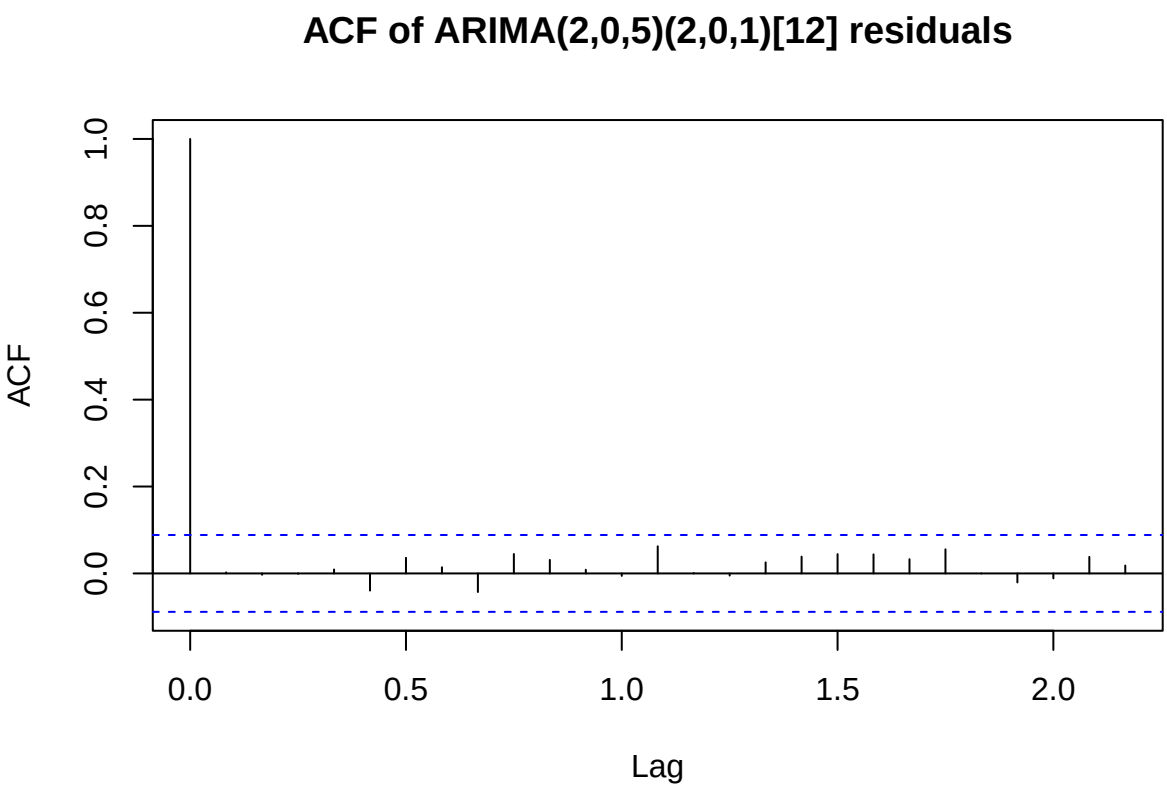
```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
summary(montreal_transformed_nphi_auto) # 0.0048783
```

```
## Series: montreal_train_ts
## ARIMA(2,0,5)(2,0,1)[12] with non-zero mean
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2933  -0.4683  -1.1324  0.3576  0.0775  -0.0112  0.1526  0.7129  0.0270
## s.e.  0.1615   0.1643   0.1629  0.1522  0.0680   0.0678  0.0628  0.2571  0.0723
##          sma1    mean
##       -0.6604  0.0034
## s.e.   0.2622  0.0007
##
## sigma^2 = 2.434e-05:  log likelihood = 1916.27
## AIC=-3808.54   AICc=-3807.89   BIC=-3758.19
##
## Training set error measures:
```

```
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -4.221651e-05 0.004878312 0.002960714 NaN  Inf 0.7689245
##                     ACF1
## Training set 0.002456102
```

```
resid_fit <- residuals(montreal_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(2,0,5)(2,0,1)[12] residuals")
```




```
# Comparing with the other ARIMA models
manual_fit <- arima(montreal_train_ts, order = c(2,0,5),
                    seasonal = list(order = c(2,0,1), period = 12), method = "ML")
```

```
## Warning in log(s2): NaNs produced
## Warning in log(s2): NaNs produced
```

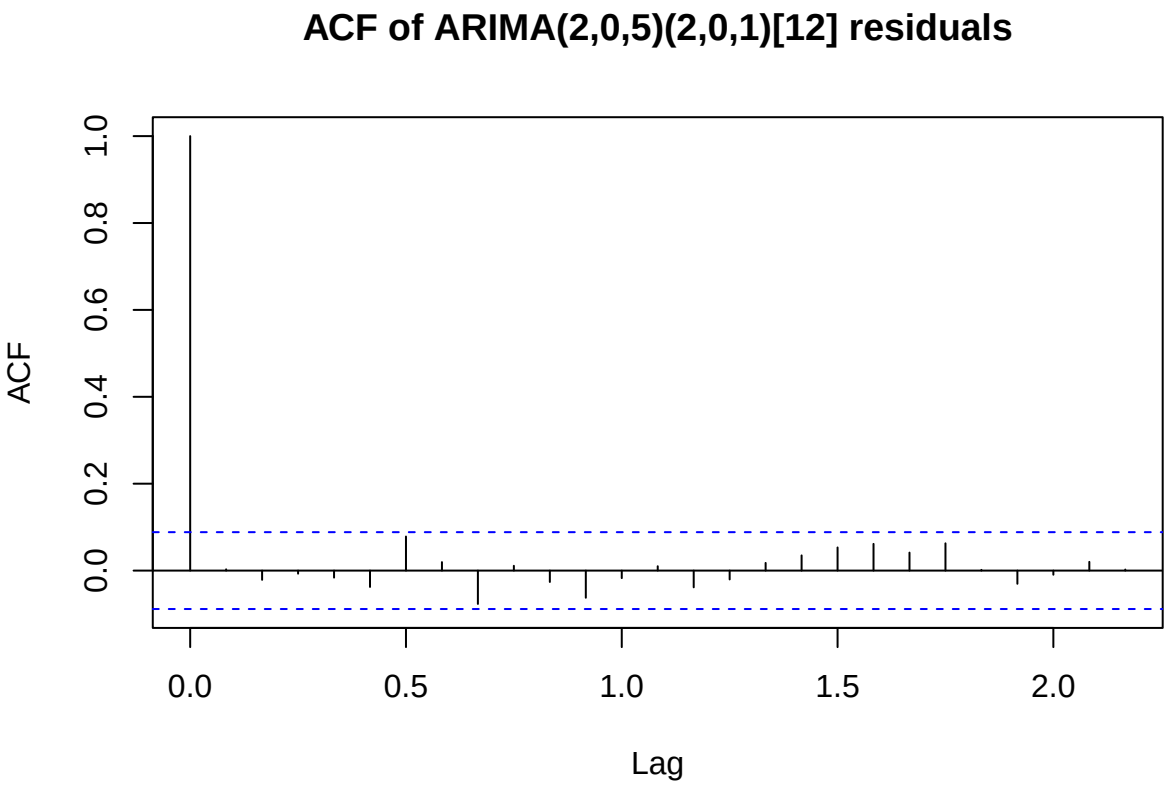
```
summary(manual_fit) # RMSE: 0.004903117
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 5), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
```

```
## Warning in sqrt(diag(x$var.coef)): NaNs produced
```

```
##          ar1     ar2      ma1      ma2    ma3     ma4     ma5    sar1    sar2
##       0.8622  0.0517  -0.6977  -0.0803  0.022  0.0013  0.1062  0.8357  0.0628
## s.e.  0.0033  0.0051      NaN      NaN    NaN     NaN  0.0123  0.0084  0.0091
##          sma1  intercept
##       -0.8601     0.0032
## s.e.   0.0067     0.0013
##
## sigma^2 estimated as 2.404e-05:  log likelihood = 1913.86,  aic = -3803.71
##
## Training set error measures:
##                        ME       RMSE         MAE MPE MAPE      MASE       ACF1
## Training set 1.504883e-05 0.00490298 0.002904972 NaN  Inf 0.7677291 0.00289282
```

```
acf(residuals(manual_fit), main = "ACF of ARIMA(2,0,5)(2,0,1)[12] residuals")
```



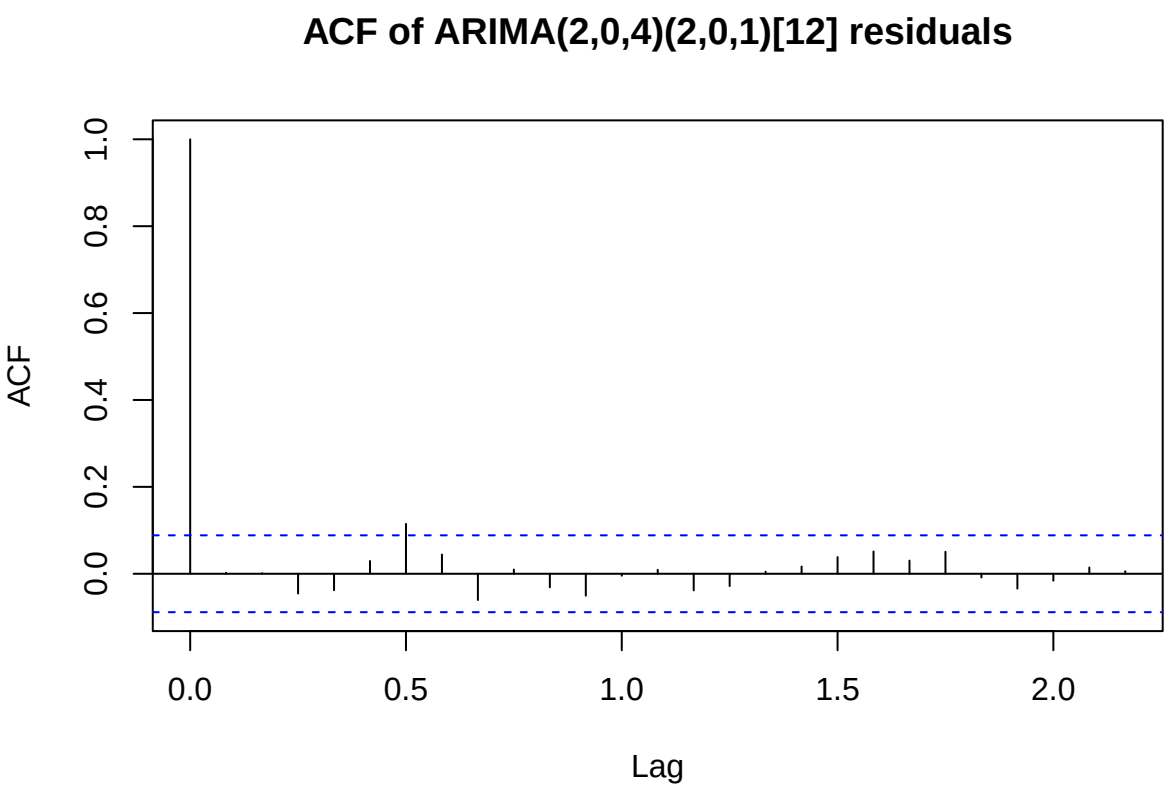

```
test1 <- arima(montreal_train_ts, order = c(2,0,4),
               seasonal = list(order = c(2,0,1), period = 12), method = "ML")
summary(test1) # RMSE: 0.004919859
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 4), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
```

```
## Warning in sqrt(diag(x$var.coef)): NaNs produced
```

```
##          ar1     ar2     ma1      ma2      ma3     ma4    sar1    sar2     sma1
##       0.1009  0.8396  0.0708  -0.7734  -0.0067  0.0520  0.8286  0.0792  -0.8625
## s.e.     NaN     NaN     NaN      NaN   0.0474  0.0536  0.1540  0.0648   0.1467
##       intercept
##          0.0043
## s.e.     0.0018
##
## sigma^2 estimated as 2.421e-05:  log likelihood = 1911.91,  aic = -3801.82
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -8.718344e-05 0.004919859 0.002886015 NaN  Inf 0.7627192
##                     ACF1
## Training set 0.002225658
```

```
acf(residuals(test1), main = "ACF of ARIMA(2,0,4)(2,0,1)[12] residuals")
```




```
test2 <- arima(montreal_train_ts, order = c(3,0,4),
               seasonal = list(order = c(2,0,1), period = 12), method = "ML")
```

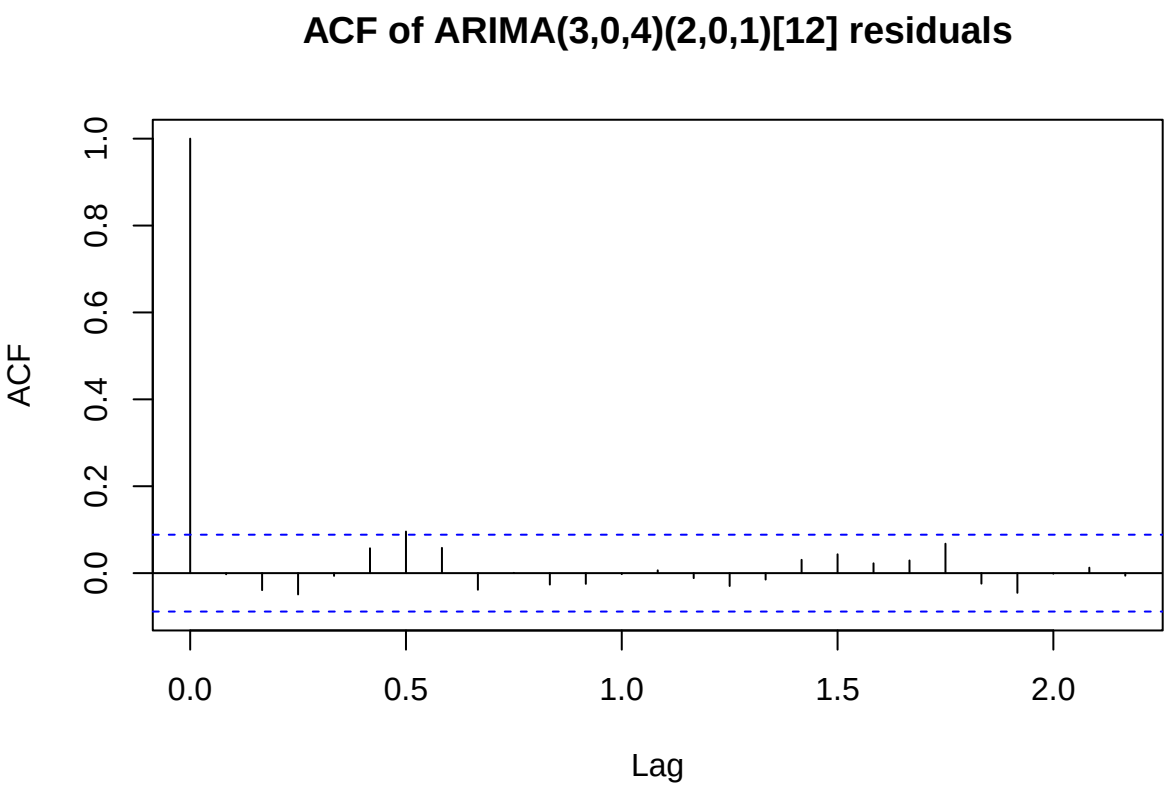
```
## Warning in arima(montreal_train_ts, order = c(3, 0, 4), seasonal = list(order =
## c(2, : possible convergence problem: optim gave code = 1
```

```
summary(test2) # RMSE: 0.004857711
```

```
##
## Call:
```

```
## arima(x = montreal_train_ts, order = c(3, 0, 4), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2     ar3      ma1     ma2      ma3      ma4    sar1
##       0.2633  -0.2965  0.9676  -0.0730  0.3999  -0.8139  -0.0725  0.8008
## s.e.  0.0164   0.0139  0.0167   0.0509  0.0328   0.0344   0.0486  0.1765
##         sar2     sma1  intercept
##       0.1004  -0.8507     0.0046
## s.e.  0.0677   0.1709     0.0021
##
## sigma^2 estimated as 2.36e-05:  log likelihood = 1915.58,  aic = -3807.16
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE     MASE
## Training set -9.309444e-05 0.004857711 0.002842069 NaN  Inf 0.751105
##                     ACF1
## Training set -0.00262493
```

```
acf(residuals(test2), main = "ACF of ARIMA(3,0,4)(2,0,1)[12] residuals")
```



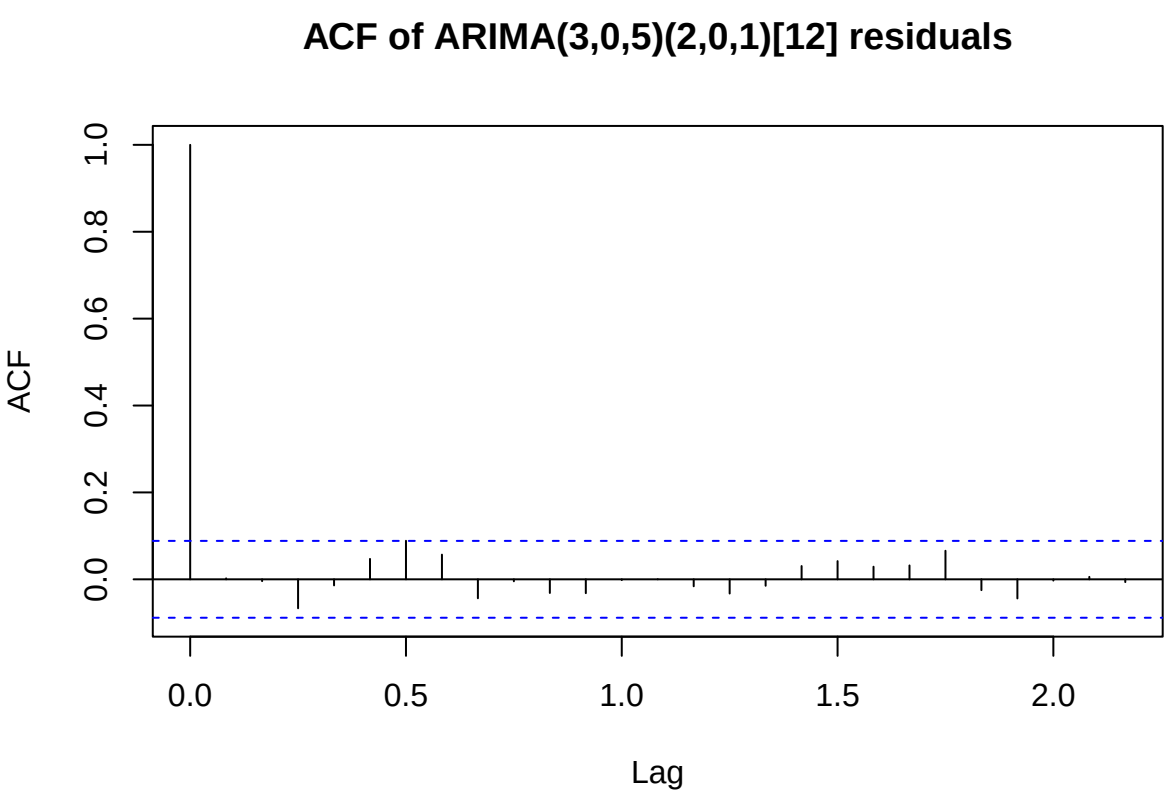

```
test3 <- arima(montreal_train_ts, order = c(3,0,5),
               seasonal = list(order = c(2,0,1), period = 12), method = "ML")
```

```
## Warning in arima(montreal_train_ts, order = c(3, 0, 5), seasonal = list(order =
## c(2, : possible convergence problem: optim gave code = 1
```

```r
summary(test3) # RMSE: 0.004852676
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(3, 0, 5), seasonal = list(order = c(2,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2     ar3      ma1     ma2      ma3      ma4     ma5
##       0.2650  -0.2938  0.9673  -0.0852  0.3589  -0.7964  -0.0709  0.0518
## s.e.  0.0181   0.0158  0.0186   0.0503  0.0494   0.0428   0.0475  0.0536
##         sar1    sar2     sma1  intercept
##       0.8062  0.1087  -0.8704     0.0047
## s.e.  0.1483  0.0654   0.1410     0.0023
##
## sigma^2 estimated as 2.355e-05:  log likelihood = 1916.06,  aic = -3806.13
##
## Training set error measures:
##                        ME        RMSE        MAE MPE MAPE      MASE
## Training set -8.636689e-05 0.004852806 0.00282837 NaN  Inf 0.7474846
##                     ACF1
## Training set 0.002177752
```

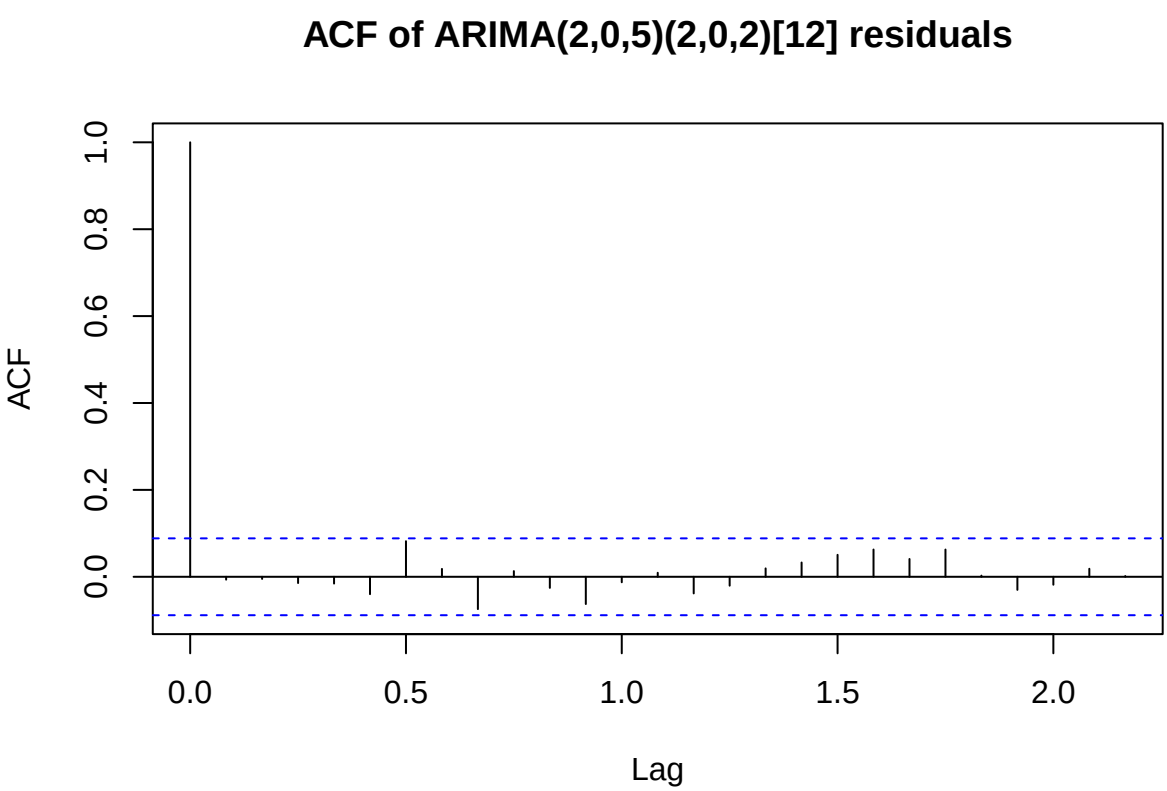
```r
acf(residuals(test3), main = "ACF of ARIMA(3,0,5)(2,0,1)[12] residuals")
```




```r
test4 <- arima(montreal_train_ts, order = c(2,0,5),
               seasonal = list(order = c(2,0,2), period = 12), method = "ML")
summary(test4) # RMSE: 0.004908301
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 5), seasonal = list(order = c(2,
##     0, 2), period = 12), method = "ML")
##
## Coefficients:

## Warning in sqrt(diag(x$var.coef)): NaNs produced

##          ar1     ar2     ma1      ma2     ma3  ma4     ma5    sar1    sar2
##       0.8194  0.0941  -0.647  -0.1375  0.0369    0  0.1048  0.6319  0.2896
## s.e.     NaN     NaN     NaN      NaN     NaN  NaN     NaN     NaN     NaN
##          sma1     sma2  intercept
##       -0.6646  -0.2117     0.0042
## s.e.      NaN      NaN     0.0012
##
## sigma^2 estimated as 2.405e-05:  log likelihood = 1913.52,  aic = -3801.04
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001607204 0.004903864 0.002935285 NaN  Inf 0.7757402
##                     ACF1
## Training set -0.00667451
```

```
acf(residuals(test4), main = "ACF of ARIMA(2,0,5)(2,0,2)[12] residuals")
```



```r
test5 <- arima(montreal_train_ts, order = c(2,0,4),
               seasonal = list(order = c(2,0,2), period = 12), method = "ML")
```

```
## Warning in arima(montreal_train_ts, order = c(2, 0, 4), seasonal = list(order =
## c(2, : possible convergence problem: optim gave code = 1
```

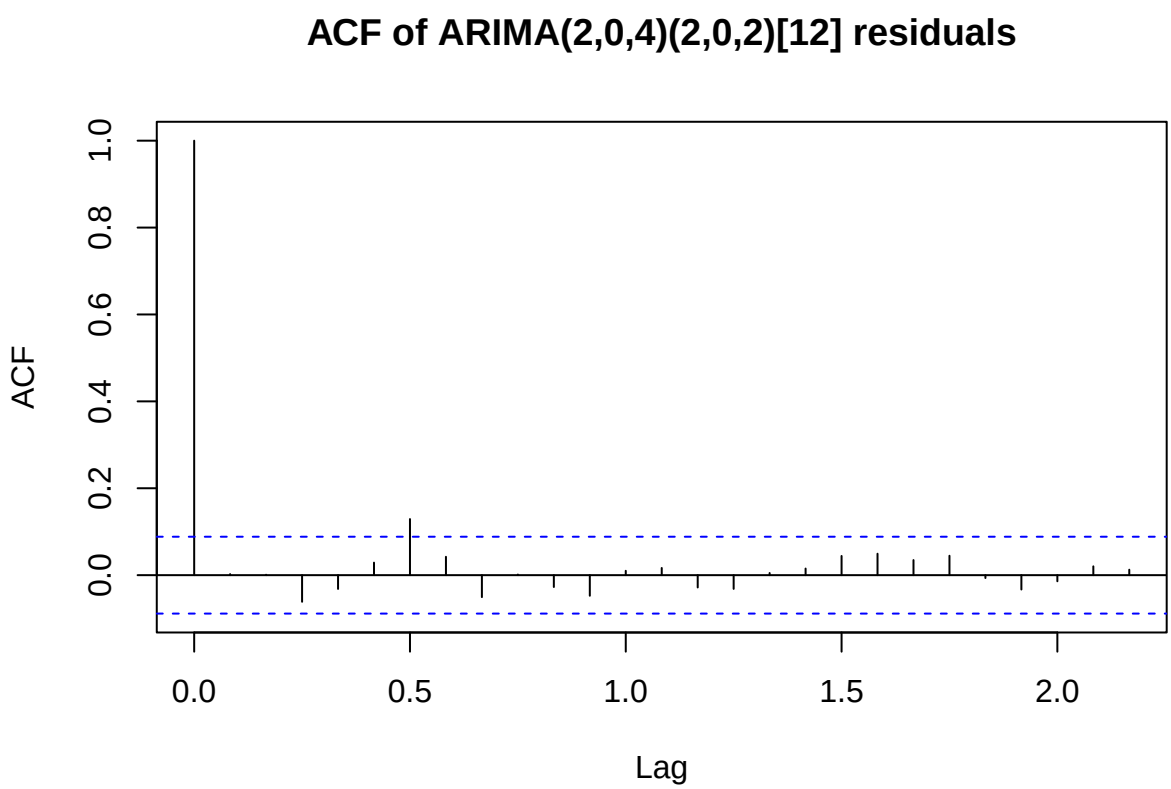
```r
summary(test5) # RMSE: 0.004891118
```

```
##
## Call:
## arima(x = montreal_train_ts, order = c(2, 0, 4), seasonal = list(order = c(2,
##     0, 2), period = 12), method = "ML")
##
## Coefficients:
```

```
## Warning in sqrt(diag(x$var.coef)): NaNs produced
```

```
##           ar1     ar2     ma1      ma2      ma3     ma4    sar1     sar2
##       -0.0227  0.9456  0.2107  -0.8658  -0.0184  0.0582  1.1899  -0.3936
## s.e.   0.0263  0.0249  0.0524   0.0561   0.0507  0.0541     NaN      NaN
##          sma1    sma2  intercept
##       -1.2437  0.5093     0.0045
## s.e.      NaN     NaN     0.0015
##
## sigma^2 estimated as 2.392e-05:  log likelihood = 1914.05,  aic = -3804.1
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001308868 0.004891118 0.002897664 NaN  Inf 0.7657977
##                     ACF1
## Training set 0.002438932
```

```r
acf(residuals(test5), main = "ACF of ARIMA(2,0,4)(2,0,2)[12] residuals")
```




```r
# Comparing AIC values
aic = cbind(manual_fit$aic, test1$aic, test2$aic, test3$aic, test4$aic, test5$aic)
colnames(aic) = c("manual_fit", "test1", "test2", "test3", "test4", "test5")
aic
```

```
##      manual_fit     test1     test2     test3     test4     test5
## [1,]   -3803.71 -3801.817 -3807.156 -3806.129 -3801.036 -3804.098
```

```r
# Best (based on lowest AIC and ACF plot): test3, ARIMA(3,0,5)(2,0,1)[12]

order <- c(3,0,5)
seasonal <- list(order = c(2,0,1), period = 12)
# Fit the model to the training data
train_arima_model <- arima(montreal_train_ts, order = order,
                           seasonal = seasonal, method = "ML", include.mean = T)
```

```
## Warning in arima(montreal_train_ts, order = order, seasonal = seasonal, :
## possible convergence problem: optim gave code = 1
```

```r
# Result the moving 1-step ahead forecast for holdout set
arima_results <- arma_fc(tsdata = montreal_transformed_nhpi,
                         ntrain = length(montreal_train_ts),
                         order = order, seasonal = seasonal, method = "ML",
                         traincoef = coef(train_arima_model), include.mean = T)
arima_results$rmse
```

```
## [1] 0.008244647
```

Doing the ARIMAX forecasts with lagged interest

```
# Method 3
montreal_fit_arimax <- forecast::auto.arima(montreal_train_ts, xreg = montreal_train_ir_xreg)
summary(montreal_fit_arimax)
```

```
## Series: montreal_train_ts
## Regression with ARIMA(2,0,5)(2,0,1)[12] errors
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2998  -0.4778  -1.1411  0.3668  0.0793  -0.0121  0.1545  0.7148  0.0216
## s.e.  0.1554   0.1577   0.1572  0.1477  0.0684   0.0681  0.0629  0.2831  0.0728
##          sma1  intercept   xreg
##       -0.6608     0.0028  1e-04
## s.e.   0.2890     0.0010  1e-04
##
## sigma^2 = 2.435e-05:  log likelihood = 1916.72
## AIC=-3807.45   AICc=-3806.69   BIC=-3752.9
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE     MASE
## Training set -2.873018e-05 0.004873876 0.002945387 NaN  Inf 0.764944
##                     ACF1
## Training set 0.003074691
```

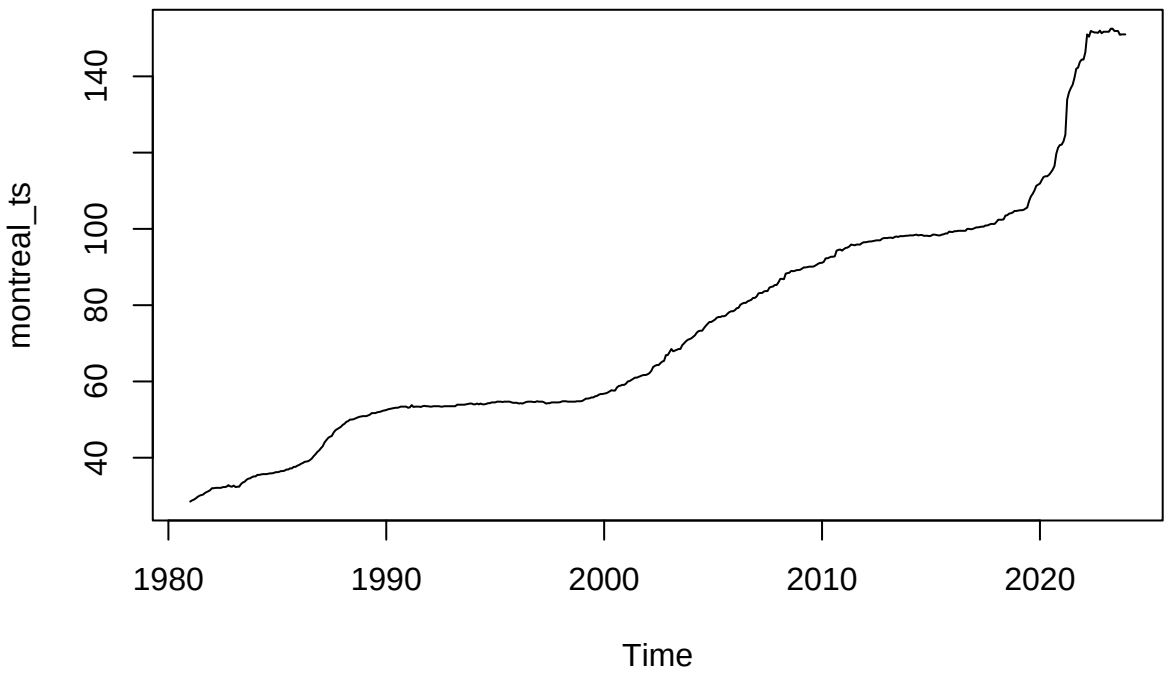
```
order <- c(3,0,5)
seasonal <- list(order = c(2,0,1), period = 12)
# Fitting a model to the training data
train_arimax_model <- arima(montreal_train_ts, order = order, seasonal = seasonal,
                            xreg = montreal_train_ir_xreg, method = "ML", include.mean = T)
```

```
## Warning in arima(montreal_train_ts, order = order, seasonal = seasonal, :
## possible convergence problem: optim gave code = 1
```

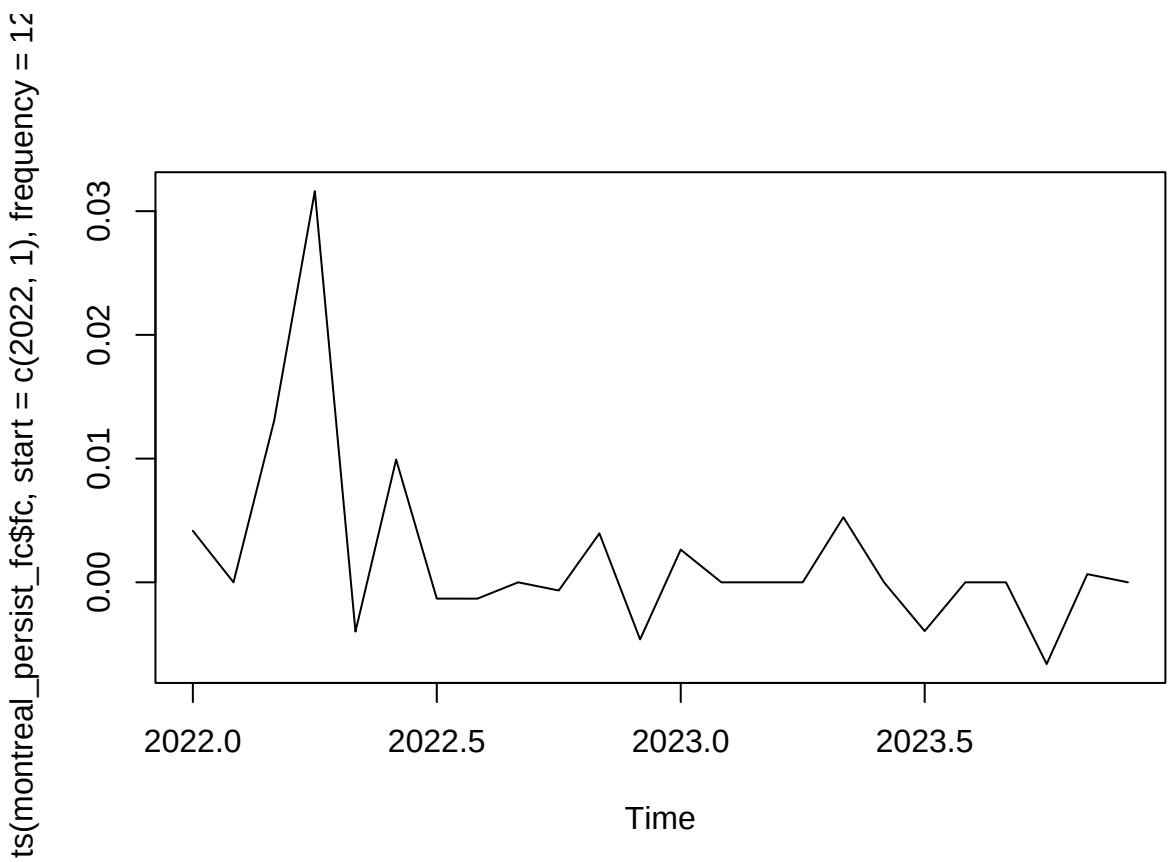
```
# Getting the result of the moving 1-step ahead forecast for holdout set
arimax_results <- armax_fc(tsdata = montreal_transformed_nhpi,
                           ntrain = length(montreal_train_ts), order = order,
                           seasonal = seasonal, xreg = montreal_irlag1[-1],
                           method = "ML", traincoef = coef(train_arimax_model), include.mean = T)
```

Viewing the results of Montreal's analyses

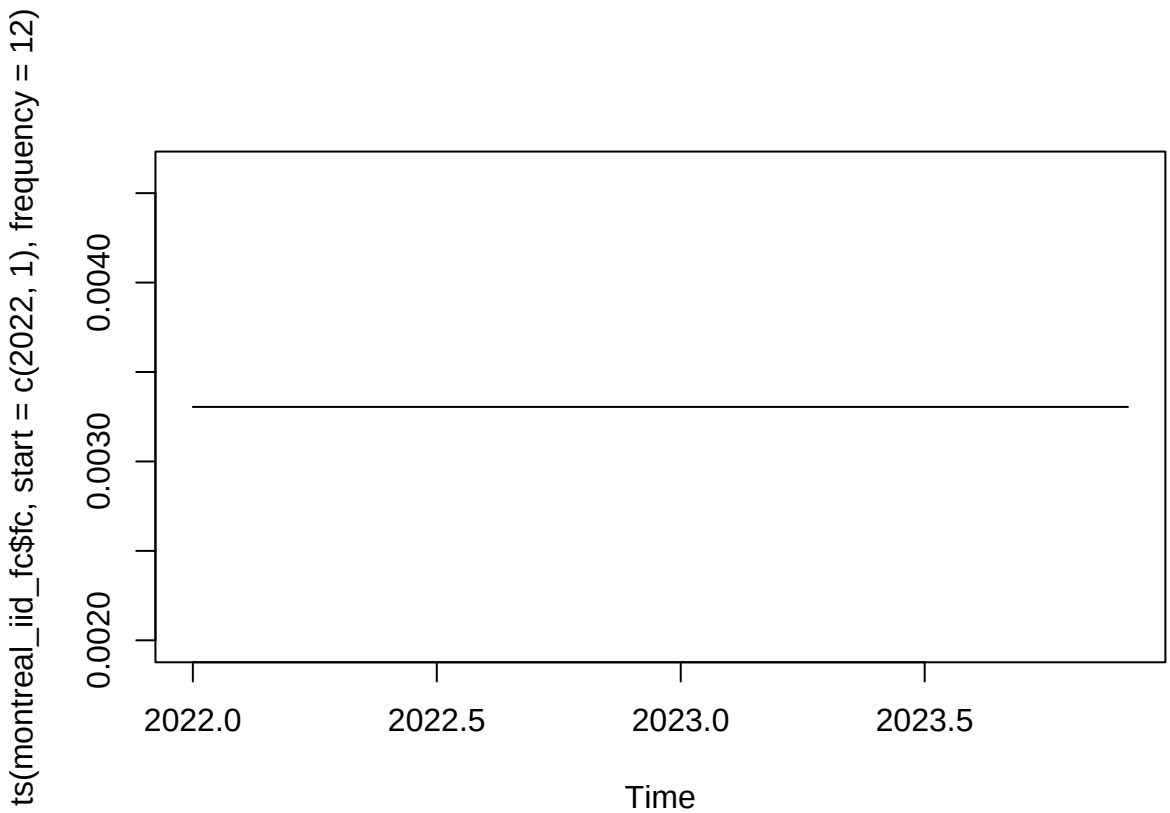
```
# 1. TIME SERIES PLOT
plot(montreal_ts)
```



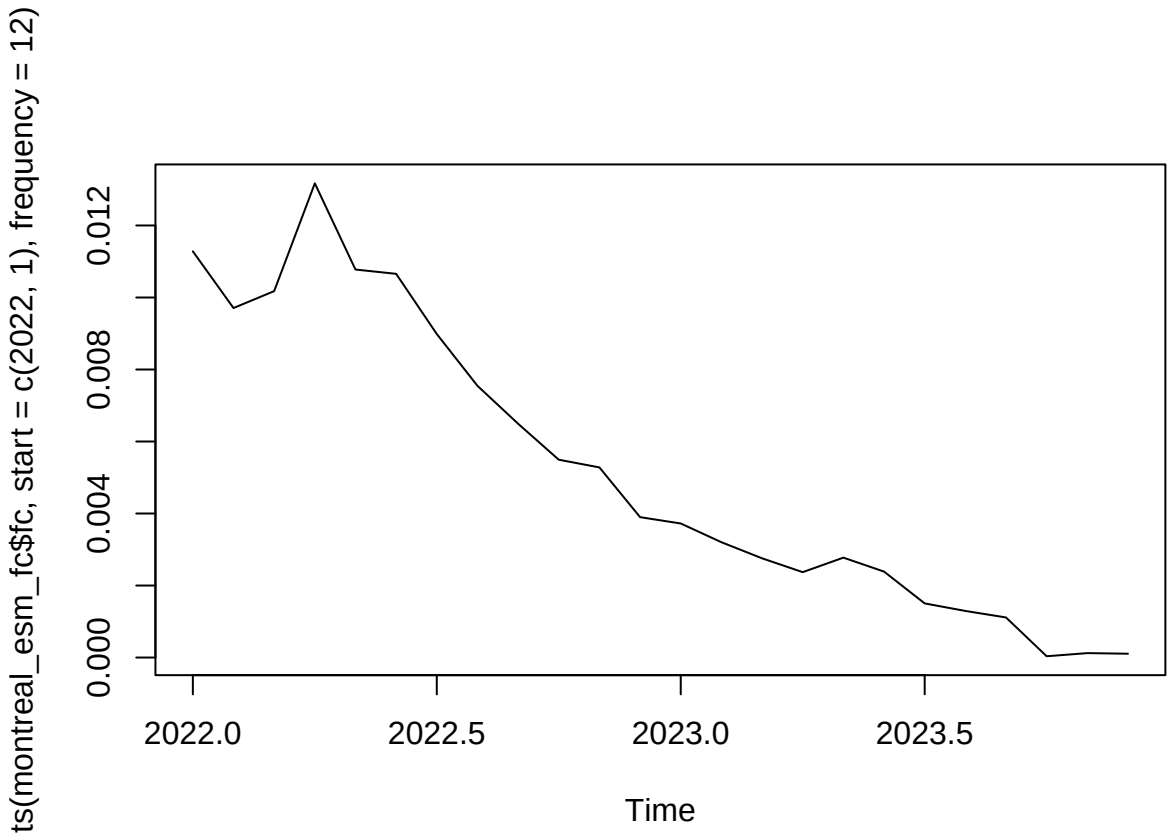

```r
# log-diff transformation
# plot(montreal_transformed_ts) # NOTE: this object doesn't exist
# ts plot from persistence
plot(ts(montreal_persist_fc$fc, start = c(2022, 1), frequency = 12))
```



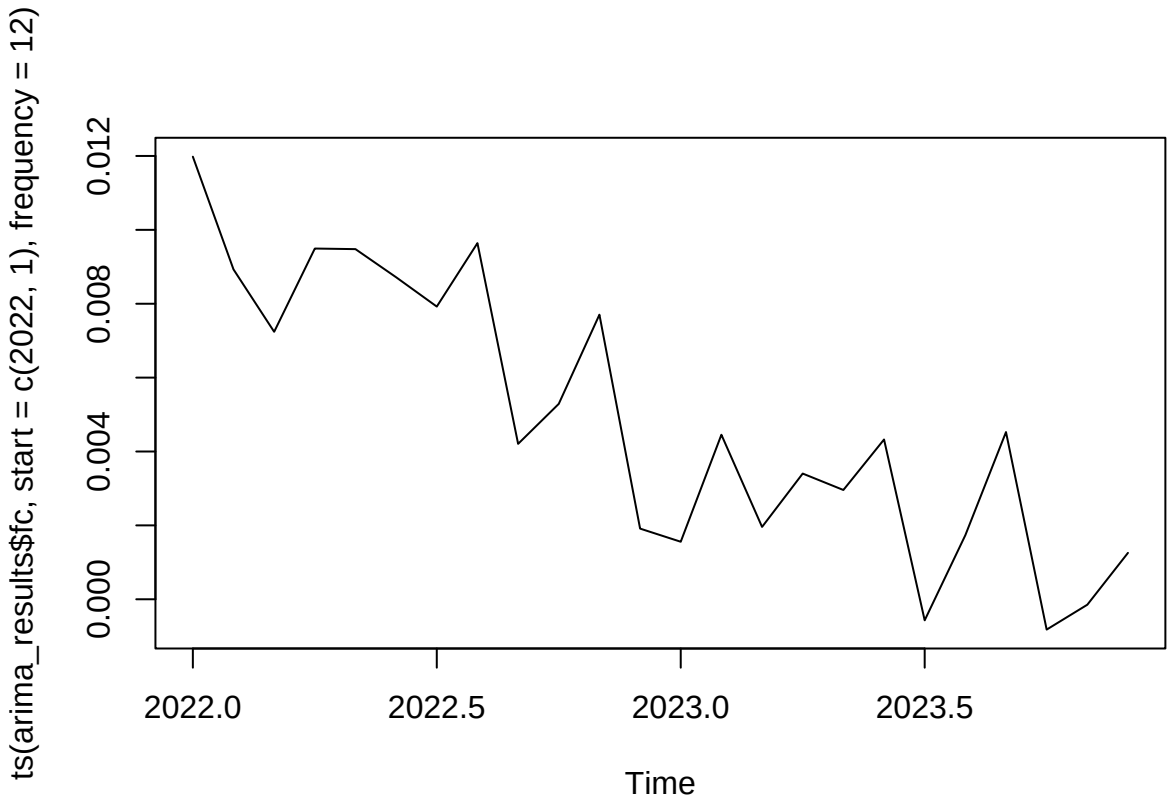
```r
# ts plot from average
plot(ts(montreal_iid_fc$fc, start = c(2022, 1), frequency = 12))
```



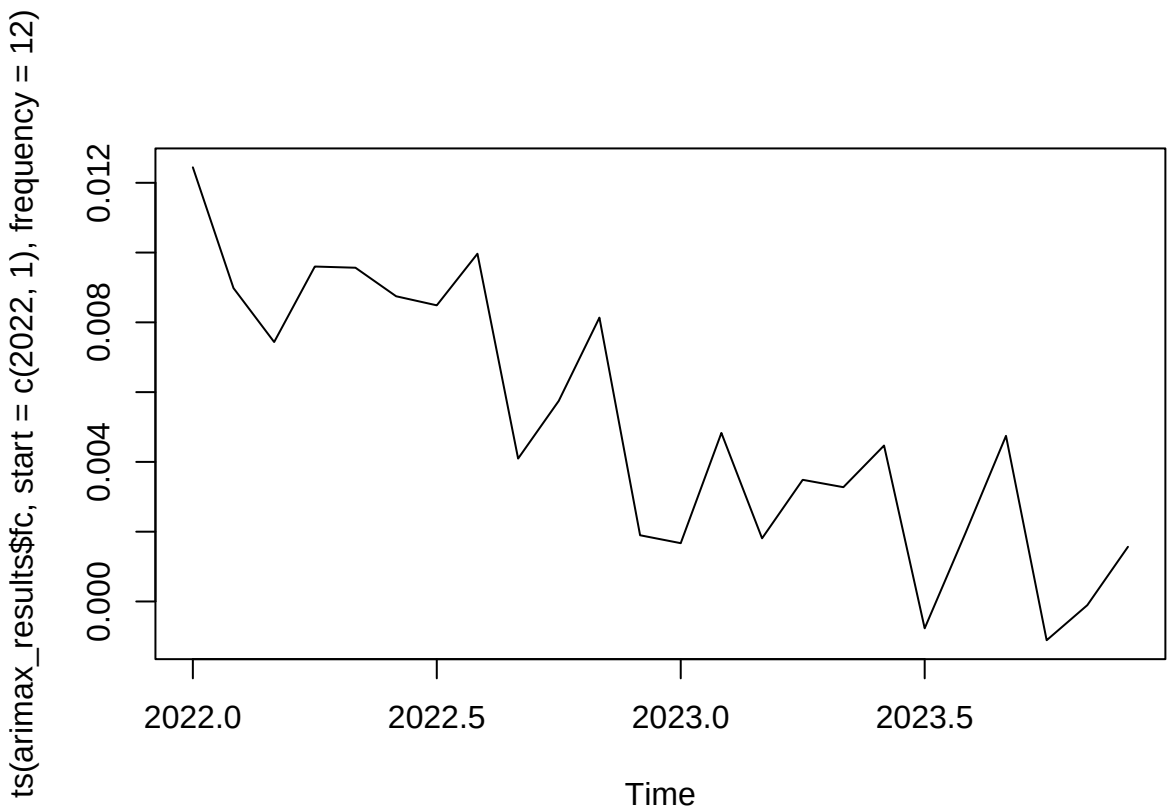

```r
# ts plot from simple exponential smoothing
plot(ts(montreal_esm_fc$fc, start = c(2022, 1), frequency = 12))
```



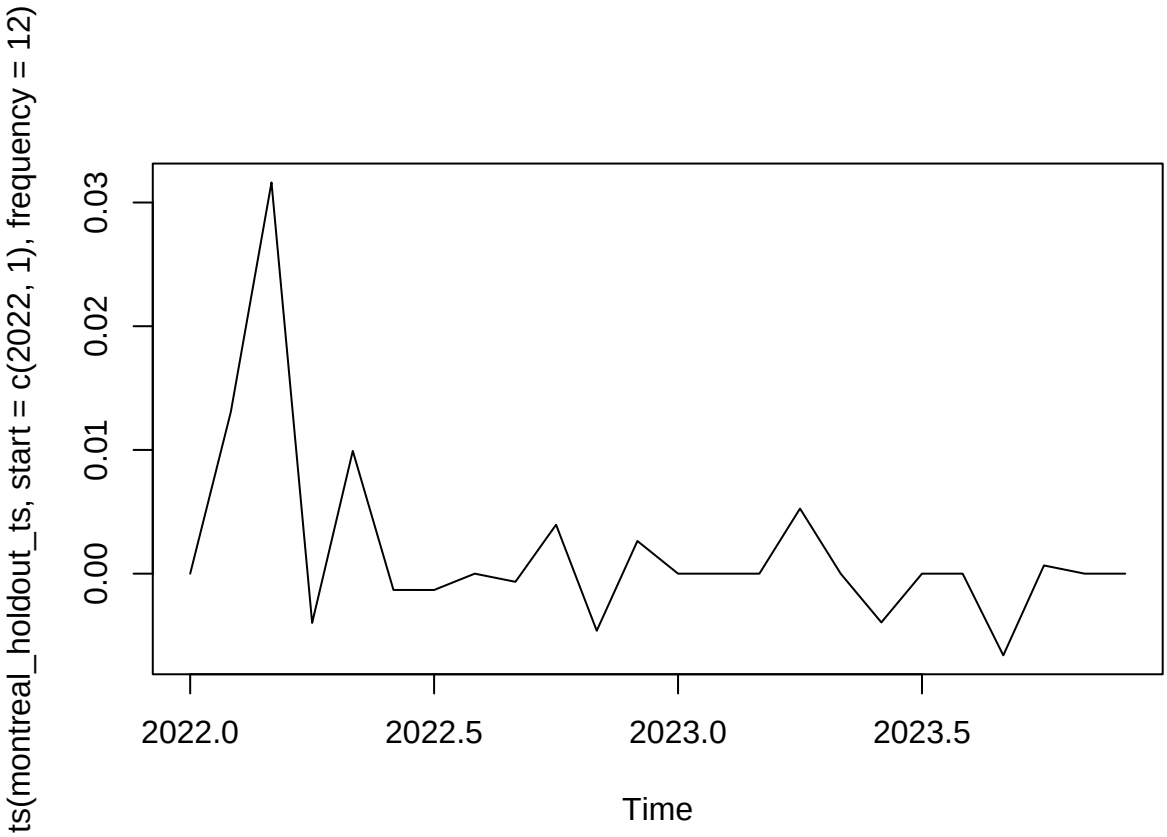

```r
# ts plot from arima
plot(ts(arima_results$fc, start = c(2022, 1), frequency = 12))
```



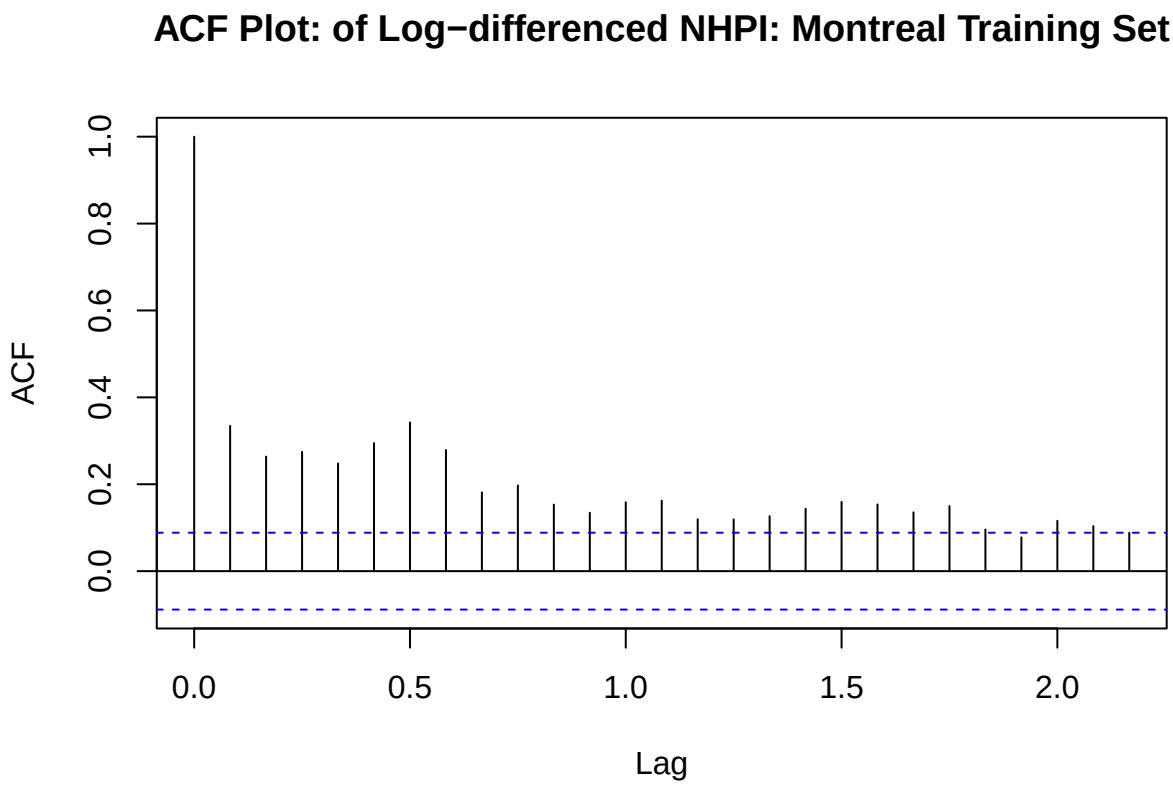

```r
# ts plot from arimax
plot(ts(arimax_results$fc, start = c(2022, 1), frequency = 12))
```



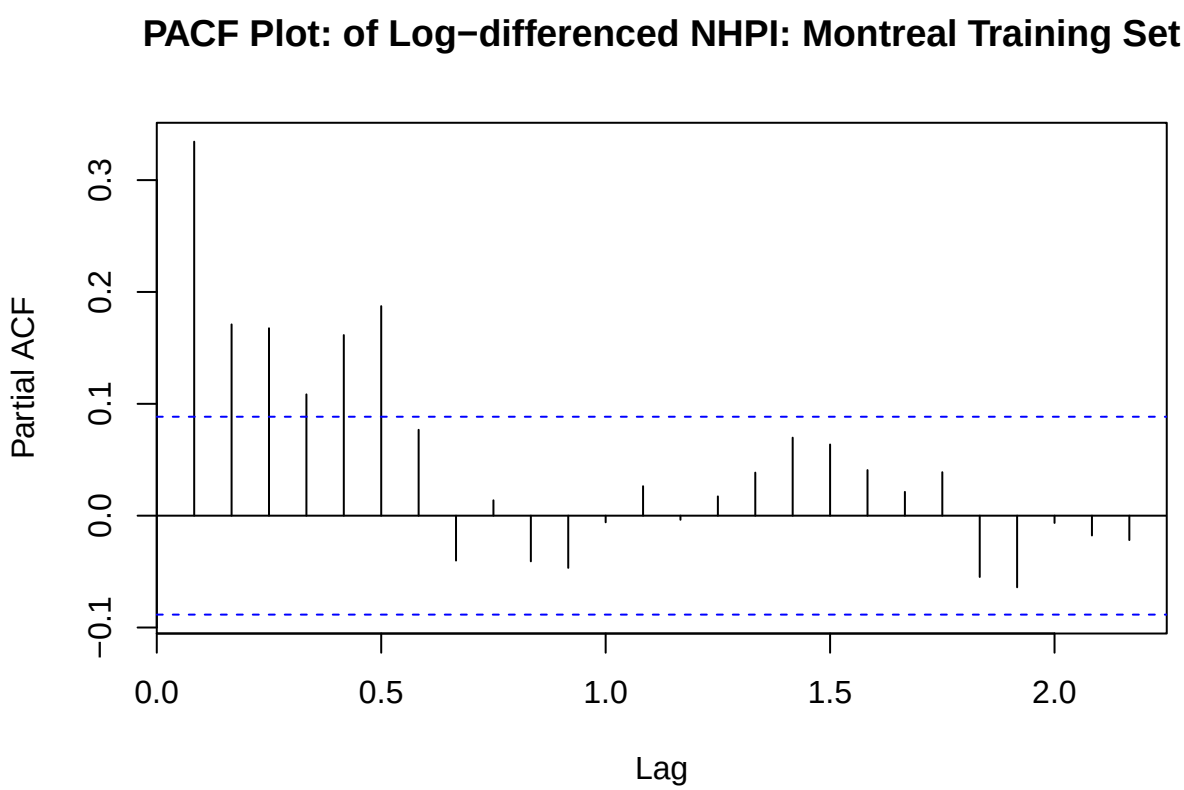
```r
# ts plot of holdout set
plot(ts(montreal_holdout_ts, start = c(2022, 1), frequency = 12))
```



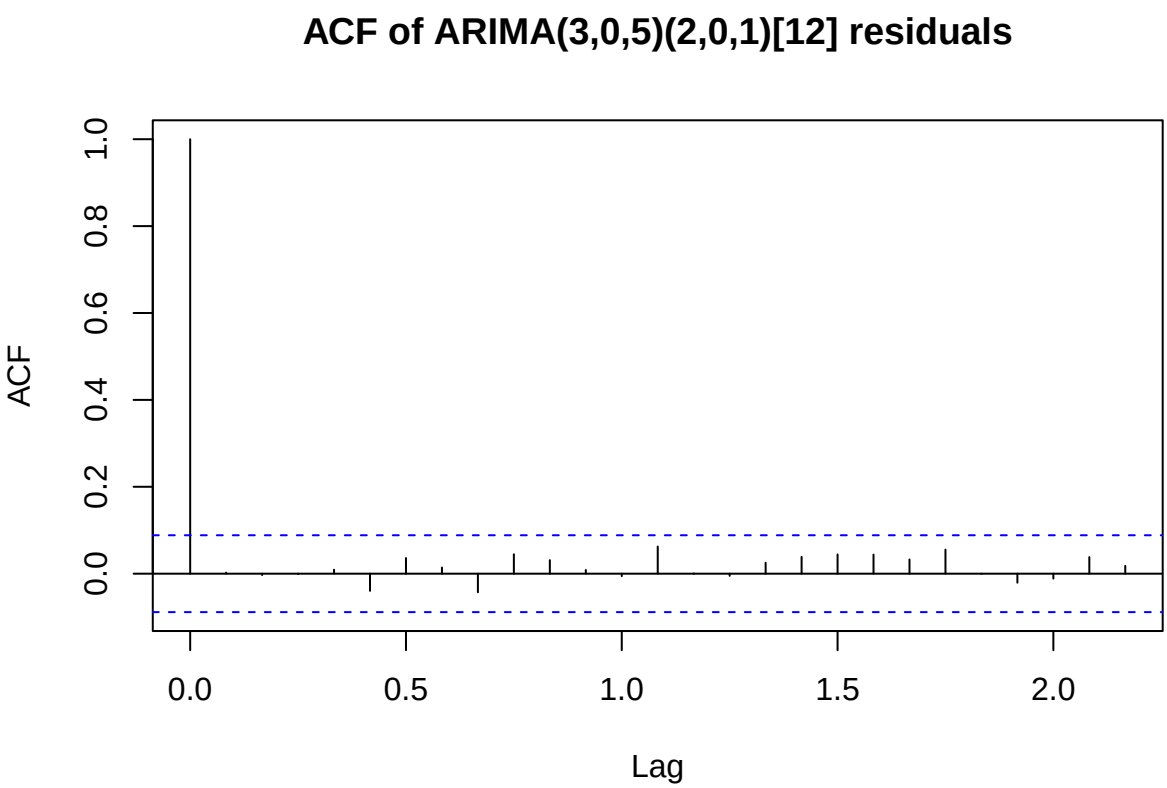

```r
# 2. ACF PLOT
# ACF and PACF of training set
acf(montreal_train_ts, main = "ACF Plot: of Log-differenced NHPI: Montreal Training Set")
```



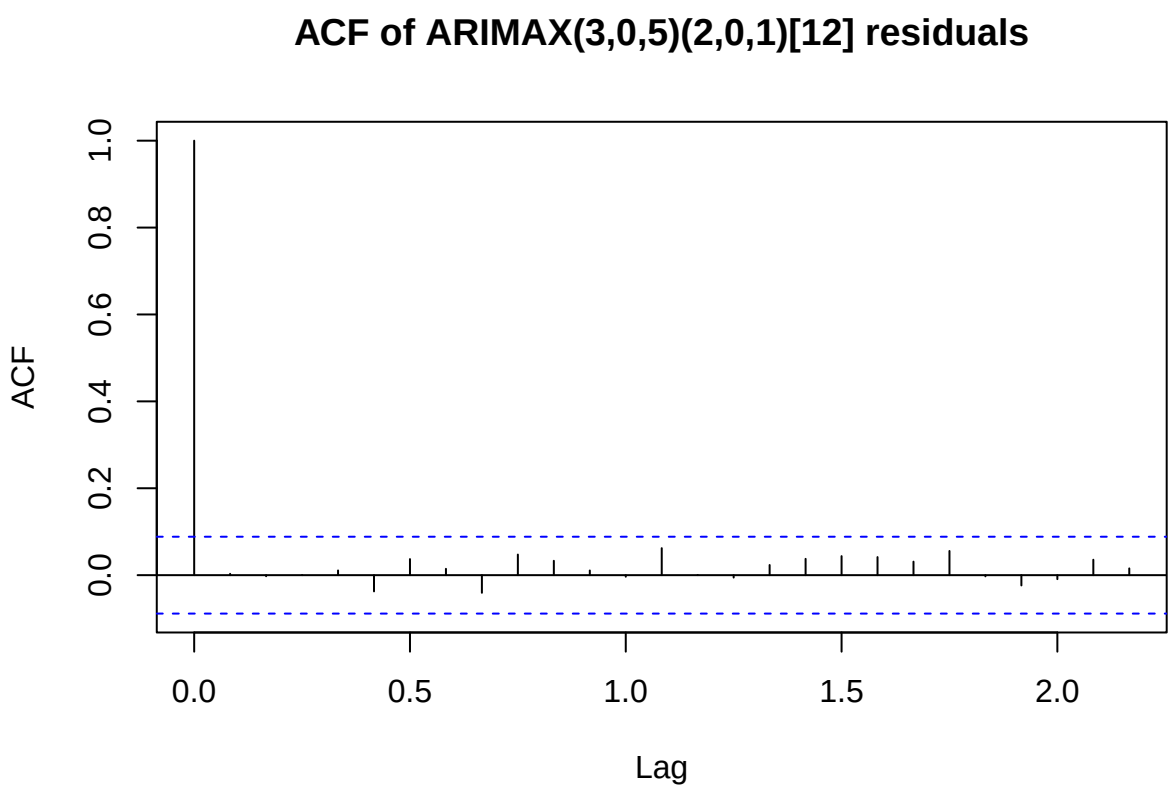

```
pacf(montreal_train_ts, main = "PACF Plot: of Log-differenced NHPI: Montreal Training Set")
```



```r
# ACF from best ARIMA model
resid_fit <- residuals(montreal_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(3,0,5)(2,0,1)[12] residuals")
```




```r
# ACF from best ARIMAX model
resid_fit <- residuals(montreal_fit_arimax)
acf(resid_fit, main = "ACF of ARIMAX(3,0,5)(2,0,1)[12] residuals")
```




```r
# 3. SUMMARY STATISTICS
summary(montreal_train_ts)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -0.012308  0.000000  0.001907  0.003305  0.005096  0.071182
```

```r
summary(montreal_holdout_ts)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0066050 -0.0008246  0.0000000  0.0018622  0.0011569  0.0316205
```

```r
# summary of best ARIMA model
summary(montreal_transformed_nphi_auto)
```

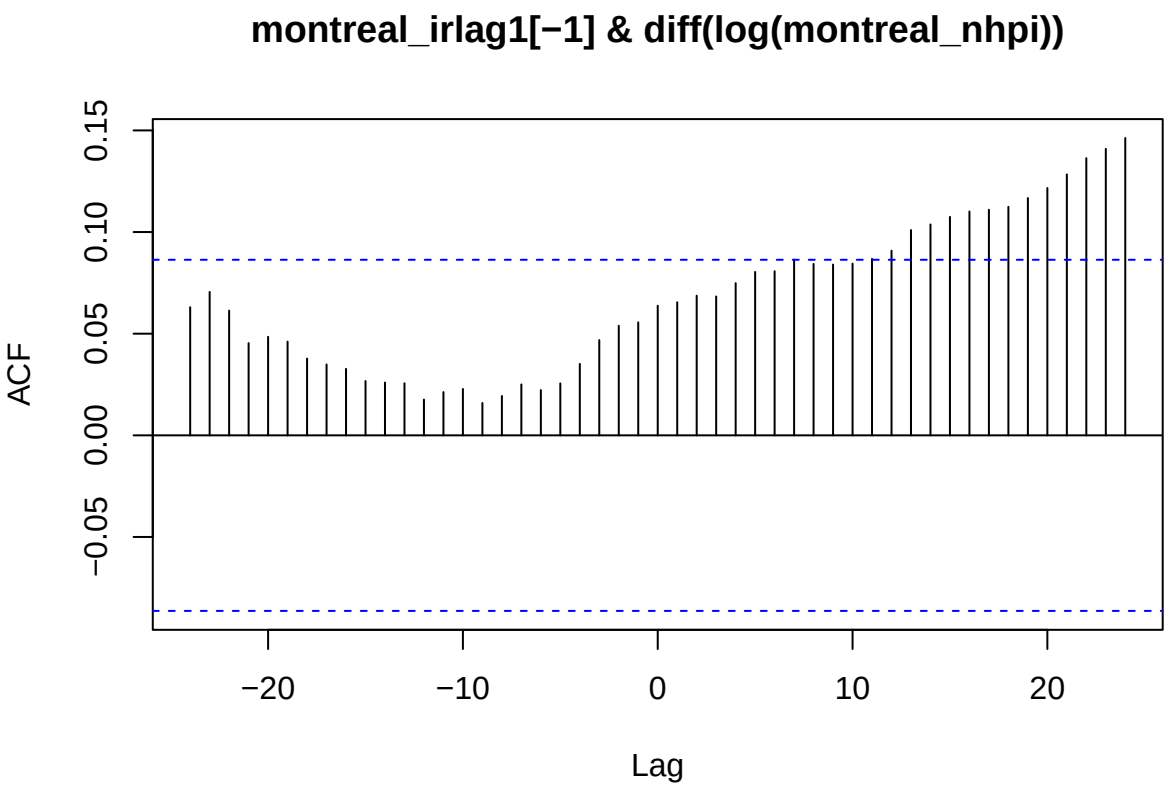
```
## Series: montreal_train_ts
## ARIMA(2,0,5)(2,0,1)[12] with non-zero mean
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2933  -0.4683  -1.1324  0.3576  0.0775  -0.0112  0.1526  0.7129  0.0270
## s.e.  0.1615   0.1643   0.1629  0.1522  0.0680   0.0678  0.0628  0.2571  0.0723
##          sma1    mean
##       -0.6604  0.0034
## s.e.   0.2622  0.0007
##
## sigma^2 = 2.434e-05:  log likelihood = 1916.27
## AIC=-3808.54   AICc=-3807.89   BIC=-3758.19
```

```
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -4.221651e-05 0.004878312 0.002960714 NaN  Inf 0.7689245
##                     ACF1
## Training set 0.002456102
```

```r
# summary of best ARIMAX model
summary(montreal_fit_arimax)
```

```
## Series: montreal_train_ts
## Regression with ARIMA(2,0,5)(2,0,1)[12] errors
##
## Coefficients:
##          ar1      ar2      ma1     ma2     ma3      ma4     ma5    sar1    sar2
##       1.2998  -0.4778  -1.1411  0.3668  0.0793  -0.0121  0.1545  0.7148  0.0216
## s.e.  0.1554   0.1577   0.1572  0.1477  0.0684   0.0681  0.0629  0.2831  0.0728
##          sma1  intercept   xreg
##       -0.6608     0.0028  1e-04
## s.e.   0.2890     0.0010  1e-04
##
## sigma^2 = 2.435e-05:  log likelihood = 1916.72
## AIC=-3807.45   AICc=-3806.69   BIC=-3752.9
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE     MASE
## Training set -2.873018e-05 0.004873876 0.002945387 NaN  Inf 0.764944
##                     ACF1
## Training set 0.003074691
```

```r
# CCF between log-differenced NHPI and the lagged interest rate
ccf(montreal_irlag1[-1], diff(log(montreal_nhpi)))
```




```r
# 4. RMSE
rmse = cbind(montreal_persist_fc$rmse, montreal_iid_fc$rmse,
             montreal_esm_fc$rmse, arima_results$rmse, arimax_results$rmse)
colnames(rmse) = c("persist", "iid", "esm", "arima", "arimax")
print(round(rmse, 3))
```

```
##   persist   iid   esm arima arimax
## a    0.01 0.008 0.008 0.008  0.008
```

Analyzing Toronto

Setting up

```r
# Reading the data
toronto_nhpi_data <- read.csv("Full_Cities/toronto_full.csv")
toronto_nhpi_ir_data <- read.csv("Extracted_Cities/toronto.csv")

# Extracting the NHPI
toronto_nhpi <- toronto_nhpi_data$VALUE

# Extracting the one-month lag interest rate
toronto_irlag1 <- toronto_nhpi_ir_data$IR_lag1

# Setting the time window from 1981-2023
toronto_ts <- ts(toronto_nhpi, start = c(1981,1), end = c(2023, 12), frequency = 12)
toronto_ir_ts <- ts(toronto_irlag1, start = c(1981,1), end = c(2023, 12), frequency = 12)
```

```r
# Applying a transformation for NHPI
toronto_transformed_nhpi <- diff(log(toronto_ts))

# Setting a training set from 1981-2021
toronto_train_ts <- window(toronto_transformed_nhpi, end = c(2021,12))
toronto_train_ir_xreg <- window(toronto_ir_ts, start = c(1981,2), end = c(2021,12))

# Setting a holdout set from 2022-2023
toronto_holdout_ts <- window(toronto_transformed_nhpi, start = c(2022,1))
toronto_holdout_ir_xreg <- window(toronto_ir_ts, start = c(2022,1))
```

Doing the baseline forecasts

```r
# Baseline 1
toronto_persist_fc <- persist_fc(toronto_train_ts, toronto_holdout_ts, iprint = F)

# Baseline 2
toronto_iid_fc <- iid_fc(toronto_train_ts, toronto_holdout_ts, iprint = F)
```

Doing the exponential smoothing forecasts

```r
# Method 1
ntrain <- length(toronto_train_ts)
ntotal <- length(toronto_ts)-1

toronto_transformed_nhpi_train <- toronto_transformed_nhpi[1:ntrain]
toronto_transformed_nhpi_holdout <- toronto_transformed_nhpi[(ntrain+1):ntotal]

# Simple Exponential Smoothing
fit_expsmo <- HoltWinters(toronto_train_ts, beta = F, gamma = F)
toronto_esm_fc <- esm_fc(toronto_transformed_nhpi_train, toronto_transformed_nhpi_holdout,
                         alpha = fit_expsmo$alpha, level = fit_expsmo$coefficients["a"],
                         iprint = F)

# Linear Exponential Smoothing
fit_hw <- HoltWinters(toronto_train_ts, gamma = F)
toronto_holt_fc <- lholt_fc(toronto_transformed_nhpi_train, toronto_transformed_nhpi_holdout,
                            alpha = fit_hw$alpha, beta = fit_hw$beta,
                            level = fit_hw$coefficients["a"],
                            slope = fit_hw$coefficients["b"], iprint = F)

# Additive Seasonal Exponential Smoothing
wafit <- HoltWinters(toronto_train_ts, seasonal = "additive")
```

```
## Warning in HoltWinters(toronto_train_ts, seasonal = "additive"): optimization
## difficulties: ERROR: ABNORMAL_TERMINATION_IN_LNSRCH
```

```r
toronto_aseason_fc <- aseason_fc(toronto_transformed_nhpi_train,
                                 toronto_transformed_nhpi_holdout,
                                 alpha = wafit$alpha, beta = wafit$beta,
                                 gamma = wafit$gamma, level = wafit$coef[1],
                                 slope = wafit$coef[2], season = wafit$coefficient[3:14],
                                 iprint = F)

# Comparing RMSE
rmse_esm <- cbind(toronto_esm_fc$rmse, toronto_holt_fc$rmse, toronto_aseason_fc$rmse)
colnames(rmse_esm) <- c("esm", "lholt", "aseason")
rmse_esm
```

```
##           esm       lholt     aseason
## a 0.002166508 0.002334658 0.002612987
```

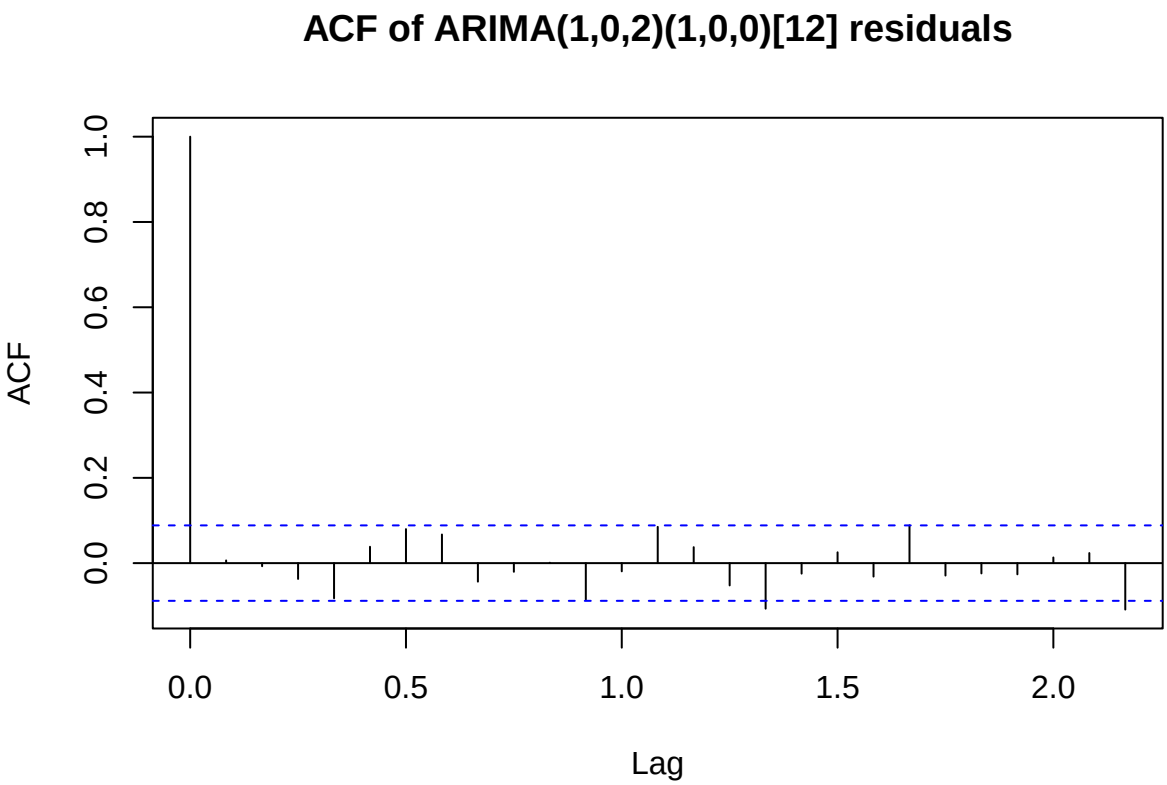
```r
# Best exponential smoothing method: Simple Exponential Smoothing
```

Doing the ARIMA forecasts

```r
# Method 2
toronto_transformed_nphi_auto <- forecast::auto.arima(toronto_train_ts)
summary(toronto_transformed_nphi_auto) # 0.0064057
```

```
## Series: toronto_train_ts
## ARIMA(1,0,2)(1,0,0)[12] with non-zero mean
##
## Coefficients:
##          ar1      ma1      ma2    sar1    mean
##       0.9161  -0.4317  -0.1386  0.1316  0.0034
## s.e.  0.0302   0.0551   0.0490  0.0487  0.0017
##
## sigma^2 = 4.146e-05:  log likelihood = 1782.61
## AIC=-3553.23   AICc=-3553.05   BIC=-3528.05
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001299236 0.006405744 0.003868432 NaN  Inf 0.6292045
##                     ACF1
## Training set 0.006342322
```

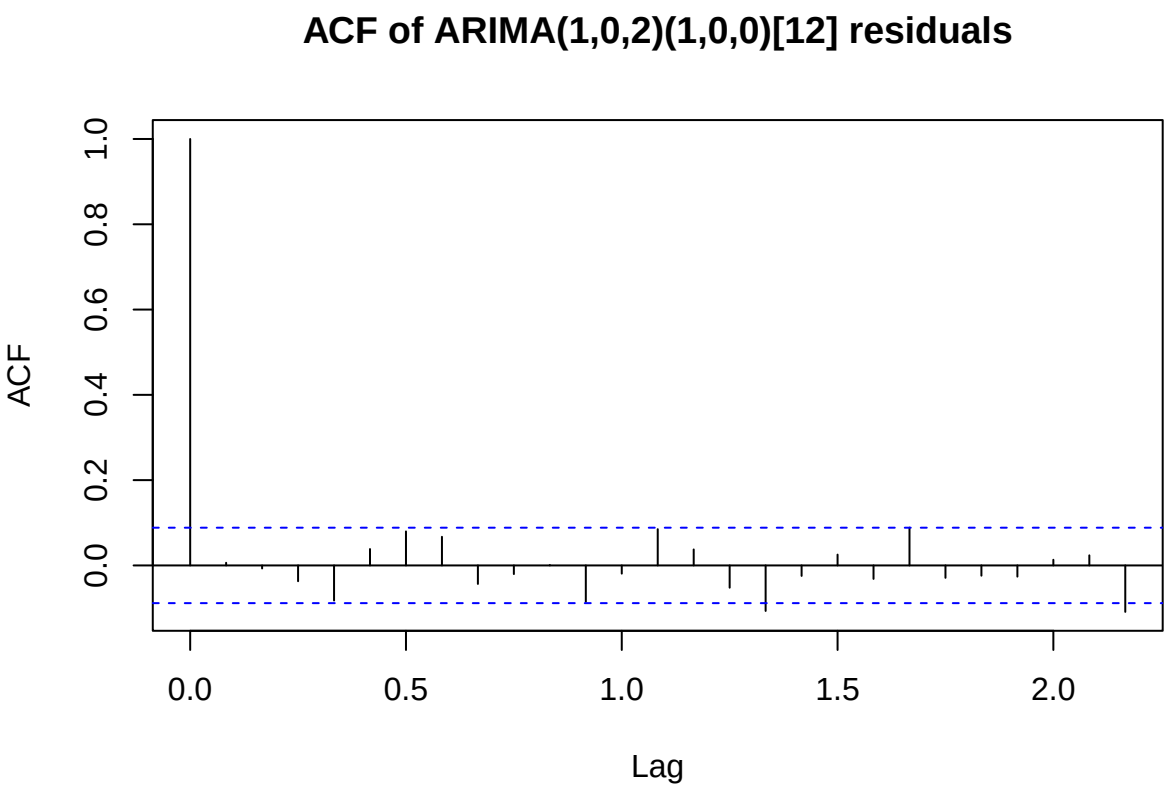
```r
resid_fit <- residuals(toronto_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(1,0,2)(1,0,0)[12] residuals")
```




```r
# Comparing with the other ARIMA models
manual_fit <- arima(toronto_train_ts, order = c(1,0,2),
                    seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(manual_fit) # RMSE: 0.006405747
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 2), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1      ma2    sar1  intercept
##       0.9160  -0.4315  -0.1388  0.1316     0.0034
## s.e.  0.0302   0.0551   0.0490  0.0487     0.0017
##
## sigma^2 estimated as 4.103e-05:  log likelihood = 1782.61,  aic = -3553.23
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001276383 0.006405747 0.003868164 NaN  Inf 0.8546019
##                     ACF1
## Training set 0.006235644
```

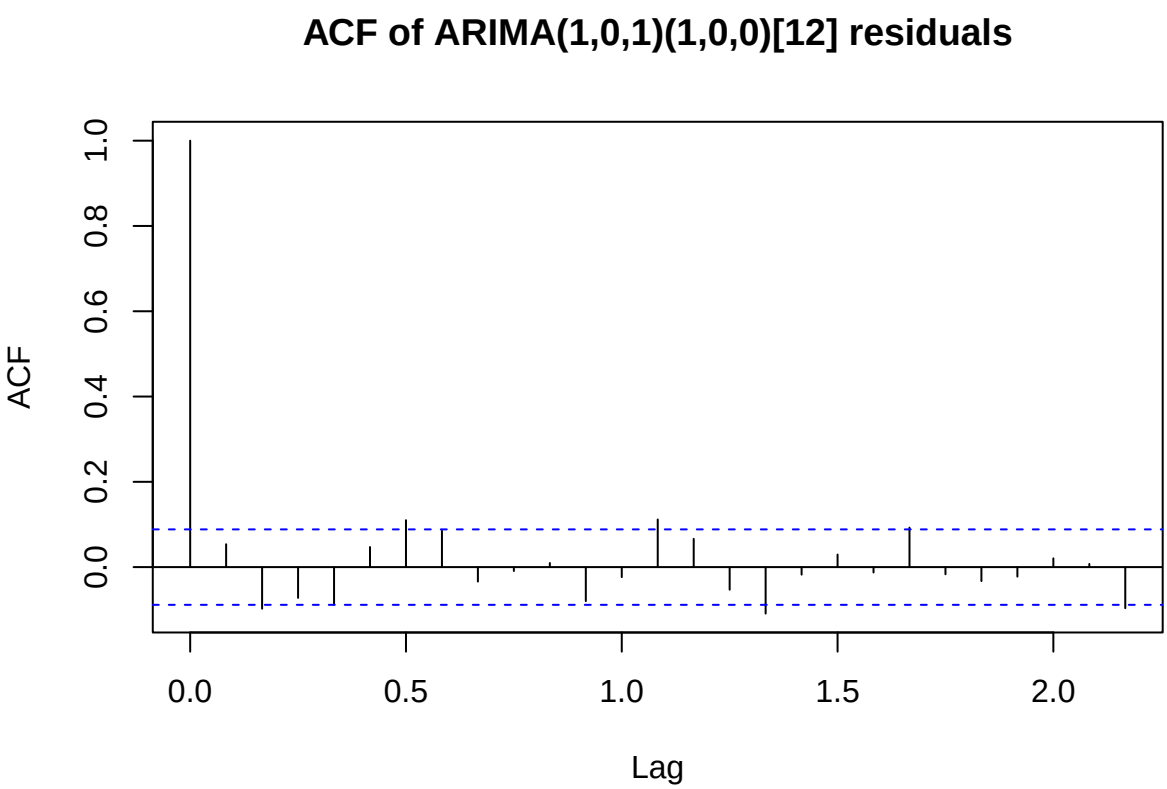
```r
acf(residuals(manual_fit), main = "ACF of ARIMA(1,0,2)(1,0,0)[12] residuals")
```




```r
test1 <- arima(toronto_train_ts, order = c(1,0,1),
               seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(test1) # RMSE: 0.006456026
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 1), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1    sar1  intercept
##       0.8636  -0.4245  0.1466     0.0033
## s.e.  0.0434   0.0829  0.0482     0.0014
##
## sigma^2 estimated as 4.168e-05:  log likelihood = 1778.8,  aic = -3547.61
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001074825 0.006456026 0.003832997 NaN  Inf 0.8468326
##                    ACF1
## Training set 0.05363762
```

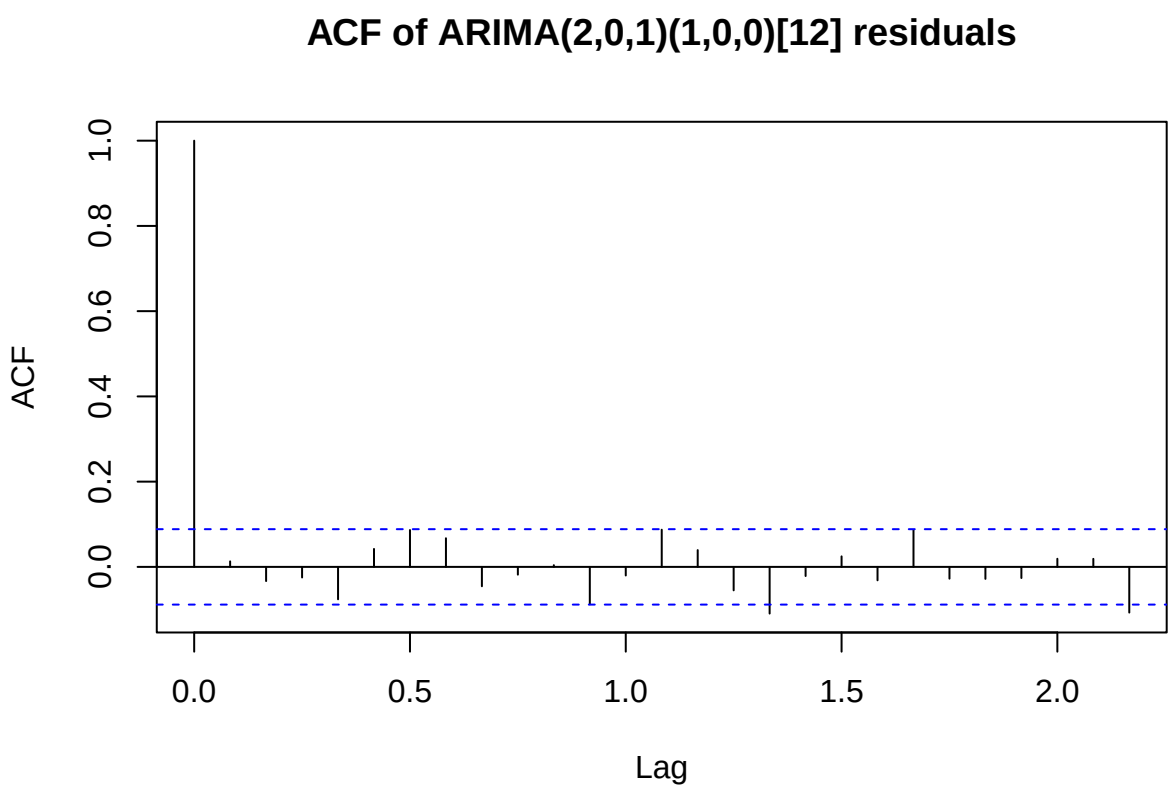
```r
acf(residuals(test1), main = "ACF of ARIMA(1,0,1)(1,0,0)[12] residuals")
```




```r
test2 <- arima(toronto_train_ts, order = c(2,0,1),
               seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(test2) # RMSE: 0.006410017
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(2, 0, 1), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2      ma1    sar1  intercept
##       1.1623  -0.2244  -0.6839  0.1292     0.0034
## s.e.  0.0959   0.0782   0.0792  0.0489     0.0017
##
## sigma^2 estimated as 4.109e-05:  log likelihood = 1782.3,  aic = -3552.6
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE     MASE       ACF1
## Training set -0.0001286044 0.006410017 0.003868191 NaN  Inf 0.854608 0.01298068
```

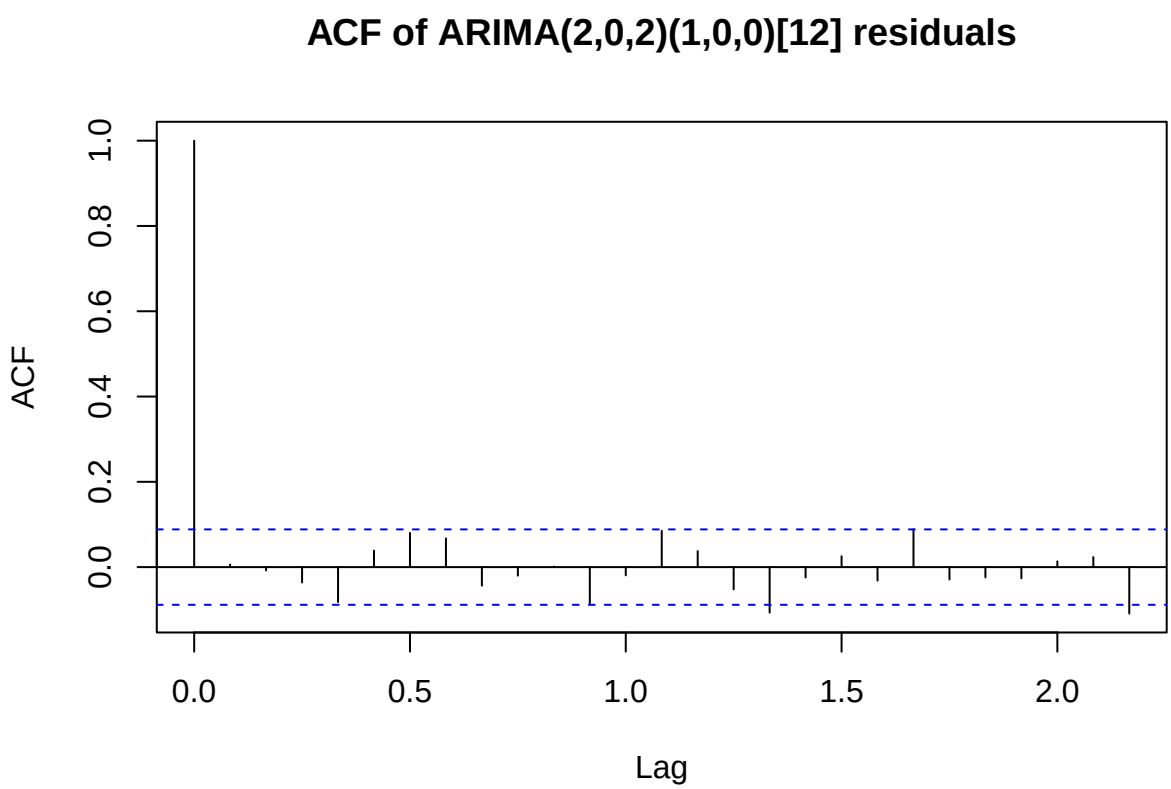
```r
acf(residuals(test2), main = "ACF of ARIMA(2,0,1)(1,0,0)[12] residuals")
```




```r
test3 <- arima(toronto_train_ts, order = c(2,0,2),
               seasonal = list(order = c(1,0,0), period = 12), method = "ML")
summary(test3) # RMSE: 0.006405754
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(2, 0, 2), seasonal = list(order = c(1,
##     0, 0), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ar2      ma1      ma2    sar1  intercept
##       0.9255  -0.0089  -0.4408  -0.1336  0.1316     0.0035
## s.e.  0.3533   0.3160   0.3501   0.1757  0.0491     0.0017
##
## sigma^2 estimated as 4.103e-05:  log likelihood = 1782.61,  aic = -3551.23
##
## Training set error measures:
##                        ME        RMSE        MAE MPE MAPE      MASE
## Training set -0.0001365831 0.006405754 0.00386938 NaN  Inf 0.8548707
##                     ACF1
## Training set 0.006071496
```

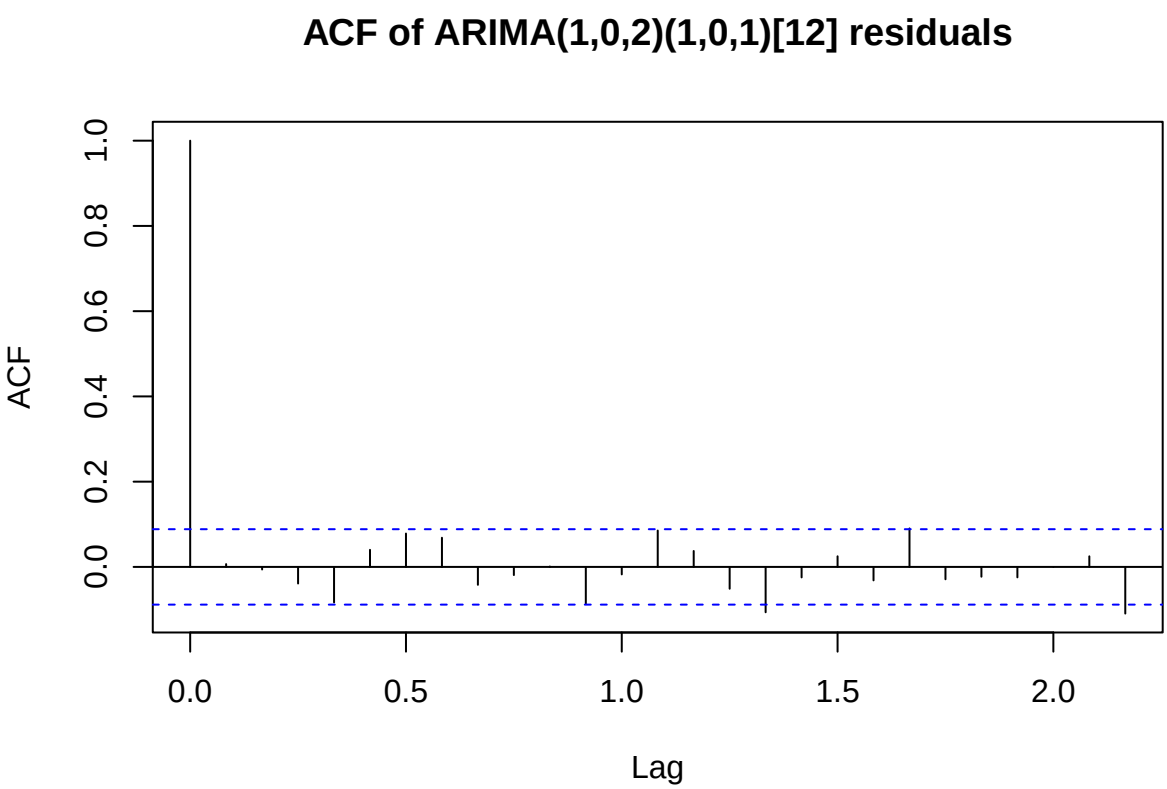
```r
acf(residuals(test3), main = "ACF of ARIMA(2,0,2)(1,0,0)[12] residuals")
```




```r
test4 <- arima(toronto_train_ts, order = c(1,0,2),
               seasonal = list(order = c(1,0,1), period = 12), method = "ML")
summary(test4) # RMSE: 0.006404998
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 2), seasonal = list(order = c(1,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1      ma2    sar1     sma1  intercept
##       0.9160  -0.4317  -0.1376  0.2437  -0.1140     0.0035
## s.e.  0.0303   0.0551   0.0491  0.3689   0.3775     0.0017
##
## sigma^2 estimated as 4.102e-05:  log likelihood = 1782.67,  aic = -3551.33
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001376317 0.006404998 0.003866659 NaN  Inf 0.8542695
##                     ACF1
## Training set 0.006611957
```

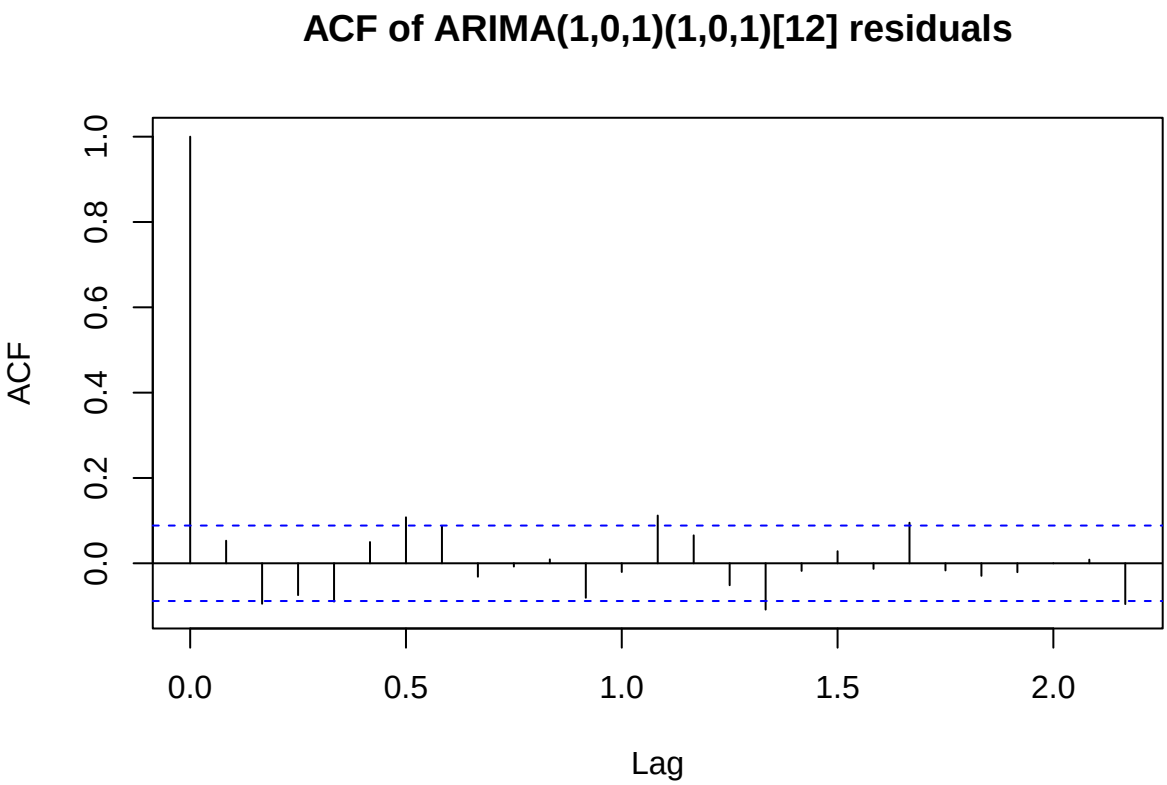
```r
acf(residuals(test4), main = "ACF of ARIMA(1,0,2)(1,0,1)[12] residuals")
```




```r
test5 <- arima(toronto_train_ts, order = c(1,0,1),
               seasonal = list(order = c(1,0,1), period = 12), method = "ML")
summary(test5) # RMSE: 0.006454211
```

```
##
## Call:
## arima(x = toronto_train_ts, order = c(1, 0, 1), seasonal = list(order = c(1,
##     0, 1), period = 12), method = "ML")
##
## Coefficients:
##          ar1      ma1    sar1     sma1  intercept
##       0.8634  -0.4227  0.3043  -0.1614     0.0033
## s.e.  0.0434   0.0830  0.2853   0.2956     0.0015
##
## sigma^2 estimated as 4.166e-05:  log likelihood = 1778.93,  aic = -3545.87
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001083458 0.006454211 0.003830568 NaN  Inf 0.8462959
##                    ACF1
## Training set 0.05289892
```

```r
acf(residuals(test5), main = "ACF of ARIMA(1,0,1)(1,0,1)[12] residuals")
```




```r
# Comparing AIC values
aic = cbind(manual_fit$aic, test1$aic, test2$aic, test3$aic, test4$aic, test5$aic)
colnames(aic) = c("manual_fit", "test1", "test2", "test3", "test4", "test5")
aic
```

```
##      manual_fit     test1     test2    test3     test4     test5
## [1,]  -3553.228 -3547.608 -3552.598 -3551.23 -3551.334 -3545.868
```

```r
# Best (based on lowest AIC and ACF plot): manual_fit, ARIMA(1,0,2)(1,0,0)[12]

order <- c(1,0,2)
seasonal <- list(order = c(1,0,0), period = 12)
# Fitting the model to the training data
train_arima_model <- arima(toronto_train_ts, order = order,
                           seasonal = seasonal, method = "ML", include.mean = T)
# Getting the result of the moving 1-step ahead forecast for holdout set
arima_results <- arma_fc(tsdata = toronto_transformed_nhpi,
                         ntrain = length(toronto_train_ts), order = order,
                         seasonal = seasonal, method = "ML",
                         traincoef = coef(train_arima_model), include.mean = T)
arima_results$rmse
```

```
## [1] 0.00204875
```

Doing the ARIMAX forecasts with lagged interest

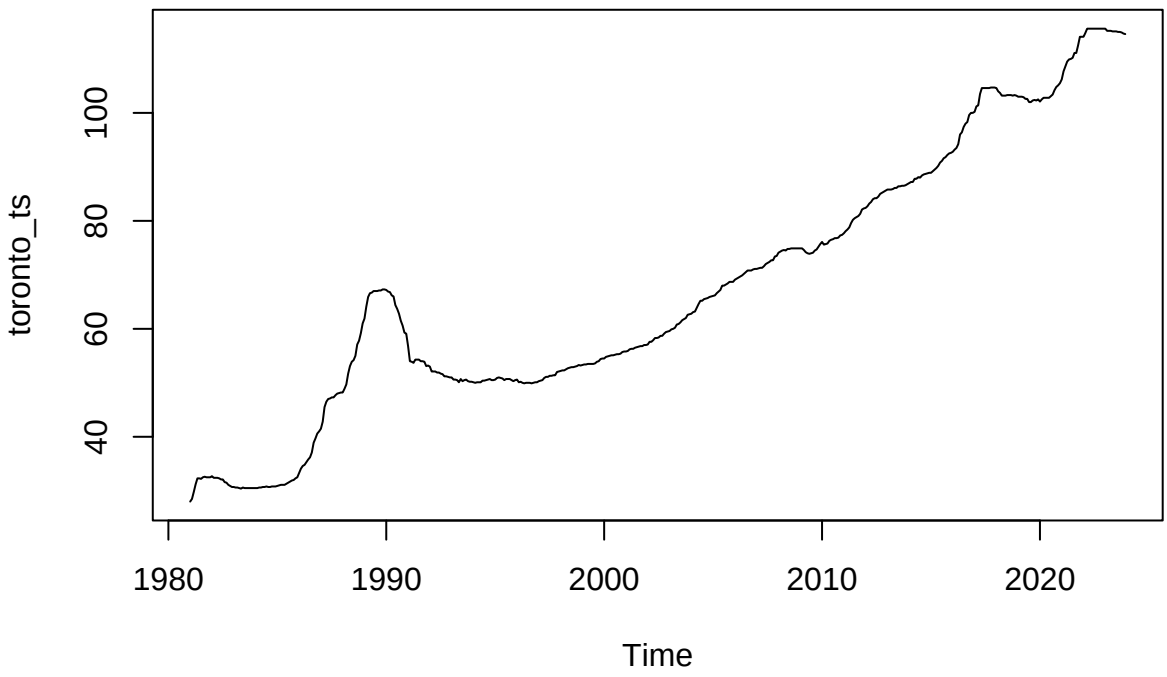
```r
# Method 3
toronto_fit_arimax <- forecast::auto.arima(toronto_train_ts, xreg = toronto_train_ir_xreg)
summary(toronto_fit_arimax)
```

```
## Series: toronto_train_ts
## Regression with ARIMA(1,0,2)(1,0,0)[12] errors
##
## Coefficients:
##          ar1      ma1      ma2    sar1  intercept    xreg
##       0.9257  -0.4415  -0.1406  0.1330     0.0051  -3e-04
## s.e.  0.0329   0.0563   0.0488  0.0489     0.0028   4e-04
##
## sigma^2 = 4.147e-05:  log likelihood = 1783
## AIC=-3551.99   AICc=-3551.76   BIC=-3522.62
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001599343 0.006400238 0.003875959 NaN  Inf 0.6304288
##                     ACF1
## Training set 0.006084236
```

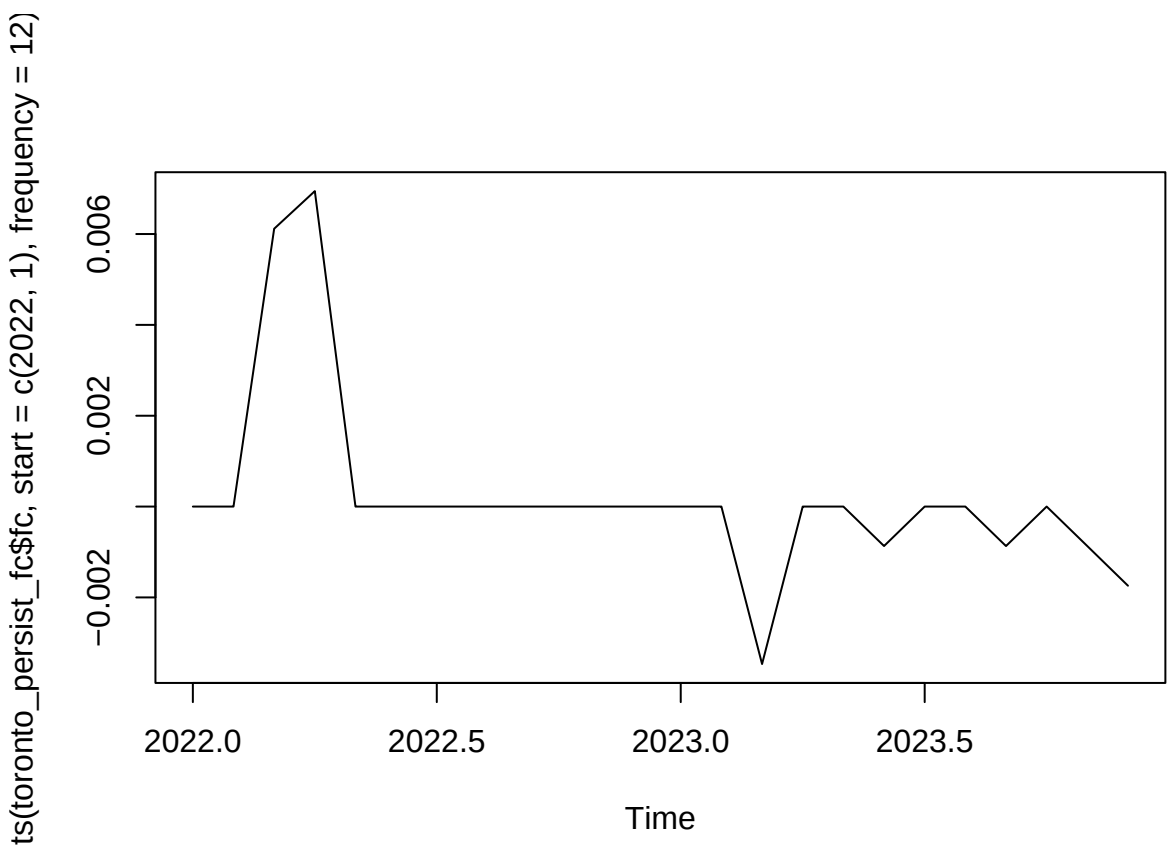
```r
order <- c(1,0,2)
seasonal <- list(order = c(1,0,0), period = 12)
# Fitting a model to the training data
train_arimax_model <- arima(toronto_train_ts, order = order, seasonal = seasonal, xreg = toronto_train_
# Getting the result of the moving 1-step ahead forecast for holdout set
arimax_results <- armax_fc(tsdata = toronto_transformed_nhpi, ntrain = length(toronto_train_ts), order =
```

Viewing the results of Toronto's analyses

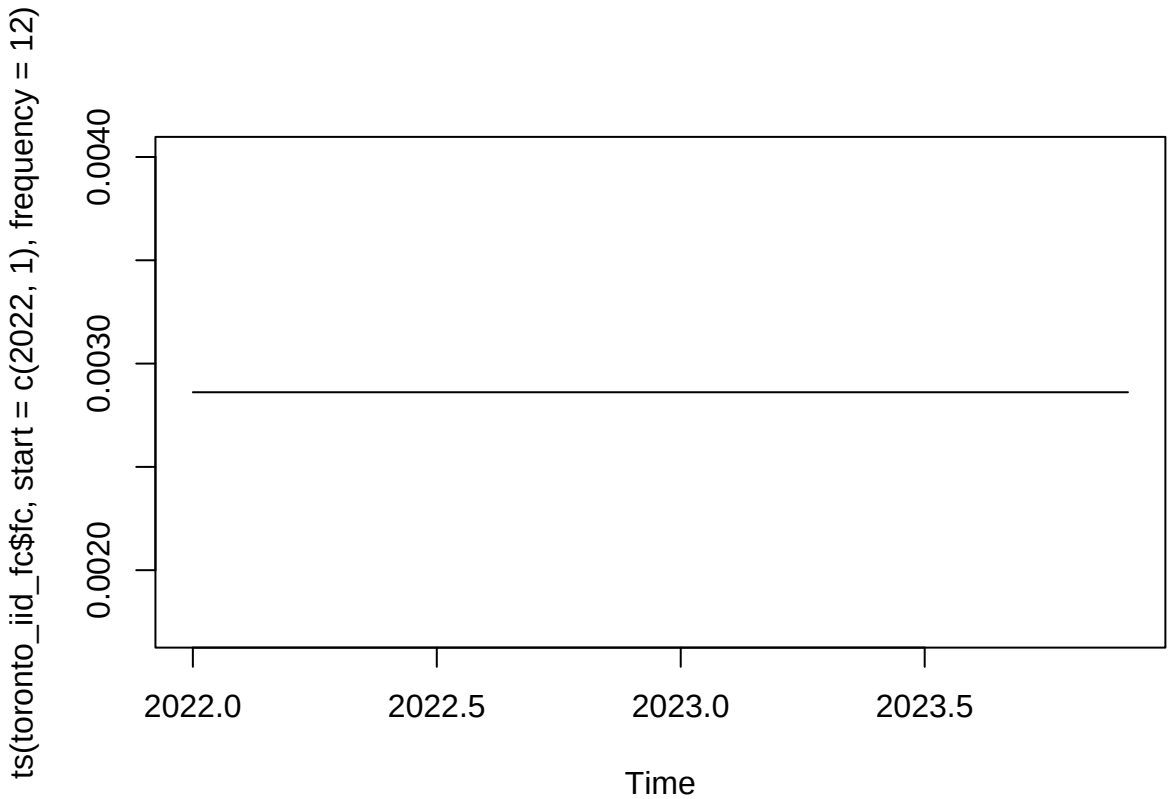
```r
# 1. TIME SERIES PLOT
plot(toronto_ts)
```



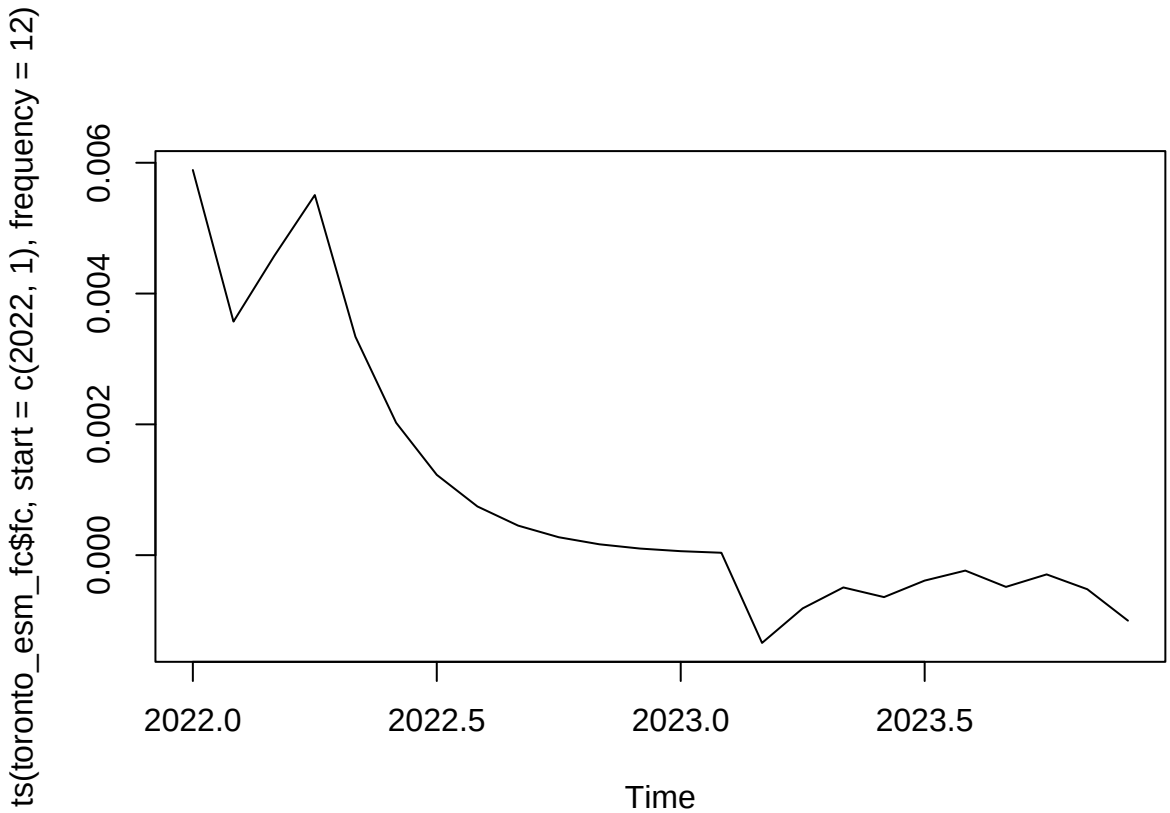

```r
# log-diff transformation
# plot(toronto_transformed_ts) # NOTE: this object doesn't exist
# ts plot from persistence
plot(ts(toronto_persist_fc$fc, start = c(2022, 1), frequency = 12))
```



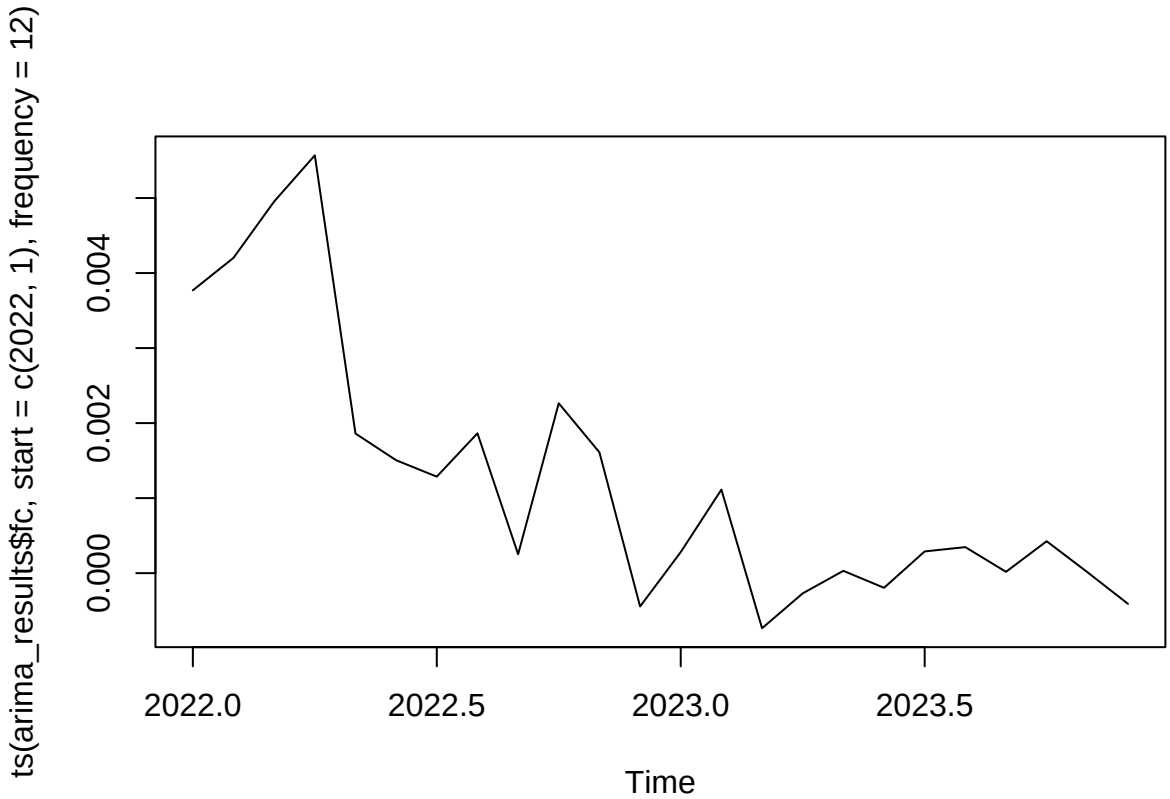
```r
# ts plot from average
plot(ts(toronto_iid_fc$fc, start = c(2022, 1), frequency = 12))
```



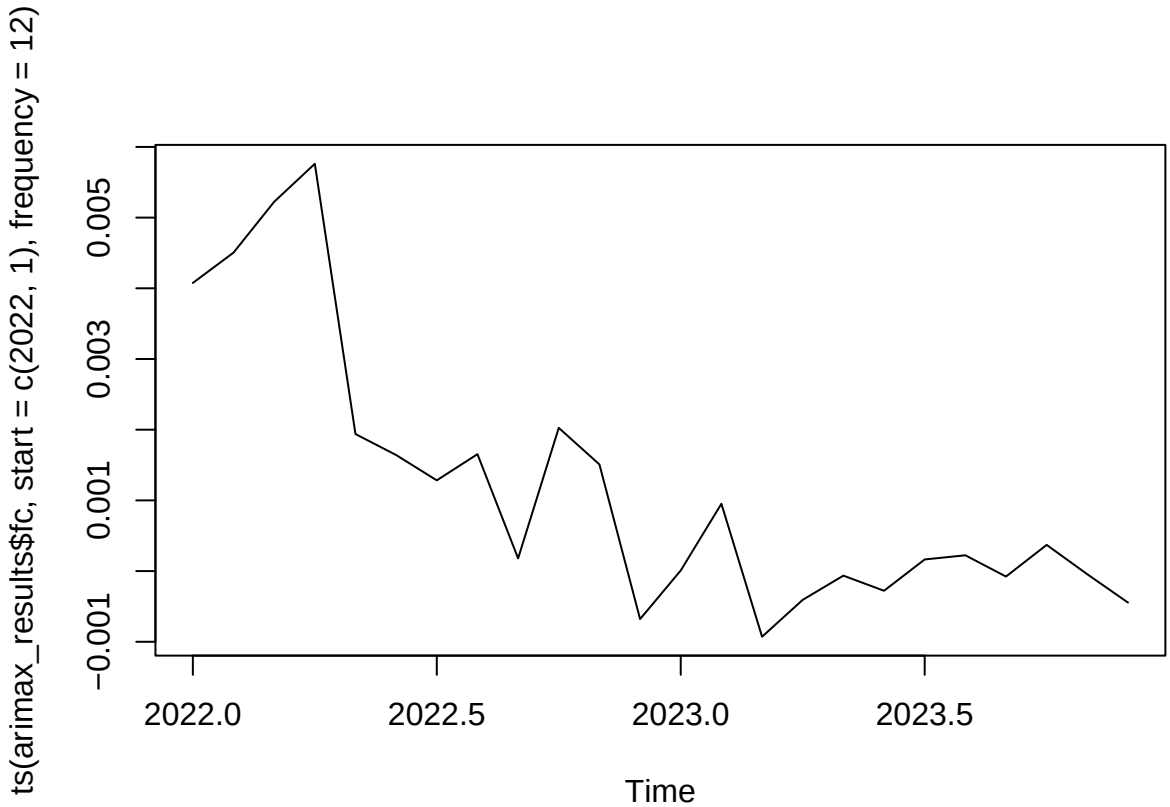

```r
# ts plot from simple exponential smoothing
plot(ts(toronto_esm_fc$fc, start = c(2022, 1), frequency = 12))
```



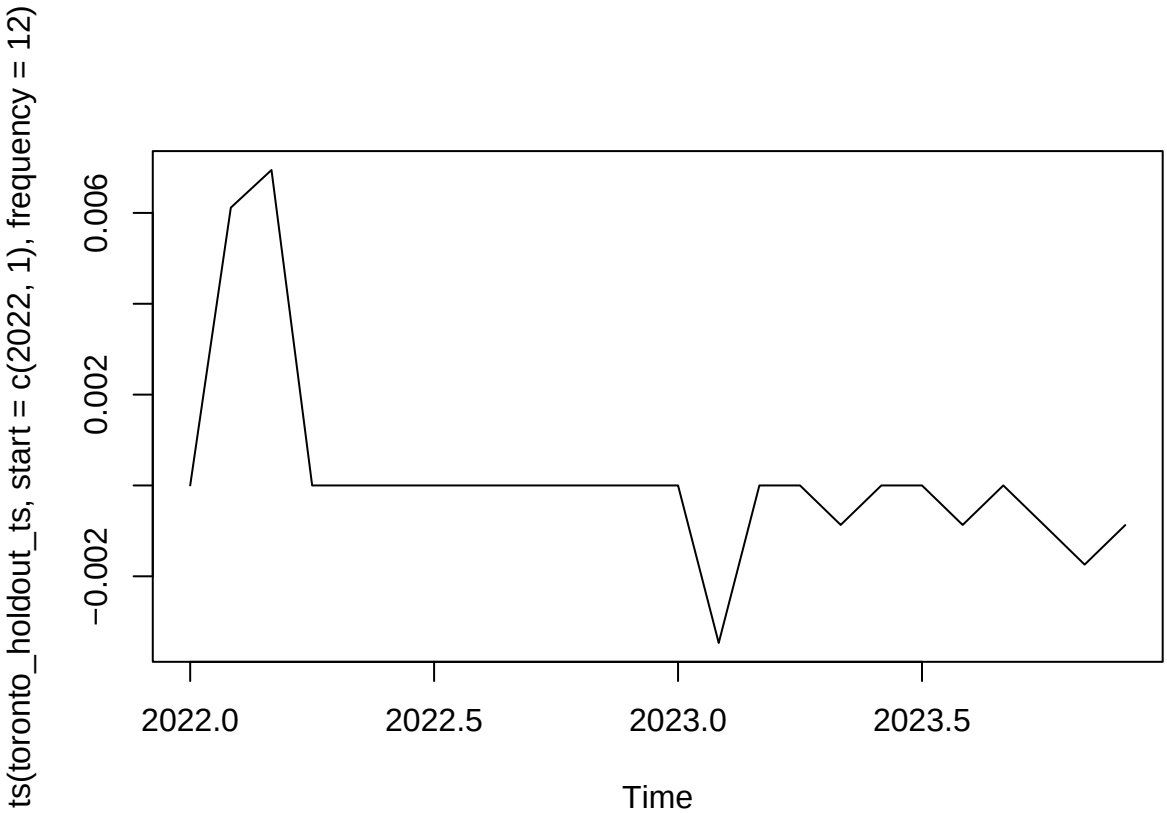

```r
# ts plot from ARIMA
plot(ts(arima_results$fc, start = c(2022, 1), frequency = 12))
```



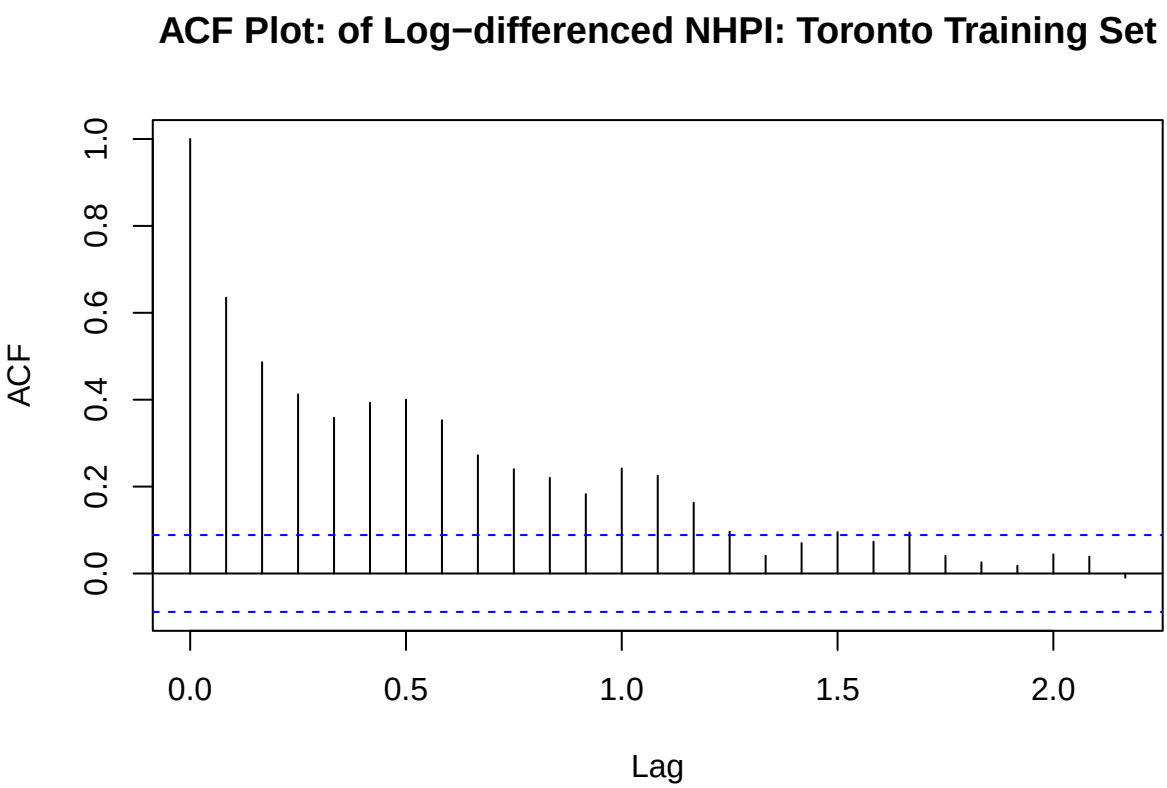
```r
# ts plot from ARIMAX
plot(ts(arimax_results$fc, start = c(2022, 1), frequency = 12))
```



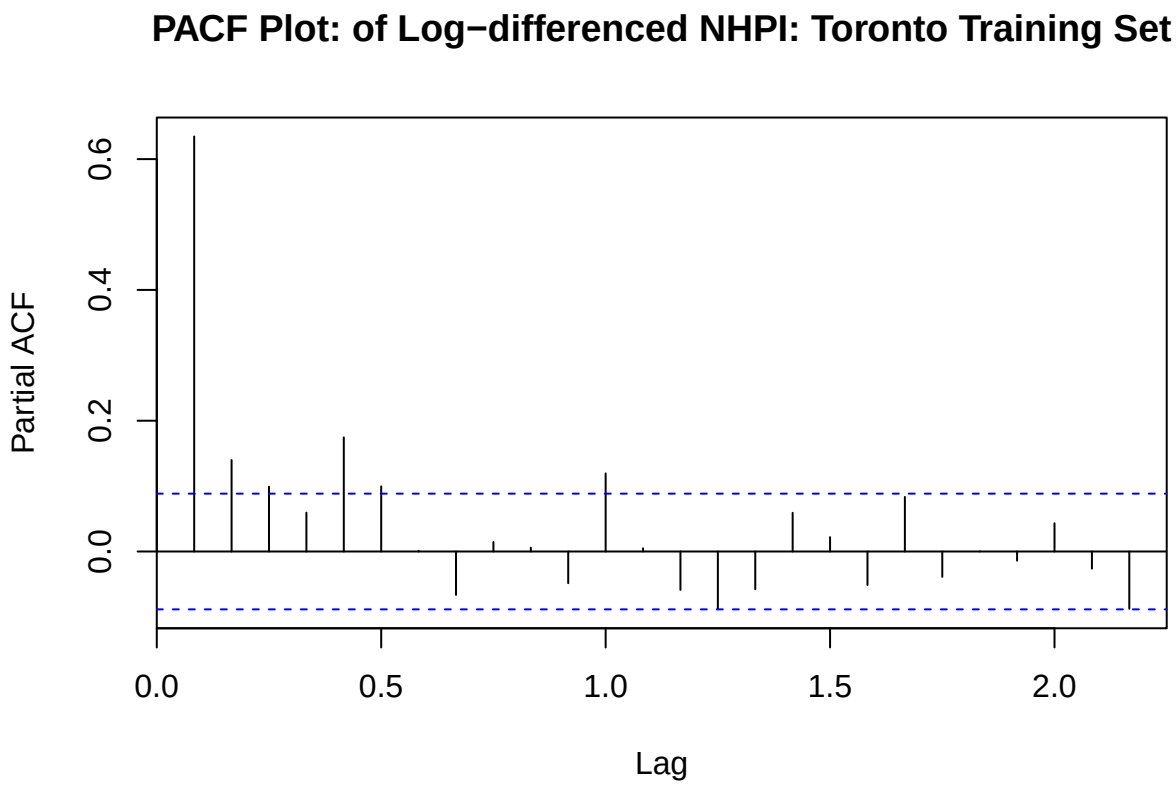

```r
# ts plot of holdout set
plot(ts(toronto_holdout_ts, start = c(2022, 1), frequency = 12))
```



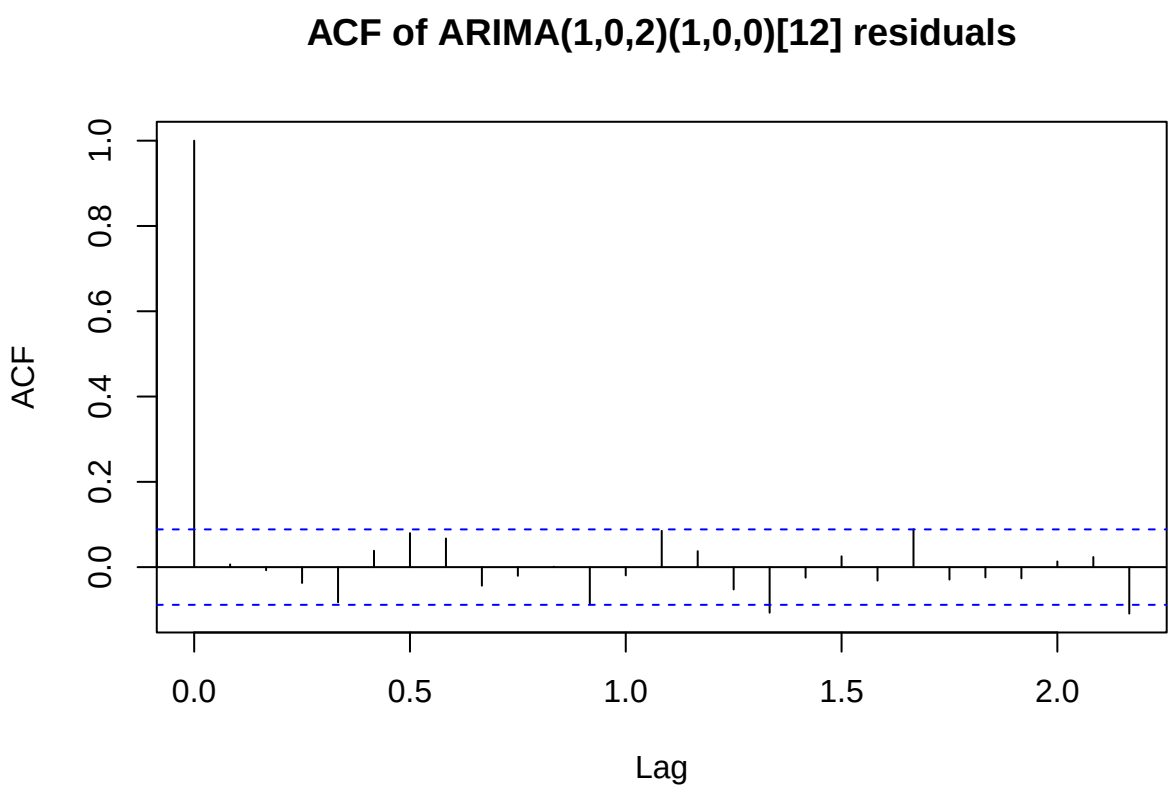

```
# 2. ACF PLOT
# ACF and PACF of training set
acf(toronto_train_ts, main = "ACF Plot: of Log-differenced NHPI: Toronto Training Set")
```



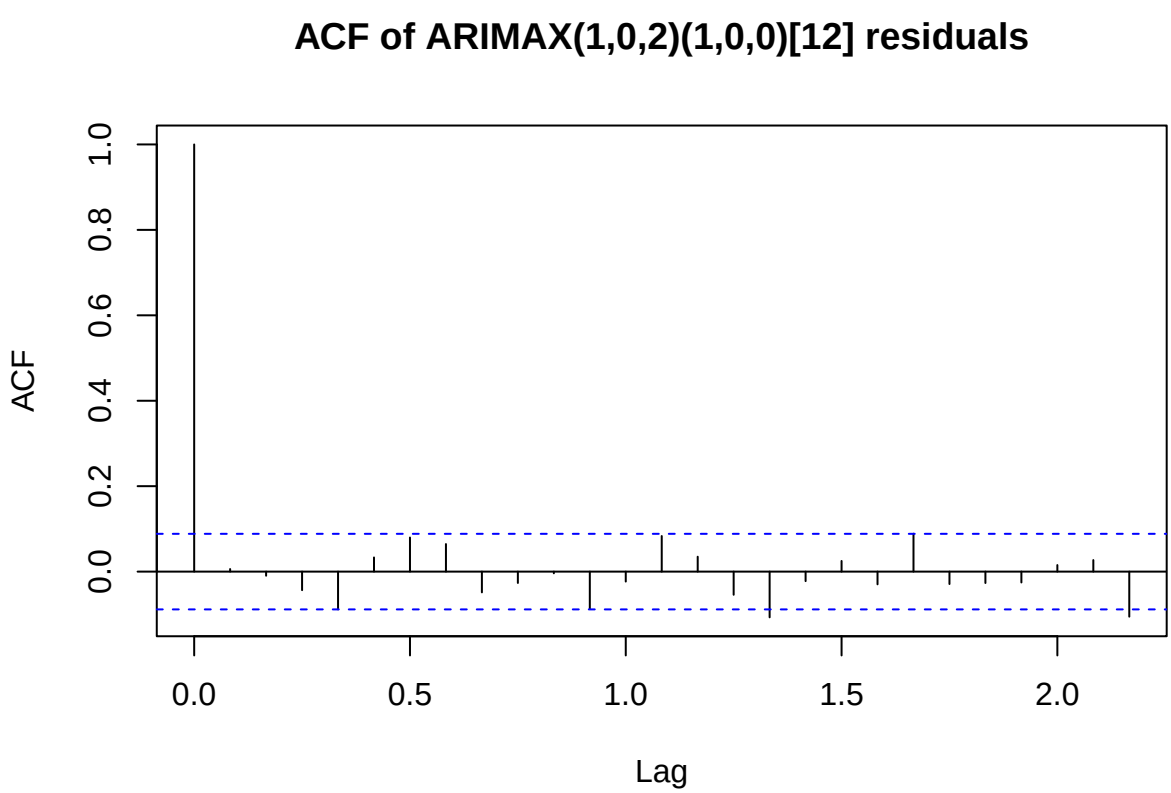
```r
pacf(toronto_train_ts, main = "PACF Plot: of Log-differenced NHPI: Toronto Training Set")
```




```r
# ACF from best ARIMA model
resid_fit <- residuals(toronto_transformed_nphi_auto)
acf(resid_fit, main = "ACF of ARIMA(1,0,2)(1,0,0)[12] residuals")
```




```r
# ACF from best ARIMAX model
resid_fit <- residuals(toronto_fit_arimax)
acf(resid_fit, main = "ACF of ARIMAX(1,0,2)(1,0,0)[12] residuals")
```



```r
# 3. SUMMARY STATISTICS
summary(toronto_train_ts)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -0.050552  0.000000  0.001820  0.002861  0.004292  0.061174
```

```r
summary(toronto_holdout_ts)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0034662 -0.0002171  0.0000000  0.0001822  0.0000000  0.0069445
```

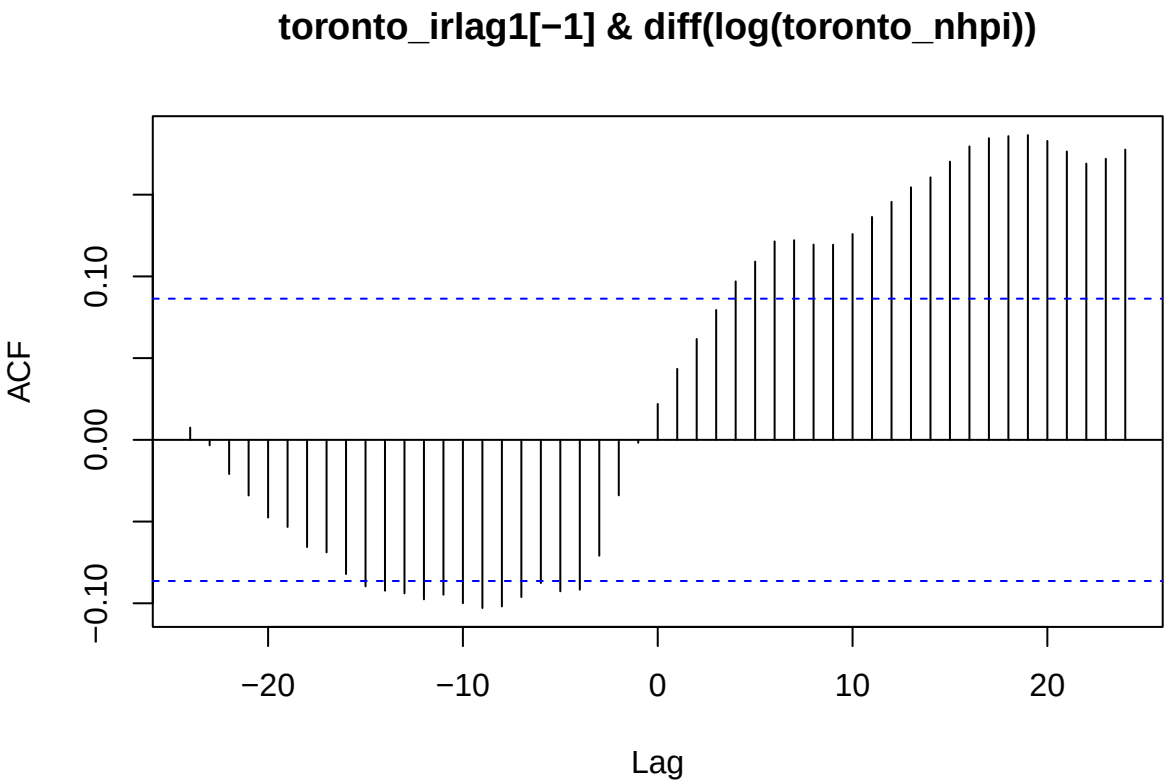
```r
# summary of best ARIMA model
summary(toronto_transformed_nphi_auto)
```

```
## Series: toronto_train_ts
## ARIMA(1,0,2)(1,0,0)[12] with non-zero mean
##
## Coefficients:
##          ar1      ma1      ma2    sar1    mean
##       0.9161  -0.4317  -0.1386  0.1316  0.0034
## s.e.  0.0302   0.0551   0.0490  0.0487  0.0017
##
## sigma^2 = 4.146e-05:  log likelihood = 1782.61
## AIC=-3553.23   AICc=-3553.05   BIC=-3528.05
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001299236 0.006405744 0.003868432 NaN  Inf 0.6292045
##                     ACF1
## Training set 0.006342322
```

```r
# summary of best ARIMAX model
summary(toronto_fit_arimax)
```

```
## Series: toronto_train_ts
## Regression with ARIMA(1,0,2)(1,0,0)[12] errors
##
## Coefficients:
##          ar1      ma1      ma2    sar1  intercept    xreg
##       0.9257  -0.4415  -0.1406  0.1330     0.0051  -3e-04
## s.e.  0.0329   0.0563   0.0488  0.0489     0.0028   4e-04
##
## sigma^2 = 4.147e-05:  log likelihood = 1783
## AIC=-3551.99   AICc=-3551.76   BIC=-3522.62
##
## Training set error measures:
##                         ME        RMSE         MAE MPE MAPE      MASE
## Training set -0.0001599343 0.006400238 0.003875959 NaN  Inf 0.6304288
##                     ACF1
## Training set 0.006084236
```

```r
# ccf between log-differenced NHPI and the lagged interest rate
ccf(toronto_irlag1[-1], diff(log(toronto_nhpi)))
```




```r
# 4. RMSE
rmse = cbind(toronto_persist_fc$rmse, toronto_iid_fc$rmse,
             toronto_esm_fc$rmse, arima_results$rmse, arimax_results$rmse)
colnames(rmse) = c("persist", "iid", "esm", "arima", "arimax")
print(round(rmse, 3))
```

```
##   persist   iid   esm arima arimax
## a   0.002 0.003 0.002 0.002  0.002
```

Analyzing Vancouver

Setting up

```r
# Reading the data
vancouver_nhpi_data <- read.csv("Full_Cities/vancouver_full.csv")
vancouver_nhpi_ir_data <- read.csv("Extracted_Cities/vancouver.csv")

# Extracting the NHPI
vancouver_nhpi <- vancouver_nhpi_data$VALUE

# Extracting the one-month lag interest rate
vancouver_irlag1 <- vancouver_nhpi_ir_data$IR_lag1
```

```r
# Setting the time window from 1981-2023
vancouver_ts <- ts(vancouver_nhpi, start = c(1981,1), end = c(2023,12), frequency = 12)
vancouver_ir_ts <- ts(vancouver_irlag1, start = c(1981,1), end = c(2023,12), frequency = 12)

# Applying a transformation for NHPI
vancouver_transformed_nhpi <- diff(log(vancouver_ts))

# Setting a training set from 1981-2021
vancouver_train_ts <- window(vancouver_transformed_nhpi, end = c(2021,12))
vancouver_train_ir_xreg <- window(vancouver_ir_ts, start = c(1981,2), end = c(2021,12))

# Setting a holdout set from 2022-2023
vancouver_holdout_ts <- window(vancouver_transformed_nhpi, start = c(2022,1))
vancouver_holdout_ir_xreg <- window(vancouver_ir_ts, start = c(2022,1))
```

Doing the baseline forecasts

```r
# Baseline 1: Naive/Persistence
vancouver_persist_fc <- persist_fc(vancouver_train_ts,vancouver_holdout_ts, iprint = FALSE)

# Baseline 2: Training-Set Average
vancouver_iid_fc <- iid_fc(vancouver_train_ts, vancouver_holdout_ts, iprint = FALSE)
```

Doing the exponential smoothing forecasts

```r
library(forecast)

# Method 1
ntrain <- length(vancouver_train_ts)
ntotal <- length(vancouver_transformed_nhpi)

vancouver_transformed_nhpi_train <- vancouver_transformed_nhpi[1:ntrain]
vancouver_transformed_nhpi_holdout <- vancouver_transformed_nhpi[(ntrain+1):ntotal]

# Simple Exponential Smoothing
fit_expsmo <- HoltWinters(vancouver_train_ts, beta = FALSE, gamma = FALSE)
vancouver_esm_fc <- esm_fc(vancouver_transformed_nhpi_train,
                           vancouver_transformed_nhpi_holdout,
                           alpha = fit_expsmo$alpha,
                           level = fit_expsmo$coefficients["a"],
                           iprint = FALSE)

# Linear Exponential Smoothing
fit_hw <- HoltWinters(vancouver_train_ts, gamma = FALSE)
vancouver_holt_fc <- lholt_fc(vancouver_transformed_nhpi_train,
                              vancouver_transformed_nhpi_holdout,
                              alpha = fit_hw$alpha,
                              beta = fit_hw$beta,
                              level = fit_hw$coefficients["a"],
```

```r
                              slope = fit_hw$coefficients["b"],
                              iprint = FALSE)

# Additive Seasonal Exponential Smoothing (using ETS for stability)
# We switched to ets() instead of HoltWinters() because it failed to converage for the Vancouver series
vancouver_aseason_fit <- ets(vancouver_train_ts, model = "AAA")
summary(vancouver_aseason_fit)
```

```
## ETS(A,Ad,A)
##
## Call:
##  ets(y = vancouver_train_ts, model = "AAA")
##
##   Smoothing parameters:
##     alpha = 0.239
##     beta  = 1e-04
##     gamma = 1e-04
##     phi   = 0.98
##
##   Initial states:
##     l = -0.0072
##     b = 1e-04
##     s = 0.0014 -0.0011 -0.002 -0.0019 0.001 -0.0031
##            0.0026 -8e-04 -7e-04 0.0017 0 0.0031
##
##   sigma:  0.0079
##
##       AIC      AICc       BIC
## -1687.583 -1686.134 -1612.047
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE      ACF1
## Training set 9.313466e-05 0.007801818 0.005032244 NaN  Inf 0.6916746 0.1151713
```

```r
vancouver_aseason_forecast <- forecast(vancouver_aseason_fit,
                                       h = length(vancouver_holdout_ts))

vancouver_aseason_rmse <- sqrt(mean((vancouver_holdout_ts -
                                     vancouver_aseason_forecast$mean)^2))

# Comparing RMSE
rmse_esm_vancouver <- cbind(vancouver_esm_fc$rmse,
                            vancouver_holt_fc$rmse,
                            vancouver_aseason_rmse)
colnames(rmse_esm_vancouver) <- c("esm", "lholt", "aseason")
rmse_esm_vancouver
```

```
##           esm     lholt     aseason
## a 0.004590305 0.0049477 0.007257523
```

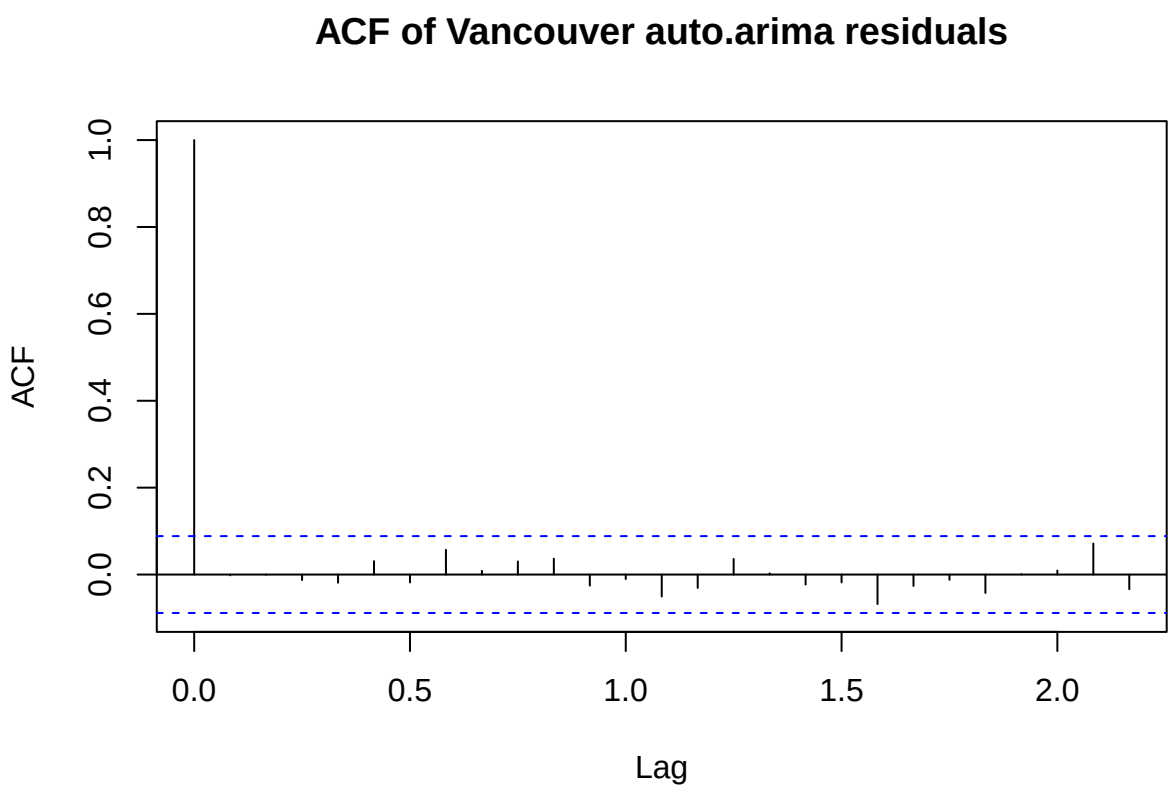
```r
# Best exponential smoothing method: Simple Exponential Smoothing
```

Doing the ARIMA forecasts

```
# Method 2
vancouver_auto_arima <- forecast::auto.arima(vancouver_train_ts)
summary(vancouver_auto_arima)
```

```
## Series: vancouver_train_ts
## ARIMA(2,0,1)(2,0,0)[12] with zero mean
##
## Coefficients:
##          ar1     ar2      ma1    sar1     sar2
##       0.5860  0.1859  -0.2732  0.0847  -0.0397
## s.e.  0.1429  0.0876   0.1451  0.0517   0.0535
##
## sigma^2 = 5.782e-05:  log likelihood = 1701.14
## AIC=-3390.28   AICc=-3390.1   BIC=-3365.1
##
## Training set error measures:
##                       ME        RMSE         MAE MPE MAPE      MASE        ACF1
## Training set 0.00019155 0.007565047 0.004533597 NaN  Inf 0.6231362 -0.00185311
```

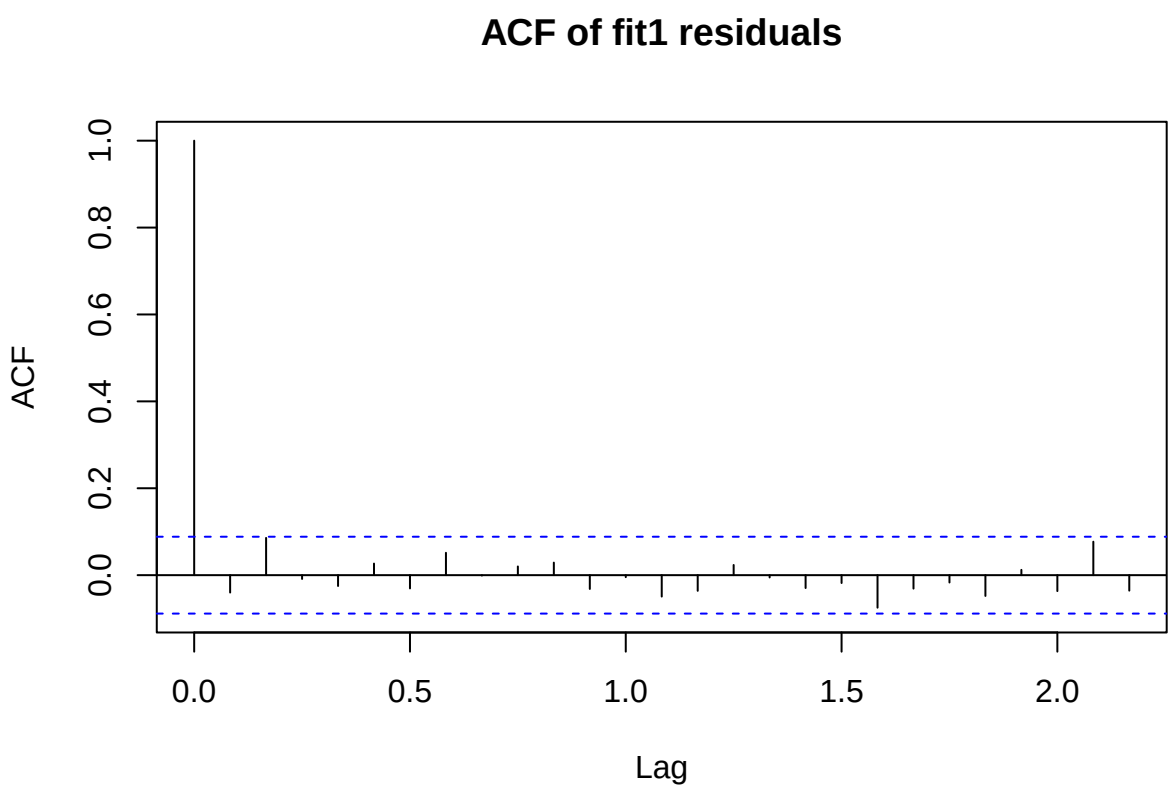
```
# Check residual ACF
acf(residuals(vancouver_auto_arima),
    main = "ACF of Vancouver auto.arima residuals")
```



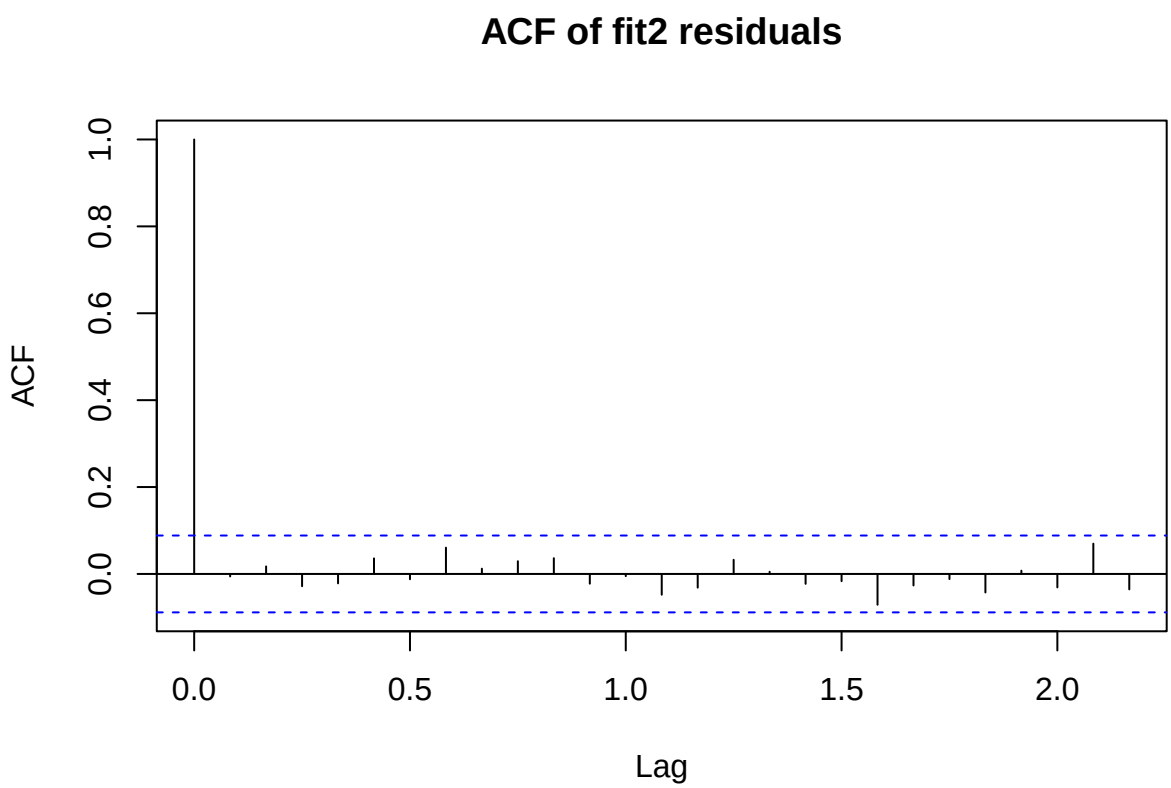
```r
# Fit several nearby candidate ARIMA models manually

fit1 <- arima(vancouver_train_ts,
              order = c(1,0,1),
              seasonal = list(order = c(1,0,0), period = 12),
              method = "ML")

fit2 <- arima(vancouver_train_ts,
              order = c(1,0,2),
              seasonal = list(order = c(1,0,0), period = 12),
              method = "ML")

fit3 <- arima(vancouver_train_ts,
              order = c(2,0,1),
              seasonal = list(order = c(1,0,0), period = 12),
              method = "ML")

fit4 <- arima(vancouver_train_ts,
              order = c(2,0,2),
              seasonal = list(order = c(1,0,0), period = 12),
              method = "ML")

fit5 <- arima(vancouver_train_ts,
              order = c(1,0,2),
              seasonal = list(order = c(1,0,1), period = 12),
              method = "ML")

# Compare AIC values
aic_vancouver <- cbind(fit1$aic, fit2$aic, fit3$aic, fit4$aic, fit5$aic)
colnames(aic_vancouver) <- c("fit1", "fit2", "fit3", "fit4", "fit5")
aic_vancouver
```

```
##           fit1     fit2      fit3      fit4      fit5
## [1,] -3387.783 -3389.46 -3390.148 -3388.239 -3387.991
```

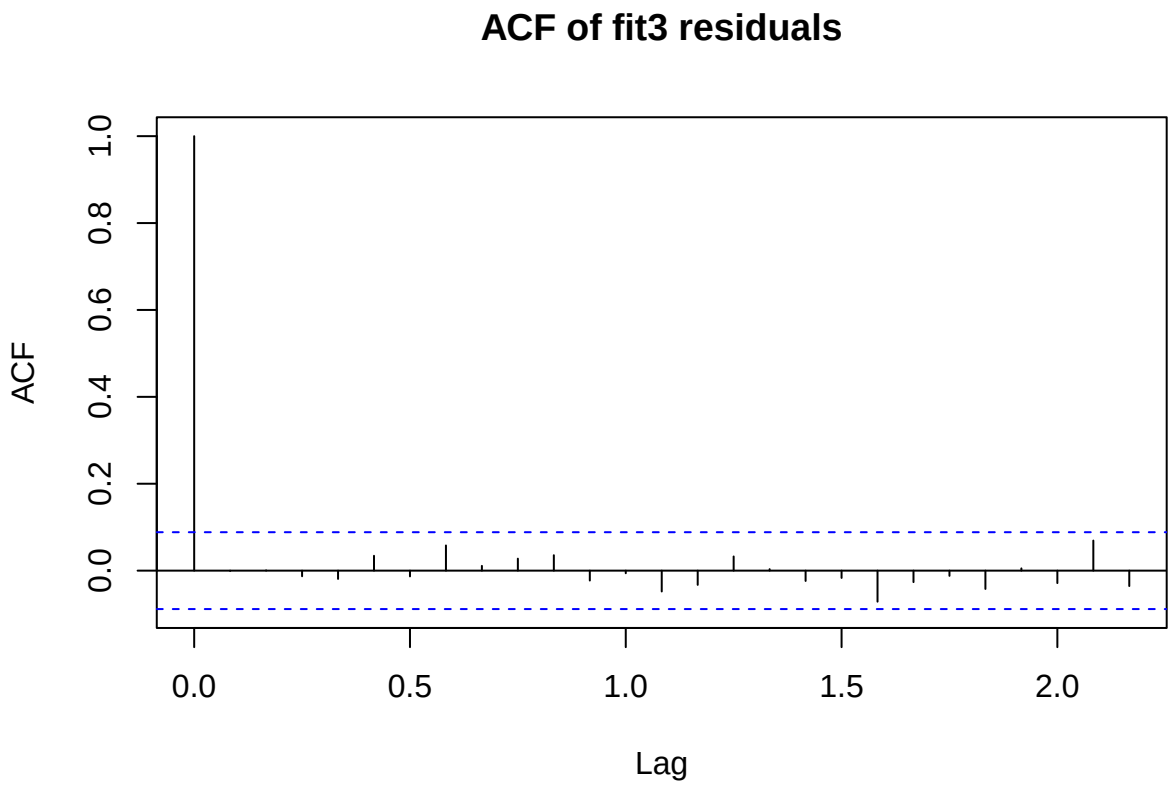
```r
# Check residual ACFs
acf(residuals(fit1), main = "ACF of fit1 residuals")
```



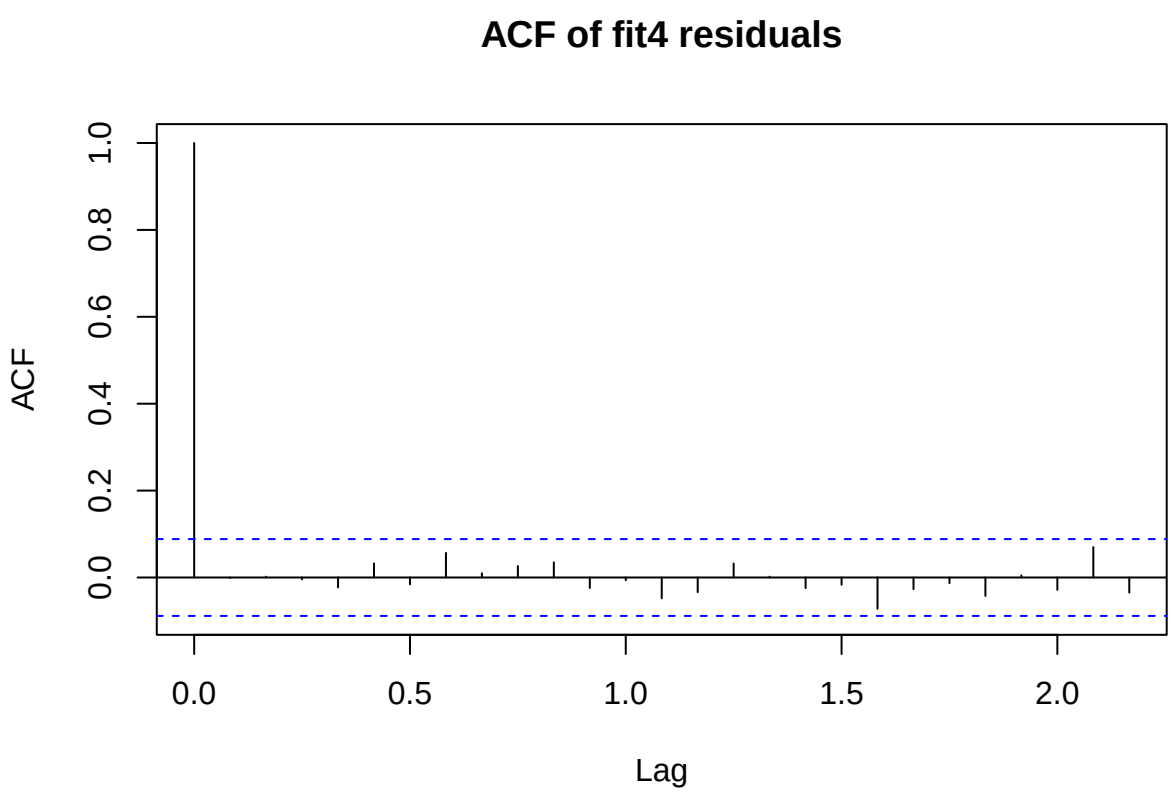

```
acf(residuals(fit2), main = "ACF of fit2 residuals")
```



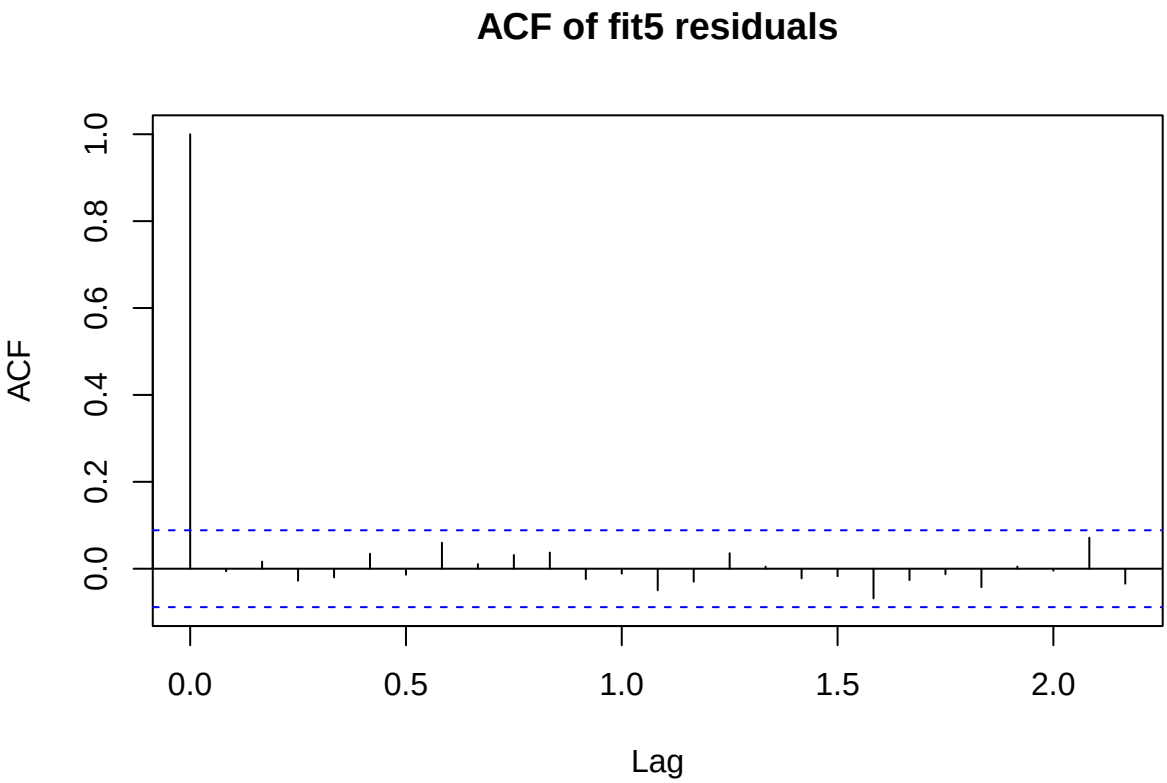
```r
acf(residuals(fit3), main = "ACF of fit3 residuals")
```




```r
acf(residuals(fit4), main = "ACF of fit4 residuals")
```



```r
acf(residuals(fit5), main = "ACF of fit5 residuals")
```




```r
# Choose the best candidate based on lowest AIC and ACF plot
# fit3 has smallest aic, so we choose fit3
order <- c(2,0,1)
seasonal <- list(order = c(1,0,0), period = 12)

# Fit the selected model on the training set
train_arima_model <- arima(vancouver_train_ts,
                           order = order,
                           seasonal = seasonal,
                           method = "ML",
                           include.mean = TRUE)

# Moving 1-step ahead forecasts for the holdout set
vancouver_arima_results <- arma_fc(tsdata = vancouver_transformed_nhpi,
                                   ntrain = length(vancouver_train_ts),
                                   order = order,
                                   seasonal = seasonal,
                                   method = "ML",
                                   traincoef = coef(train_arima_model),
                                   include.mean = TRUE,
                                   iprint = FALSE)

vancouver_arima_results$rmse
```

```
## [1] 0.004518064
```

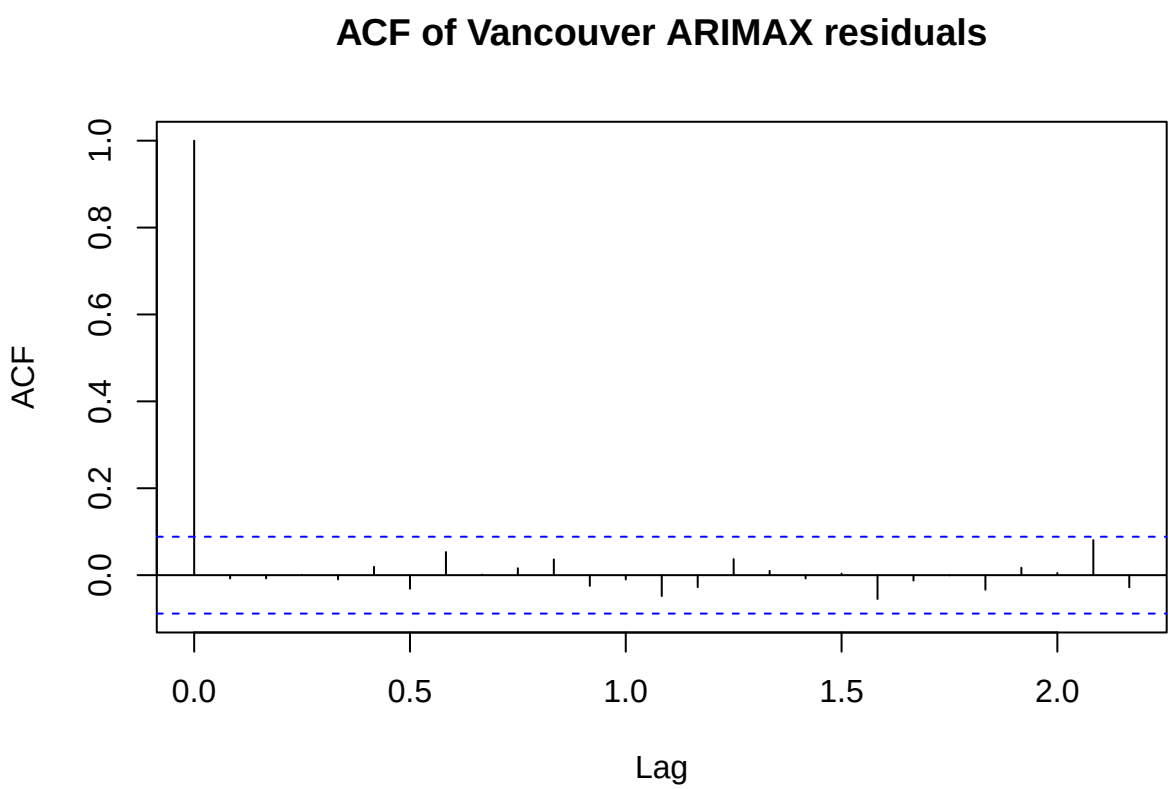
```r
#0.004518064
```

Doing the ARIMAX forecasts with lagged interest

```r
# Method 3
vancouver_fit_arimax <- forecast::auto.arima(vancouver_train_ts,
                                             xreg = vancouver_train_ir_xreg)
summary(vancouver_fit_arimax)
```

```
## Series: vancouver_train_ts
## Regression with ARIMA(1,0,3)(2,0,0)[12] errors
##
## Coefficients:
##          ar1      ma1     ma2      ma3    sar1     sar2  intercept    xreg
##       0.8510  -0.5583  0.1035  -0.0536  0.0849  -0.0216     0.0049  -8e-04
## s.e.  0.0591   0.0764  0.0541   0.0536  0.0522   0.0540     0.0019   3e-04
##
## sigma^2 = 5.695e-05:  log likelihood = 1706.4
## AIC=-3394.81   AICc=-3394.43   BIC=-3357.04
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -6.790793e-05 0.007484566 0.004592843 NaN  Inf 0.6312796
##                      ACF1
## Training set -0.007682121
```

```r
acf(residuals(vancouver_fit_arimax),
    main = "ACF of Vancouver ARIMAX residuals")
```



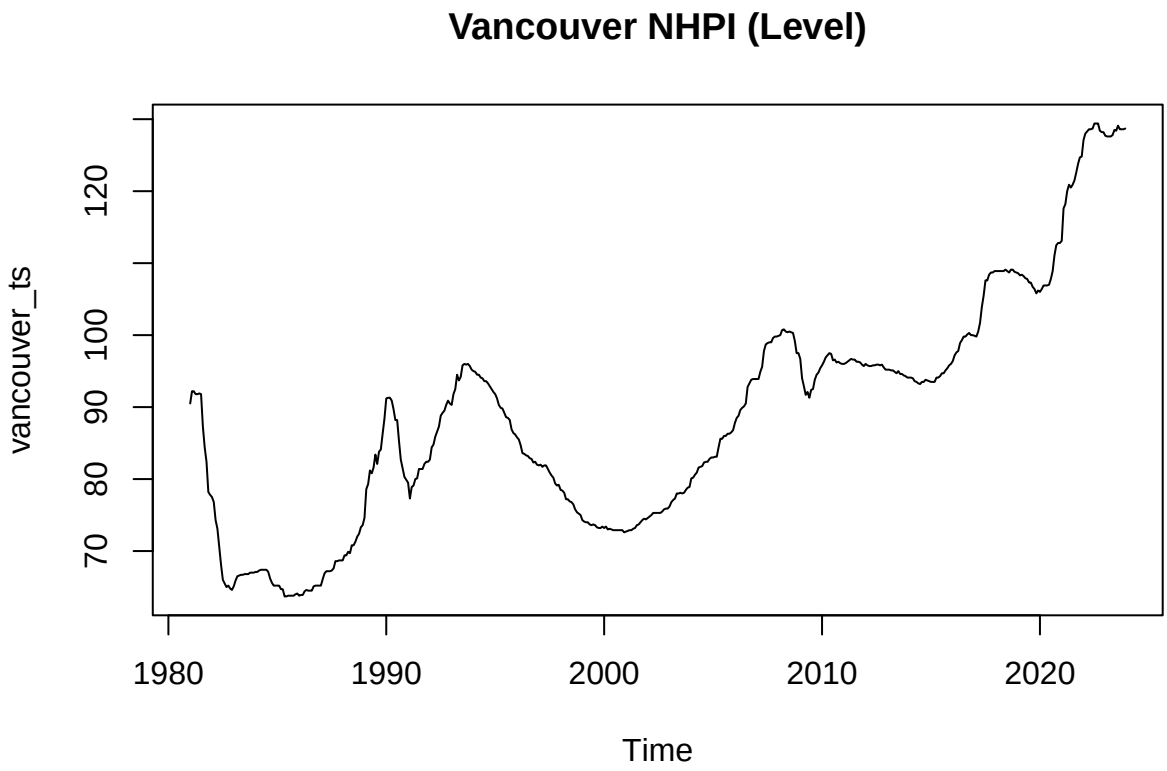

```
order <- c(2,0,1)
seasonal <- list(order = c(1,0,0), period = 12)

# Use the same ARIMA order as chosen above as a starting point
train_arimax_model_vancouver <- arima(vancouver_train_ts,
                                       order = order,
                                       seasonal = seasonal,
                                       xreg = vancouver_train_ir_xreg,
                                       method = "ML",
                                       include.mean = TRUE)

# Getting the result of the moving 1-step ahead forecast for holdout set
vancouver_arimax_results <- armax_fc(tsdata = vancouver_transformed_nhpi, ntrain = length(vancouver_trai
                                      order = order,
                                      seasonal = seasonal,
                                      xreg = vancouver_irlag1[-1],
                                      method = "ML",
                                      traincoef = coef(train_arimax_model_vancouver), include.mean = TRU

vancouver_arimax_results$rmse
```

```
## [1] 0.004444058
```

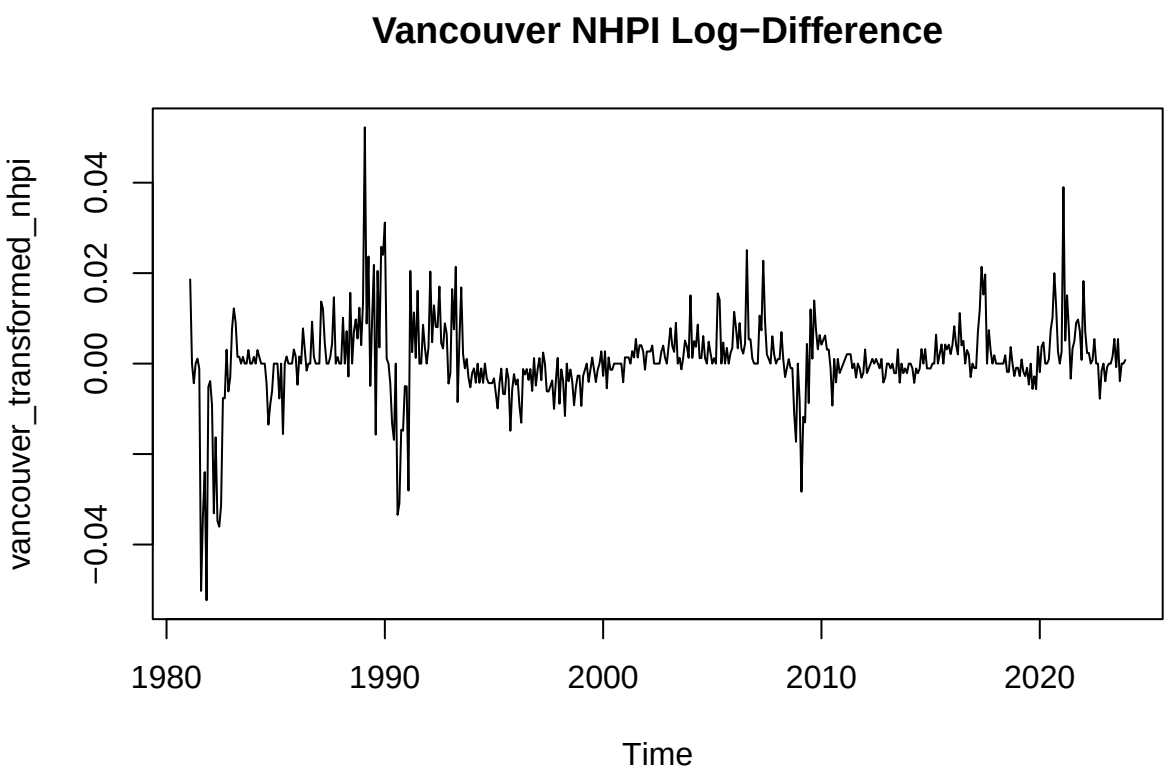
```
#0.004444058
```

Viewing the results of Vancouver's analyses

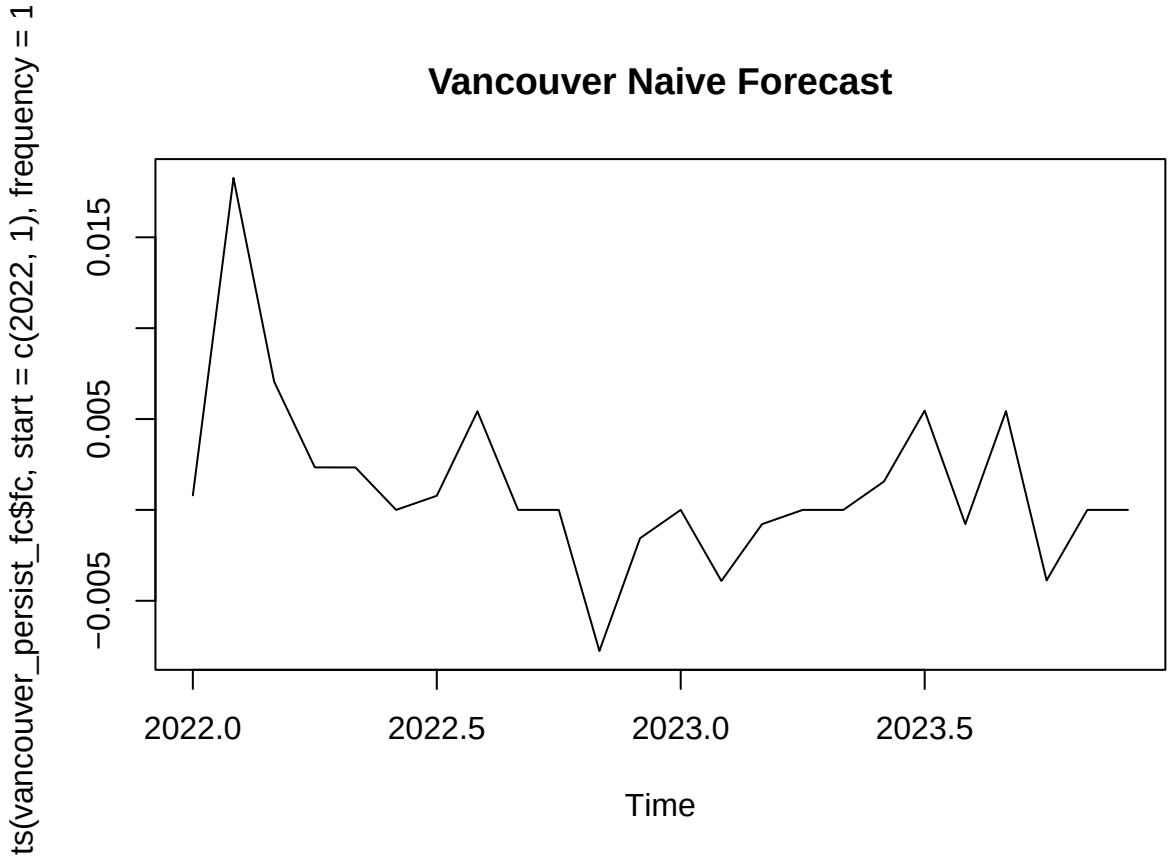
```
# 1. TIME SERIES PLOT
plot(vancouver_ts, main = "Vancouver NHPI (Level)")
```



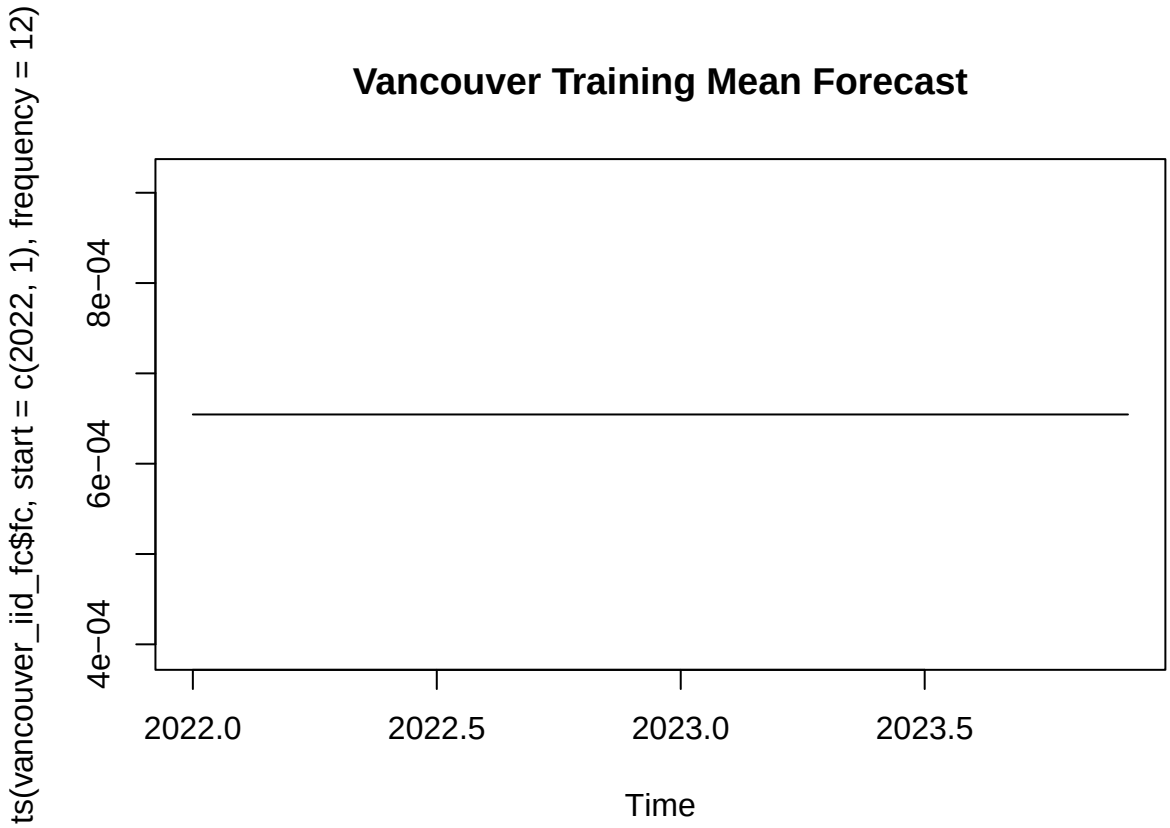

```
plot(vancouver_transformed_nhpi, main = "Vancouver NHPI Log-Difference")
```



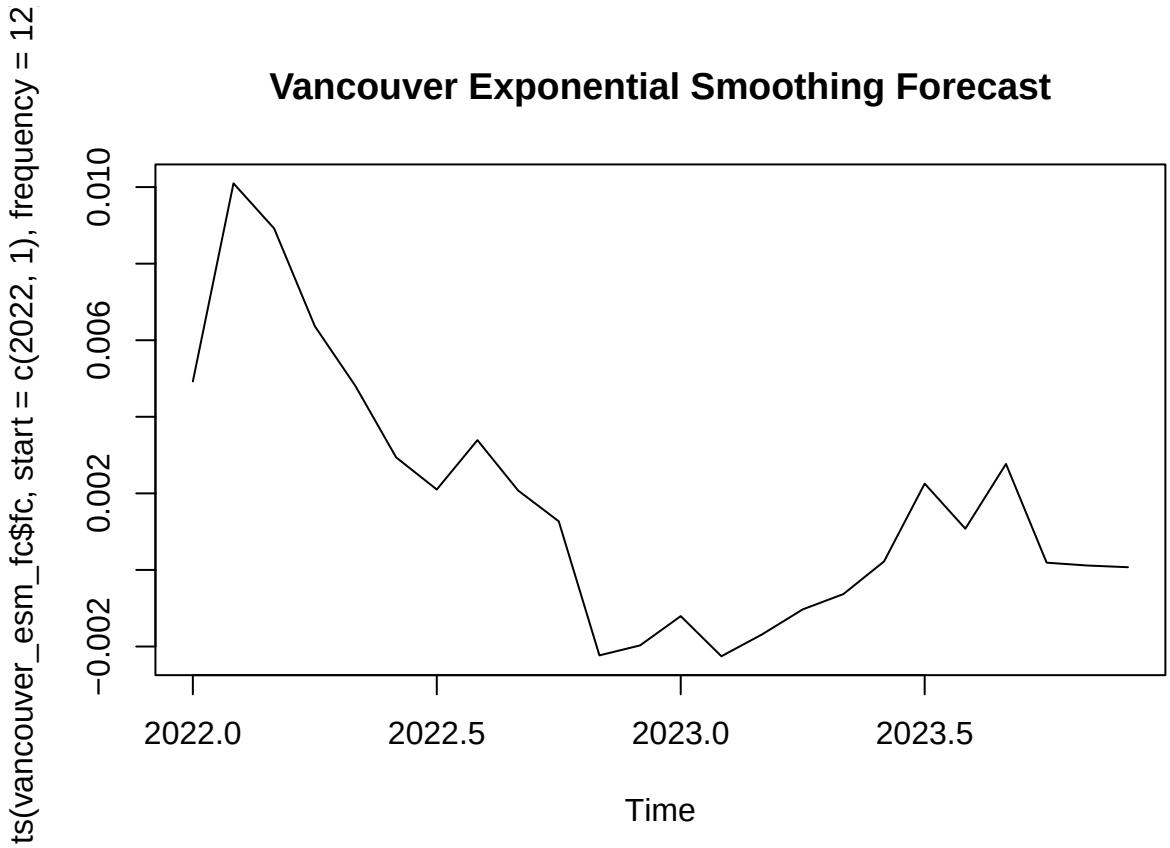

```
# Forecast plots
plot(ts(vancouver_persist_fc$fc, start = c(2022,1), frequency = 12),
     main = "Vancouver Naive Forecast")
```



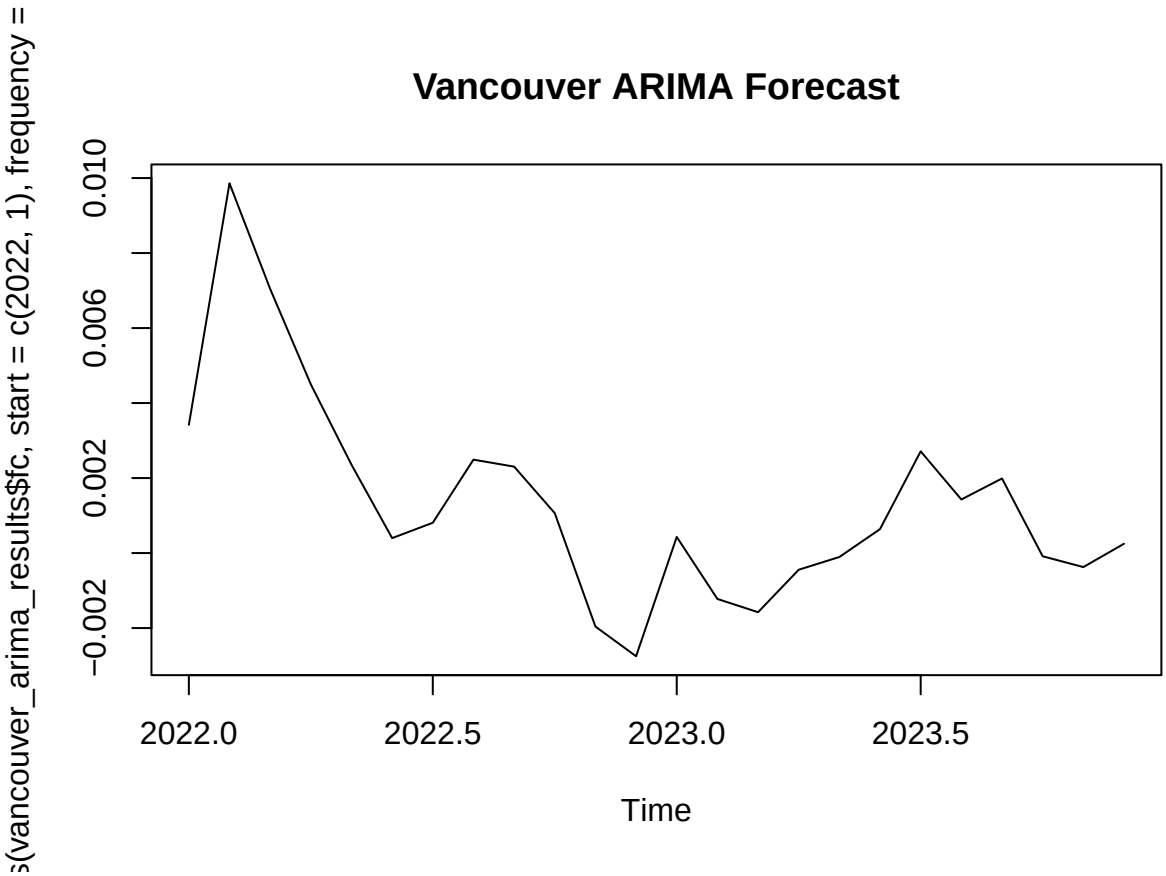

```
plot(ts(vancouver_iid_fc$fc, start = c(2022,1), frequency = 12),
     main = "Vancouver Training Mean Forecast")
```



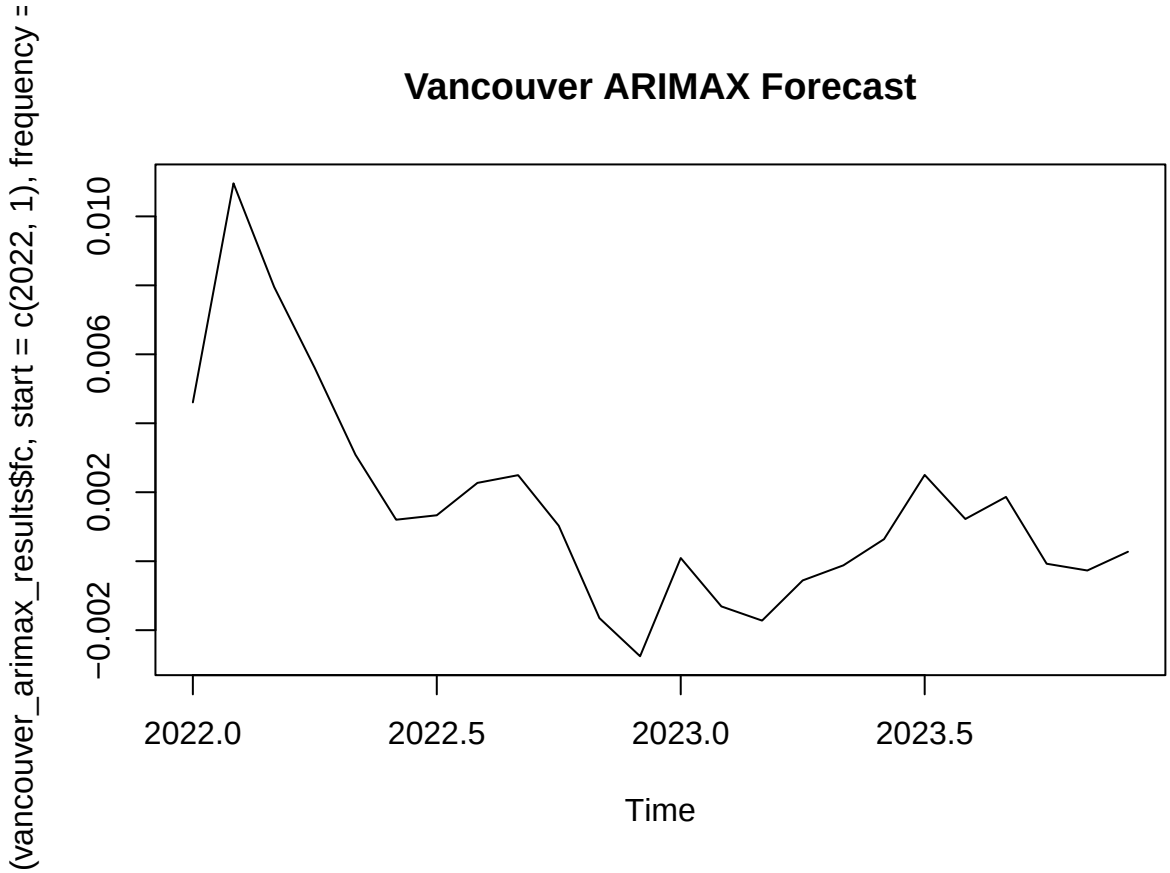

```
plot(ts(vancouver_esm_fc$fc, start = c(2022,1), frequency = 12),
     main = "Vancouver Exponential Smoothing Forecast")
```



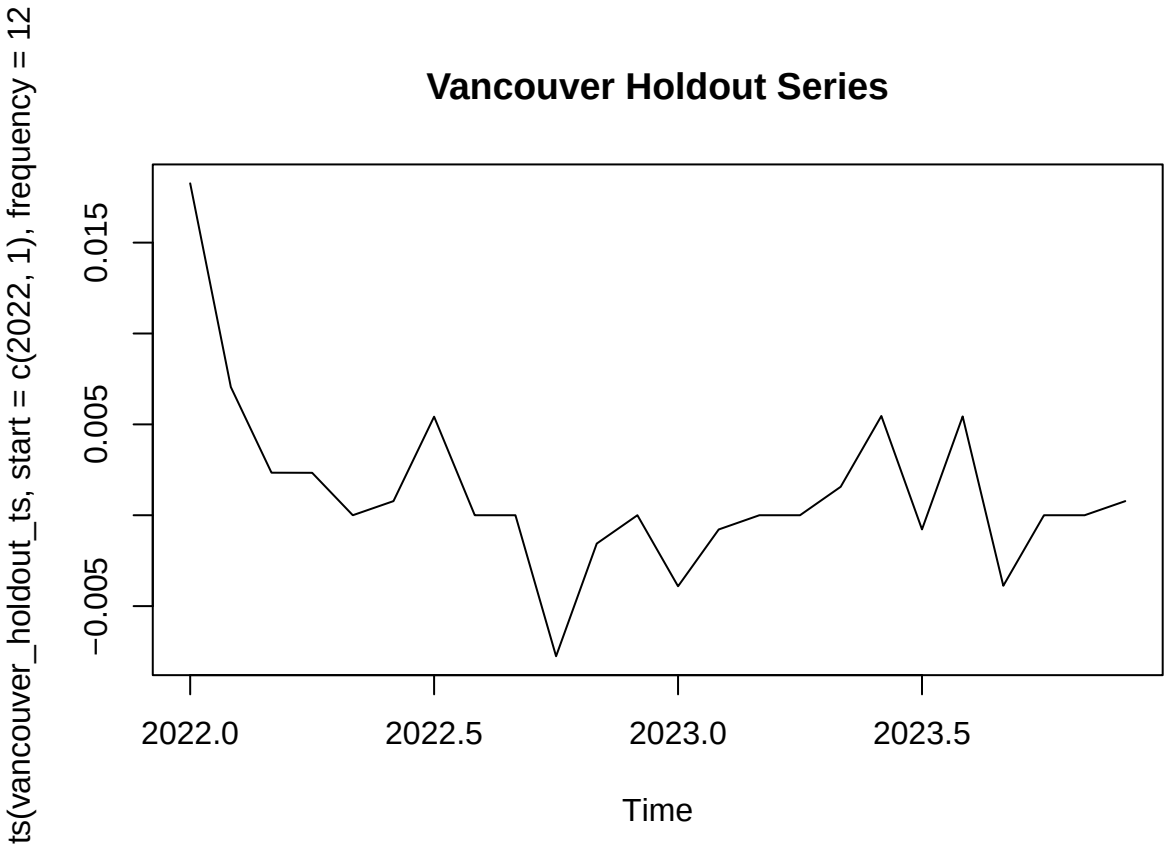

```
plot(ts(vancouver_arima_results$fc, start = c(2022,1), frequency = 12),
     main = "Vancouver ARIMA Forecast")
```



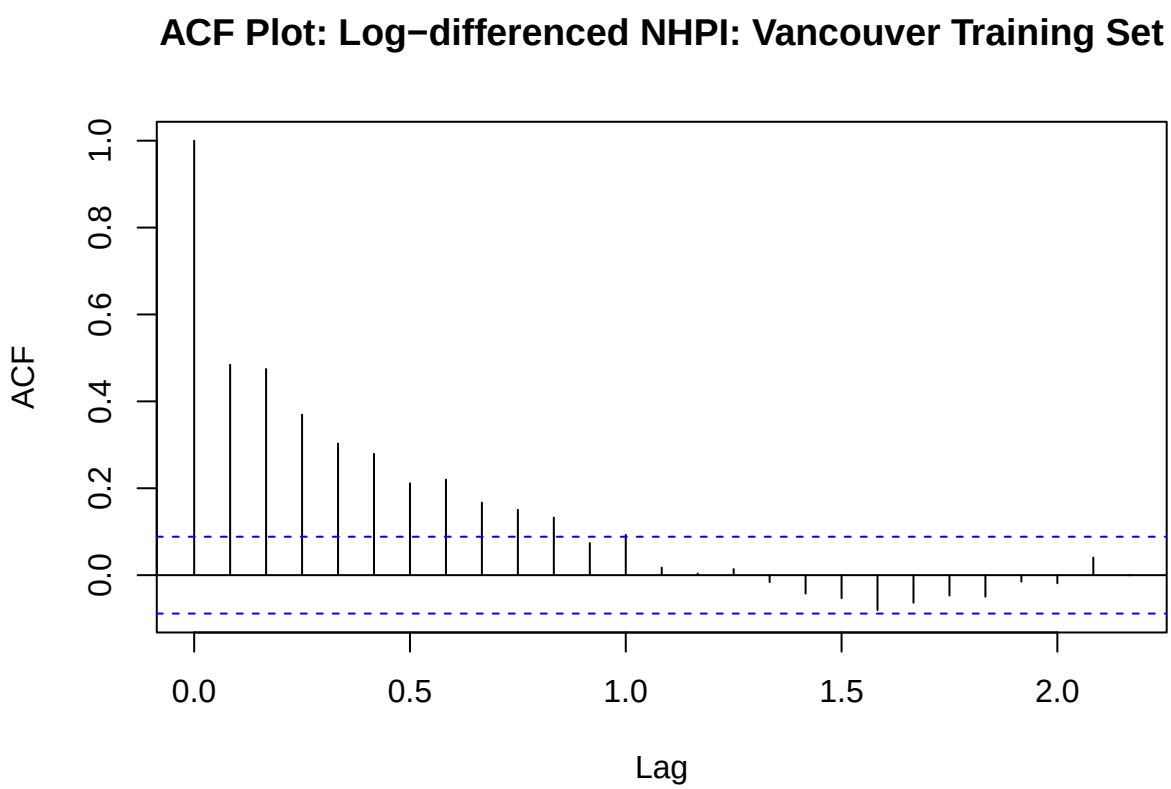

```
plot(ts(vancouver_arimax_results$fc, start = c(2022,1), frequency = 12),
     main = "Vancouver ARIMAX Forecast")
```



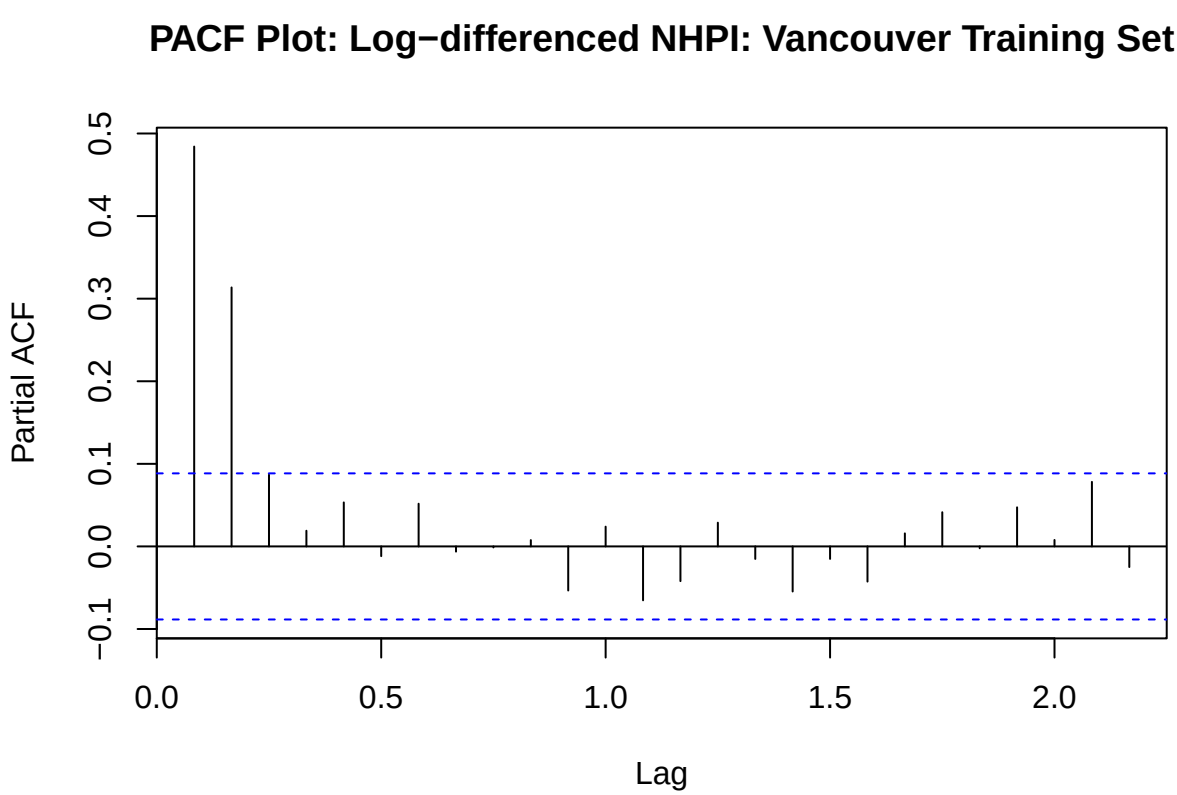

```
plot(ts(vancouver_holdout_ts, start = c(2022,1), frequency = 12),
     main = "Vancouver Holdout Series")
```



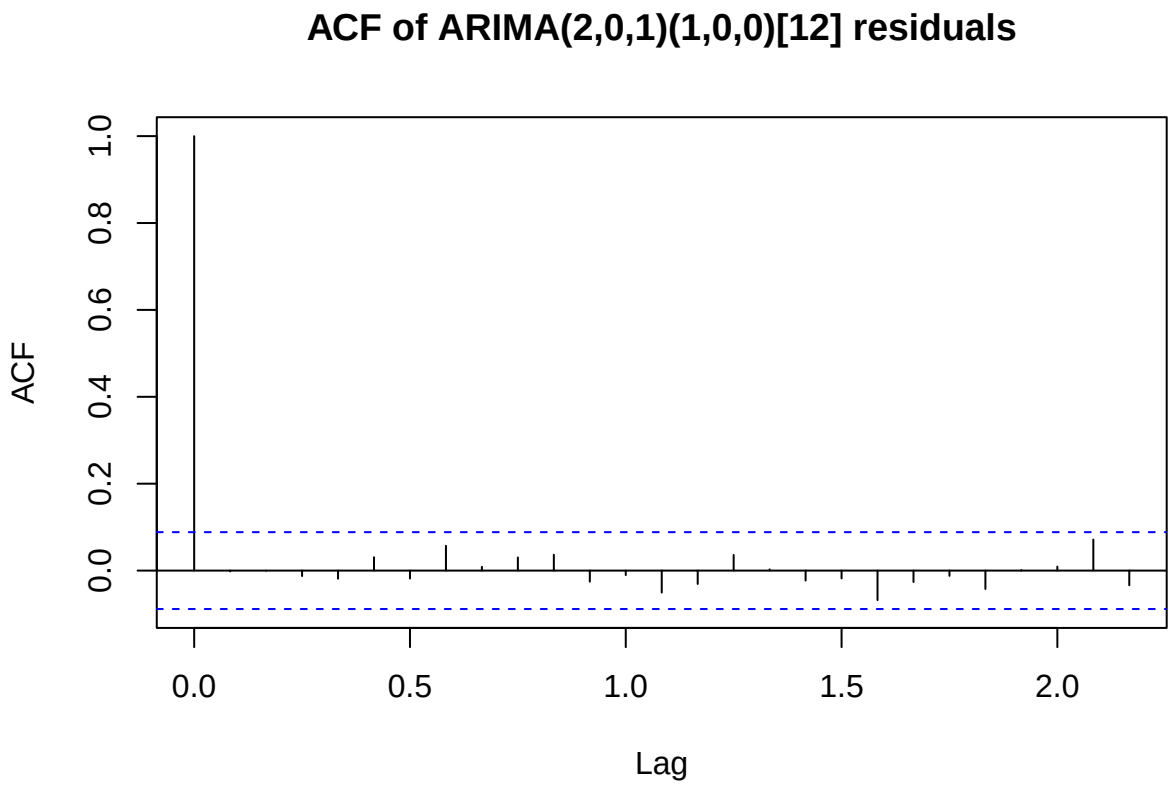

```r
# 2. ACF PLOT
acf(vancouver_train_ts,
    main = "ACF Plot: Log-differenced NHPI: Vancouver Training Set")
```



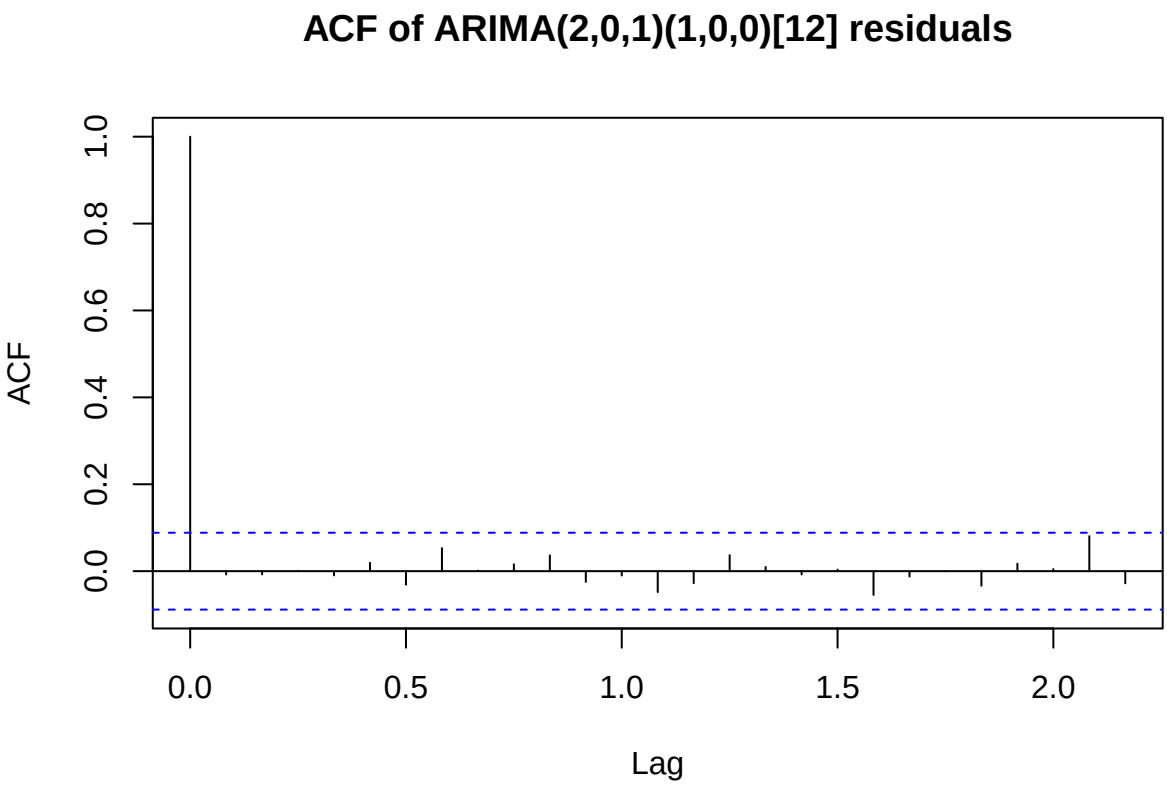

```
pacf(vancouver_train_ts,
     main = "PACF Plot: Log-differenced NHPI: Vancouver Training Set")
```



```r
acf(residuals(vancouver_auto_arima),
    main = "ACF of ARIMA(2,0,1)(1,0,0)[12] residuals")
```




```r
acf(residuals(vancouver_fit_arimax),
    main = "ACF of ARIMA(2,0,1)(1,0,0)[12] residuals")
```




```r
# 3. SUMMARY STATISTICS
summary(vancouver_train_ts)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0523158 -0.0020822  0.0000000  0.0006545  0.0035279  0.0522312
```

```r
summary(vancouver_holdout_ts)
```

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -0.0077580 -0.0001946  0.0000000  0.0012822  0.0023369  0.0182617
```

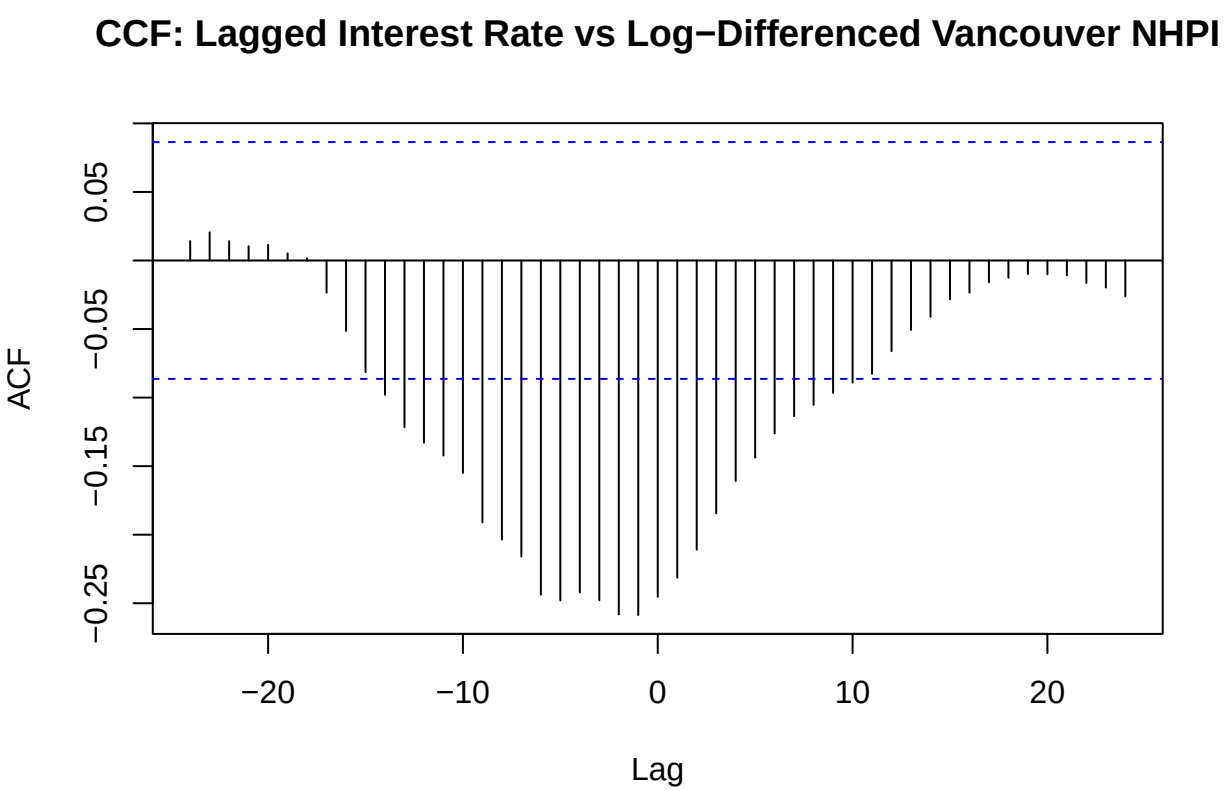
```r
summary(vancouver_auto_arima)
```

```
## Series: vancouver_train_ts
## ARIMA(2,0,1)(2,0,0)[12] with zero mean
##
## Coefficients:
##          ar1     ar2      ma1    sar1     sar2
##       0.5860  0.1859  -0.2732  0.0847  -0.0397
## s.e.  0.1429  0.0876   0.1451  0.0517   0.0535
##
## sigma^2 = 5.782e-05:  log likelihood = 1701.14
## AIC=-3390.28   AICc=-3390.1   BIC=-3365.1
##
## Training set error measures:
##                       ME        RMSE         MAE MPE MAPE      MASE        ACF1
## Training set 0.00019155 0.007565047 0.004533597 NaN  Inf 0.6231362 -0.00185311
```

```r
summary(vancouver_fit_arimax)
```

```
## Series: vancouver_train_ts
## Regression with ARIMA(1,0,3)(2,0,0)[12] errors
##
## Coefficients:
##          ar1      ma1     ma2      ma3    sar1     sar2  intercept    xreg
##       0.8510  -0.5583  0.1035  -0.0536  0.0849  -0.0216     0.0049  -8e-04
## s.e.  0.0591   0.0764  0.0541   0.0536  0.0522   0.0540     0.0019   3e-04
##
## sigma^2 = 5.695e-05:  log likelihood = 1706.4
## AIC=-3394.81   AICc=-3394.43   BIC=-3357.04
##
## Training set error measures:
##                        ME        RMSE         MAE MPE MAPE      MASE
## Training set -6.790793e-05 0.007484566 0.004592843 NaN  Inf 0.6312796
##                      ACF1
## Training set -0.007682121
```

```r
# CCF between log-differenced NHPI and lagged interest rate
ccf(vancouver_irlag1[-1], diff(log(vancouver_nhpi)),
    main = "CCF: Lagged Interest Rate vs Log-Differenced Vancouver NHPI")
```




```r
# 4. RMSE
vancouver_rmse <- cbind(vancouver_persist_fc$rmse,
                        vancouver_iid_fc$rmse,
                        vancouver_esm_fc$rmse,
```

```r
                        vancouver_arima_results$rmse,
                        vancouver_arimax_results$rmse)
colnames(vancouver_rmse) <- c("persist", "iid", "esm", "arima", "arimax")
print(round(vancouver_rmse, 3))
```

```
##   persist   iid   esm arima arimax
## a   0.006 0.005 0.005 0.005  0.004
```